

AirNav Risk Analysis Manual

A practitioner guide to twenty-six safety risk assessment methods for air navigation service provision

- Bowtie
- JHA
- FHA
- Fatigue modelling
- Collision risk (Reich)
- Occurrence risk (ERC / RAT)
- RBD and Markov
- Simulation (RTS / FTS)
- Investigation (SOAM / HFACS / Tripod)
- HAZID
- FMEA / FMECA
- STPA
- FRAM
- Event tree analysis
- GSN safety cases
- SWIFT
- Expert judgement
- Wildlife strike risk (species)
- HAZOP
- LOPA
- FTA
- Bayesian Belief Networks
- HRA (HEART / CARA)
- Common cause analysis
- Hierarchical task analysis
- Security risk

Document control

Item	Detail
Title	AirNav Risk Analysis Manual
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Ed.	Date	Description	Prepared	Approved
0.1	Sep 2026	First draft: framework and 12 method chapters	SQRM drafting team	—
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0.6	Sep 2026	Appendix E added: investigation expert system — ORLIO case library, fuzzy AcciMap inference, Bayesian network with ranked clues, AI-assisted drafting	SQRM drafting team	—
0.7	Oct 2026	Appendix F added: ATSB safety investigation analysis (tests for existence, influence, importance; safety issues on the AirNav 5×5 or the ATSB 6×6; safety action; review). Appendix E aligned to ATSB terms and codes	SQRM drafting team	—
0.8	Oct 2026	Appendix F: F.10 worked example from KNKT final report KNKT.24.10.22.04 (ground collision, Kualanamu) and F.11 investigation report structure (ORLIO sections, corrective actions by the hierarchy of controls); §15.5 Samn-Perelli fatigue check with action thresholds; Appendix G worked example of surveillance radar performance degradation (all relevant methods by phase and hazard type)	SQRM drafting team	—

How to use this manual

Part A sets out the common framework that every AirNav safety risk assessment must follow: the risk management process, the risk classification scheme, how to choose a method and how to run a workshop. Part B contains one chapter per method. Every method chapter follows the same pattern — *at a glance*, key concepts, step-by-step procedure, worksheet template, a worked ANSP example, tips and pitfalls, and key references — so that an assessor can pick up any chapter on its own. Part C explains how to combine outputs into a Safety Assessment Report and keep them alive in the hazard log. Appendices C and D give complete, ready-to-use method chains for two common types of change — a new AIM system and a major organisational change — as examples of how the methods combine.

All numbers in the worked examples (failure rates, probabilities, frequencies, conditional probability tables) are **illustrative**. They were chosen to demonstrate the arithmetic and must not be quoted as AirNav data. Real assessments must use AirNav occurrence data, equipment reliability records, manufacturer data or documented expert judgement, and state their source.

Words such as **shall** indicate a requirement of this manual; **should** indicates recommended practice; **may** indicates an option.

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1. Introduction

1.1 Purpose

An air navigation service provider (ANSP) manages risk every time it changes a system, a procedure, its airspace or its organisation, and every time it learns from an occurrence. ICAO Annex 19 requires service providers to implement a Safety Management System (SMS) in which hazards are identified and the associated safety risks are assessed and mitigated. ICAO does not prescribe which analytical method to use; the choice is left to the service provider, provided the method is appropriate to the nature and complexity of the change or the question being asked.

This manual gives AirNav staff a single, consistent reference for twenty-six methods that are either widely used in ATM or that bring a capability the classic tools lack. It is written for safety specialists, facilitators and the operational and technical subject-matter experts who take part in assessments.

1.2 Regulatory and guidance basis

Source	Relevance to this manual
ICAO Annex 19 — Safety Management [R1]	Framework of four SMS components; hazard identification and safety risk assessment and mitigation are mandatory elements of the Safety Risk Management (SRM) component.
ICAO Doc 9859 — Safety Management Manual, 4th ed. 2018 [R2]	Guidance on hazard identification, safety risk probability and severity, the risk matrix and tolerability, and management of change. Basis of Chapter 3.
CASR Part 19 — Safety Management System, issued by Minister of Transportation Regulation PM 2 Tahun 2026 [R3]	National SMS requirement for Indonesian aviation service providers, structured around safety policy and objectives, safety risk management, safety assurance and safety promotion.
ICAO Annex 11 and Doc 9966 [R4, R5]; ICAO/CANSO/IFATCA Fatigue Management Guide for ATS Providers [R6]	Requirements and guidance for managing ATC fatigue by prescriptive limits and/or a Fatigue Risk Management System. Basis of Chapter 15.
EUROCONTROL SAM and ESARR 4 [R7, R8]	Widely used ATM safety assessment lifecycle (FHA → PSSA → SSA) and ATM severity classification; good practice reference for Chapter 12.
FAA ATO SMS Manual and SRM Guidance [R9]	ANSP example of an SRM process that uses a tool-box of hazard analysis methods.
ISO 31000:2018 and IEC 31010:2019 [R10, R11]	Generic risk management principles and the catalogue of risk assessment techniques.
Method standards [R12–R67]	Bowtie (UK CAA), HAZID (ISO 17776), HAZOP (IEC 61882), JHA (OSHA 3071), FMEA/FMECA (IEC 60812), LOPA (CCPS), FHA (SAE ARP4761A, EUROCONTROL SAM), STPA (Leveson & Thomas), FTA (IEC 61025), fatigue models, FRAM (Hollnagel), BBN (Fenton & Neil); CRM (ICAO Doc 9689, 9574), ETA (IEC 62502), HEART and CARA, ARMS ERC and EUROCONTROL RAT, GSN Community Standard, common cause analysis (ARP4761A), RBD (IEC 61078) and Markov (IEC 61165), SWIFT, HTA, E-OCVM, Cooke's classical model, ISO/IEC 27005 and ED-205, SOAM, HFACS and Tripod Beta; wildlife hazard management (ICAO Doc 9137 Part 3, CAP 772, Allan species risk matrix).

1.3 Key definitions

Term	Definition used in this manual
Hazard	A condition or an object with the potential to cause or contribute to an aircraft incident or accident (ICAO). Example: “aircraft and vehicles moving on or near an active runway”.
Safety risk	The predicted probability and severity of the consequences or outcomes of a hazard (ICAO).
Consequence / outcome	The potential end state of a hazard, e.g. runway collision, mid-air collision, controlled flight into terrain.
Risk control (barrier, mitigation, safeguard)	A measure that reduces the probability and/or the severity of a consequence. Different methods use different names: barrier (bowtie), safeguard (HAZOP), independent protection layer (LOPA), control / safety constraint (STPA).
Safety objective / safety requirement	A statement of the maximum tolerable frequency of a hazard (objective) and the means to achieve it (requirement), used in the FHA/PSSA lifecycle.
ALARP	As low as reasonably practicable: a tolerable risk must be reduced further unless the cost of doing so is grossly disproportionate to the benefit.
Hazard log	The controlled register of identified hazards, their assessed risk, controls, owners and status (see Part C).

2. The safety risk management process

Every assessment, whatever method is used, follows the six-step process in Figure 1. The methods in Part B are tools that support steps 2 to 5; none of them replaces the process.

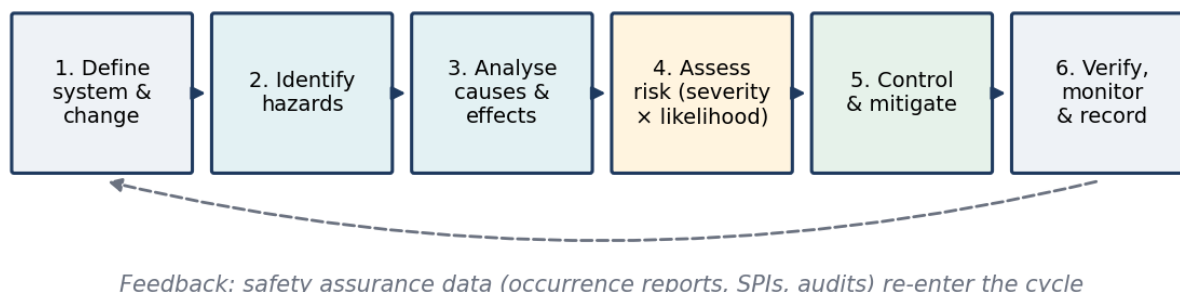


Figure 1. AirNav safety risk management process (after ICAO Doc 9859 and ISO 31000).

2.1 When a safety assessment is required

- Introduction or modification of ATM, CNS or supporting systems (e.g. new ATM automation, ADS-B, VCCS, AMHS, power systems).
- New or changed ATS procedures, airspace structure, routes, instrument flight procedures, sector configuration or opening schemes.
- Organisational changes affecting operations: manning levels, rosters and shift patterns, relocation of units, contracting of services, training arrangements.
- Temporary changes: works on the manoeuvring area, degraded modes, contingency operations, special events.
- After a serious incident or a trend in safety performance indicators (SPIs) that suggests a control is not working.

2.2 Roles

Role	Responsibilities
Change sponsor / system owner	Describes the change, provides design and operational information, owns the resulting safety requirements and actions.
Safety assessor / facilitator	Selects the method(s), prepares and runs the sessions, records results, challenges assumptions, drafts the Safety Assessment Report.
Subject-matter experts (SMEs)	Current operational ATCOs, supervisors, ATSEP/engineers, AIM and procedure designers, and where relevant pilots, airport operator and military. They bring work-as-done knowledge.
Safety Manager	Assures the quality and consistency of assessments, maintains the hazard log, checks that the risk classification scheme is applied correctly.
Risk acceptance authority	Accepts residual risk at the level set in Table 3.3, up to the Accountable Executive for the highest tolerable risks.

2.3 Describing the system

The single most common cause of a weak assessment is an unclear scope. Before any method is applied, the assessor shall document: the purpose of the change; the system boundary (people, procedures, equipment, environment and interfaces — the “SHELL” elements); the operational environment (airspace class, traffic volume and mix, weather, time of day); the assumptions made; and the before/after comparison where the change modifies an existing service.

3. Risk classification scheme

AirNav uses a 5 × 5 qualitative matrix based on the example in ICAO Doc 9859 [R2]. Severity is always assessed for the **most credible** outcome, not the worst conceivable one; where two outcomes are both credible (for example a minor loss of separation that is likely and a mid-air collision that is remote), both should be assessed and the higher risk index retained.

3.1 Likelihood

Level	Descriptor	Qualitative meaning	Indicative quantitative band (per operating hour of the affected unit/system)*
5	Frequent	Likely to occur many times; has occurred frequently	$> 10^{-3}$
4	Occasional	Likely to occur sometimes; has occurred infrequently	10^{-5} to 10^{-3}
3	Remote	Unlikely to occur, but possible; has occurred rarely	10^{-7} to 10^{-5}
2	Improbable	Very unlikely to occur; not known to have occurred	10^{-9} to 10^{-7}
1	Extremely improbable	Almost inconceivable that the event will occur	$< 10^{-9}$

Calibration. * The quantitative bands are an AirNav proposal to allow quantitative methods (FTA, LOPA, BBN) to be mapped onto the matrix. They must be calibrated against AirNav traffic exposure and approved by the Safety Review Board before use. ICAO Doc 9859 gives only qualitative descriptors.

3.2 Severity

Level	Descriptor	ATM meaning (examples)
A	Catastrophic	Accident: mid-air collision, runway collision, CFIT; multiple fatalities; equipment destroyed.
B	Hazardous	Serious incident: large reduction in safety margins, e.g. separation lost with high collision risk (ACAS RA required to avoid collision), runway incursion with narrowly avoided collision; serious injury.
C	Major	Major incident: significant reduction in safety margins, e.g. separation lost but controlled recovery; ATCO or crew workload significantly increased; total loss of a service with fallback in place.
D	Minor	Significant incident: slight reduction in margins, e.g. separation infringement with no collision risk, nuisance alerts, minor operating limitations.
E	Negligible	No immediate effect on safety; little consequence.

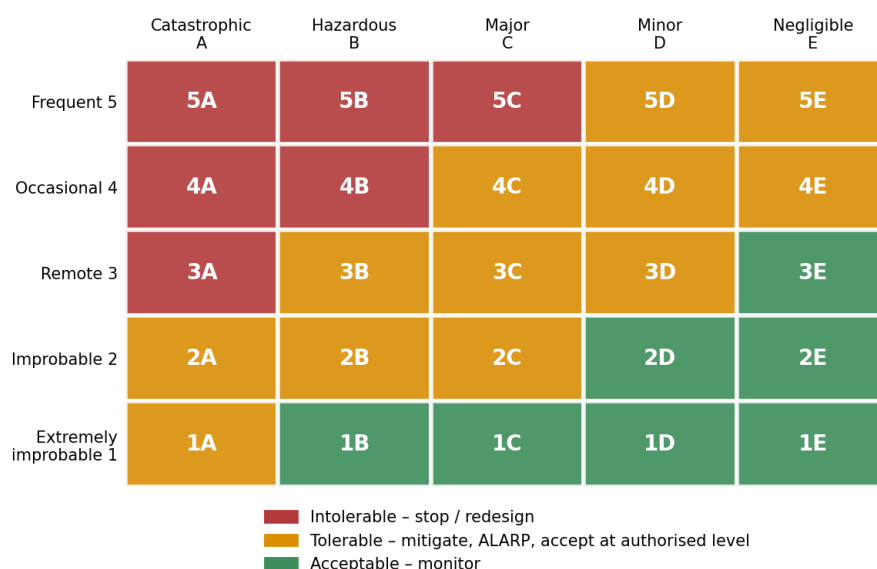


Figure 2. AirNav safety risk matrix and tolerability regions (adapted from the ICAO Doc 9859 example).

3.3 Tolerability and acceptance authority

Region	Risk indices	Required action	Acceptance authority
Intolerable	5A, 5B, 5C, 4A, 4B, 3A	Do not introduce the change or continue the operation. Redesign or add controls until the risk moves into the tolerable region.	Cannot be accepted
Tolerable (upper)	5D, 4C, 3B, 2A	Additional controls required; demonstrate ALARP; risk may be accepted only with a documented justification and monitoring plan.	Accountable Executive (President Director) on advice of Safety Review Board
Tolerable (lower)	5E, 4D, 4E, 3C, 3D, 2B, 2C, 1A	Mitigate where reasonably practicable; document ALARP argument.	Director of Operations / Director of Engineering (as applicable)
Acceptable	3E, 2D, 2E, 1B, 1C, 1D, 1E	Accept; monitor through normal safety assurance.	Unit / branch manager

To confirm. The split of the tolerable region and the named authorities are proposals for AirNav to confirm; they must match the delegations in the AirNav SMS Manual.

3.4 Quantitative safety targets

Quantitative methods (FTA, LOPA, BBN, FMECA) produce frequencies or probabilities. They must be compared with a quantitative target that is consistent with the matrix above. A well-known ATM reference is ESARR 4 [R8], which set the maximum tolerable probability of ATM *directly contributing* to an accident at 1.55×10^{-8} per flight hour for commercial air transport in the ECAC area. That value was derived from European accident statistics and traffic; AirNav should derive its own targets from its accident and traffic data, apportioned across hazards, rather than import the European figure unchanged.

Two rules apply to all quantitative work: state the unit (per flight hour, per operation, per sector hour, per year) and never compare numbers with different units; and treat results as order-of-magnitude estimates — a factor of 3 is within the uncertainty of most inputs.

4. Selecting a method

No single method answers every safety question. The methods differ in what they model (functions, components, tasks, control loops, everyday variability), in whether they quantify, and in when they are most useful in the life of a change. Table 4.1 summarises the twenty-six methods; Figure 3 maps them onto the lifecycle. Methods 18–30 were added in edition 0.2 and the wildlife method (chapter 31) in edition 0.3; several of them quantify or support the core twelve rather than stand alone (HRA supplies human error probabilities to FTA, ETA and LOPA; common cause analysis checks the independence that FTA and RBD assume; GSN structures the argument that uses all the others).

Method	Main question it answers	Type	Typical ANSP use	Effort
Bowtie	What prevents this top event, what limits its consequences, and are those barriers healthy?	Qualitative, barrier-based	Communicating major hazards (runway incursion, airborne conflict, CFIT); SMS risk register; monitoring barrier health	Medium
HAZID	What hazards could this change introduce?	Qualitative, guideword brainstorm	Early concept of a change, new unit, airspace redesign	Low
HAZOP	How could the design or procedure deviate from its intent, and with what effect?	Qualitative, systematic guidewords	Detailed procedures, coordination and data flows (AIDC/OLDI, CPDLC, AMHS)	High
JHA	What can hurt the person doing this task, step by step?	Qualitative, task-based	Engineering maintenance, work at height, RF/electrical work, airside vehicle tasks	Low
FMEA / FMECA	How can each component fail and what does each failure do to the service?	Semi-quantitative, bottom-up	CNS/ATM equipment, power, networks; maintenance strategy	Medium-High
LOPA	Is one scenario controlled by enough independent layers to meet the target?	Semi-quantitative, scenario-based	Checking barriers against a frequency target; justifying a new barrier	Medium
FHA	What happens if a function is lost, degraded or erroneous, how severe is it, and how often may it happen?	Qualitative + safety objectives	Start of the SAM-style lifecycle for systems and services	Medium
STPA	How could the control structure (people, automation, procedures) issue unsafe control actions?	Qualitative, systems-theoretic	Automation, new concepts, human-automation interaction; CAST for investigations	High
FTA	Which combinations of basic events cause this top event, and how likely is it?	Quantitative, top-down deductive	Loss of surveillance/communications; proving safety objectives in PSSA/SSA	Medium-High
Fatigue modelling	How alert will controllers be across this roster?	Quantitative, bio-mathematical	Roster design, FRMS, change of shift pattern, investigations	Low-Medium
FRAM	How does everyday performance variability combine to produce success or failure?	Qualitative, functional	Understanding complex work-as-done; Safety-II; investigations	High
BBN	How do multiple uncertain factors combine, and what do observations tell us about causes?	Quantitative, probabilistic graph	Integrated risk models, SPI analysis, human and organisational factors	High
CRM (Reich)	Is the collision risk of this separation minimum or route spacing below the target level of safety?	Quantitative, geometric and probabilistic	RVSM monitoring, route spacing, PBN airspace design	High
ETA	Given this initiating event, how do the barriers determine the outcome and how often?	Quantitative, forward inductive	Quantifying bowtie consequence side; value of safety nets	Medium
HRA (HEART / CARA)	How likely is a person to fail this task, and why?	Semi-quantitative	Human error probabilities for FTA, ETA, LOPA; procedure and HMI design	Low-Medium
Occurrence risk (ERC / RAT)	How risky was this occurrence, and should it be investigated?	Semi-quantitative, event scoring	Occurrence triage; risk-weighted SPIs	Low

Method	Main question it answers	Type	Typical ANSP use	Effort
GSN	Why does the evidence show the change is acceptably safe?	Argument structure	Safety cases; Safety Assessment Report argument	Medium
Common cause analysis	Are the items claimed independent really independent?	Qualitative, systematic checks	Redundant CNS/ATM architectures; site and room changes	Medium
RBD / Markov	How available is the service, and which part limits it?	Quantitative, reliability	Availability and continuity of voice, surveillance, nav aids	Medium
SWIFT	What could go wrong with this small or temporary change?	Qualitative, what-if prompts	Works on the manoeuvring area, temporary procedures, outages	Low
HTA	What exactly do people have to do, in what order, and where can they err?	Qualitative, task decomposition	Procedures, controller tools, training; input to HRA	Medium
Simulation (RTS / FTS)	How does the change perform with controllers or modelled traffic against safety criteria?	Empirical, statistical	Airspace and procedure changes; new tools	High
Expert judgement	What is the best defensible estimate where data are missing?	Structured elicitation	Filling data gaps in quantitative models	Medium
Security risk	Which intentional threats could affect safety, and how serious are they?	Semi-quantitative register	ATM/CNS systems, remote access, third parties	Medium
Investigation (SOAM / HFACS / Tripod)	Why did this occurrence happen, and which barriers failed?	Qualitative, causal	Occurrence investigation; feedback to bowties	Medium-High
Wildlife strike risk (species)	Which species threaten aircraft here, and where should wildlife measures be aimed?	Semi-quantitative, data-driven	Aerodrome wildlife programme; ATS bird warnings; land use within 13 km	Low-Medium

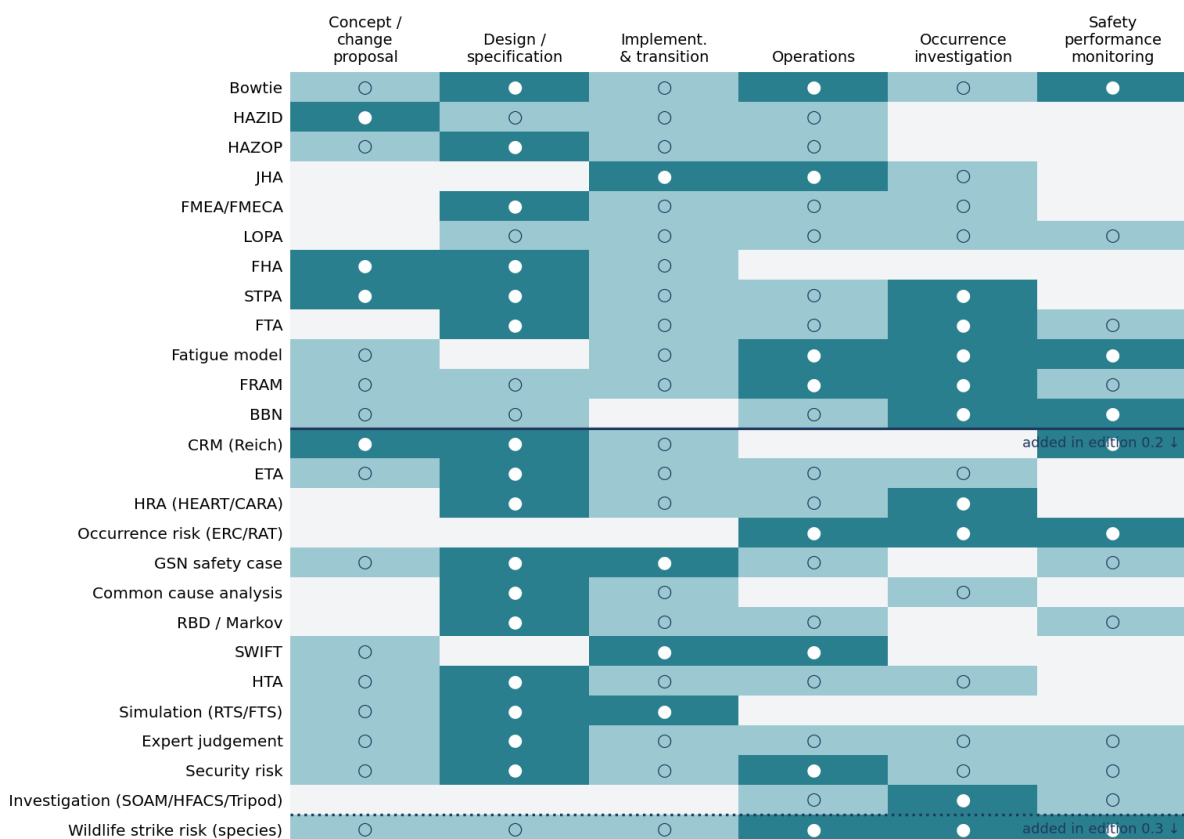


Figure 3. Where each method adds most value in the life of a change (● primary use, ○ supporting use).

4.1 Recommended combinations

Situation	Recommended chain of methods
New AIM system (acquisition, migration, transition)	The template in Appendix C: FHA and data-flow HAZOP → FMEA, FTA/CCA, STPA, HTA/HRA, security → JHA and SWIFT for installation → migration comparison → transition bowtie and shadow operation → GSN argument.
New or modified CNS/ATM system	HAZID or FHA to set hazards and safety objectives → FMEA of the architecture → FTA to show objectives are met (with common-cause analysis) → bowtie update for the affected major hazards.
New or changed ATS procedure / airspace	HAZID → HAZOP of the procedure steps and coordination → STPA if automation or new controller-pilot interactions are involved → bowtie; LOPA for any scenario that relies on few barriers.
New automation or controller tools (e.g. AMAN, CPDLC, conflict probes)	FHA → STPA (unsafe control actions, mode confusion) → FTA for quantitative objectives.
Major organisational change (units merged, posts combined, functions moved)	The template in Appendix D: organisational function map and FHA → STPA of current and new structure → organisational HAZID, HTA of merged posts, Delphi → SWIFT and transition bowtie per phase → SPI register, culture survey and reviews at 3, 6 and 12 months → GSN argument.
Roster or staffing change	Fatigue modelling of the old and new rosters → JHA-style task review of critical periods → bowtie barrier review (supervision, relief, STCA).
Engineering maintenance task	JHA, with FMEA input on the effect on service if the maintained equipment is taken out of service.
Initial phase of an ATS occurrence investigation	The expert system in Appendix E: search the ORLIO case library for similar past occurrences → rate the evidence known so far → fuzzy inference per case and the Bayesian network → collect the evidence the ranked clues point to, perishable evidence first → then the ATSB analysis (Appendix F) and SOAM/HFACS (Ch. 30).
Occurrence investigation or recurring trend	FRAM or CAST (STPA for accidents) to understand work-as-done → FTA for technical failures → BBN to combine evidence → bowtie to show which barriers failed.
Organisation-wide risk picture	Bowtie library for top hazards → BBN linking SPIs and precursors to the bowtie barriers → ERC-weighted occurrence data feeding both.
Separation minimum or route spacing	CRM for technical risk → ETA/HRA for operational (human) errors → simulation for workload → GSN argument to the regulator.
Redundant CNS/ATM service	FMEA → RBD/Markov for availability → FTA for integrity → common cause analysis of every independence claim.
Gradual degradation of a CNS/ATM system (e.g. surveillance radar)	Appendix G: HAZID, FHA and bowtie → FMEA, FTA, CCA, RBD/Markov, STPA, HAZOP, security, CRM, LOPA, ETA, HTA/HRA for the degraded mode → JHA, SWIFT, simulation and fatigue for the repair works → FRAM and BBN in operation → ERC, ATSB and SOAM for occurrences → SPI register and GSN.
Small or temporary change	SWIFT; JHA for maintenance tasks; escalate to HAZID/HAZOP if the change turns out not to be small.
Safety-related security	Security risk assessment (ISO/IEC 27005, ED-205) with STPA-Sec for control paths → safety effects into the hazard log.
Occurrence	ERC/RAT triage → investigation with SOAM/HFACS/Tripod → failed barriers linked to bowties → hazard log update.
Wildlife (bird and bat strikes)	Species risk matrix from strike data → bowtie of the wildlife strike top event with ATS and aerodrome barriers → ERC for individual strikes → SWIFT or HAZID when habitat or land use changes.

5. Conducting an assessment workshop

Most methods in this manual rely on a structured team session. The quality of the result depends far more on who is in the room and how the session is run than on the worksheet.

5.1 Preparation

- Agree the scope, system description and assumptions with the change sponsor in writing before the session.
- Choose participants: 4–8 people is ideal. Include at least one current front-line practitioner for each affected role (ATCO, supervisor, ATSEP) and, where the change affects them, airspace users and the airport operator.
- Send a pre-read (system description, drawings, draft procedures, previous assessments and relevant occurrences) at least three working days in advance.
- Prepare the worksheet template and guidewords for the chosen method; pre-populate what can be pre-populated (functions, nodes, task steps) so the session time is spent on analysis.

5.2 Facilitation

- Open with the purpose, the scope boundary, the risk classification scheme and the rule that every view is recorded.
- Keep the facilitator and the scribe as separate people; the scribe records in the tool or worksheet in view of the team.
- Watch for known biases: *anchoring* on the first estimate, *optimism* about new systems, *groupthink* and deference to seniority. Ask the most junior practitioner first when estimating likelihood.
- Distinguish controls that exist and are verified from controls that are planned or assumed. Only verified controls may be credited when estimating current risk.
- Limit sessions to about three hours; long sessions degrade quality.

5.3 Evidence and recording

For each likelihood and severity judgement, record the basis: occurrence data (with period and exposure), reliability data, analogous systems, or expert judgement (with the names or roles of those who judged). Record dissenting views. Every action shall have an owner and a due date, and every hazard shall be entered into the hazard log with a unique identifier.

6. Bowtie analysis

At a glance	
Purpose	Show, for one hazard and one top event, all the threats that can release it, the consequences that can follow, and the barriers on each path; then manage the health of those barriers.
When to use	Major operational hazards (runway incursion, airborne conflict, CFIT, loss of control, runway excursion, wake encounter); SMS risk register; communicating risk to management and staff; after an occurrence to show which barriers failed.
Inputs	Hazard and top-event definition; occurrence data; procedures and system descriptions; existing HAZID/HAZOP results. The UK CAA bowtie templates are a ready starting point [R12].
Outputs	Bowtie diagram; barrier register with type, owner, effectiveness and criticality; list of weak or missing barriers; links to SPIs.
Team and time	Facilitator + 4-6 SMEs; one to two half-day sessions per bowtie.
Key references	UK CAA bowtie guidance and CAP1329 [R12]; IEC 31010:2019 [R11]; CCPS/Energy Institute bowtie guidelines.

6.1 Key concepts

The bowtie joins a fault-tree-like left side (causes) and an event-tree-like right side (consequences) around a central **top event** — the moment control over the hazard is lost. Its strength is that it is easy to read and puts the emphasis on the **barriers** that an organisation must keep working.

Element	Meaning	ANSP example
Hazard	Normal part of the operation that has potential for harm.	Aircraft and vehicles moving on or near an active runway.
Top event	The release of the hazard; loss of control over it. Must be before damage occurs.	Unauthorised entry of an aircraft, vehicle or person onto the active runway.
Threat	A possible direct cause that can release the top event on its own.	Pilot misinterprets taxi clearance.
Consequence	Undesired outcome resulting from the top event.	Runway collision.
Prevention barrier	Stops a threat from producing the top event.	Stop bars operated H24; standard phraseology with read-back.
Recovery (mitigation) barrier	Stops the top event from developing into the consequence, or reduces severity.	A-SMGCS/RIMCAS alert and ATCO intervention.
Escalation factor	A condition that degrades a barrier.	Stop bar unserviceable; low visibility; ATCO fatigue.
Escalation-factor barrier	Keeps the escalation factor from defeating the main barrier.	Procedure for stop-bar U/S: conditional clearances prohibited, follow-me mandatory.

A barrier must be able, on its own, to stop the sequence on that path. A training course, a policy or “awareness” is usually not a barrier by itself — it is a support to a barrier (for example, RT training supports the barrier “read-back / hear-back”). Recording supports as escalation-factor barriers keeps the diagram honest.

6.2 Procedure

1. **Define the hazard and top event.** One top event per bowtie. Phrase it as a loss of control, not as damage.
2. **Identify threats.** Use occurrence data, HAZID output and the CAA template threats. Each threat must be sufficient on its own to cause the top event.
3. **Identify consequences.** Usually three to six credible outcomes; assess each with the risk matrix (Chapter 3).
4. **Identify prevention and recovery barriers** for each path, in time order from threat to consequence. Test each against the barrier criteria: effective, independent of the threat and other barriers on the path, and auditable.
5. **Add escalation factors and their barriers** for barriers that are known to degrade.
6. **Characterise each barrier:** type (hardware, software, human action, human-hardware), owner, effectiveness rating and criticality, and link it to an SPI or audit item.
7. **Review** the diagram with the team, identify weak paths (few or poor barriers) and raise actions.
8. **Keep it alive:** update after occurrences, audits and changes; use it to code occurrence reports by barrier failed [R12].

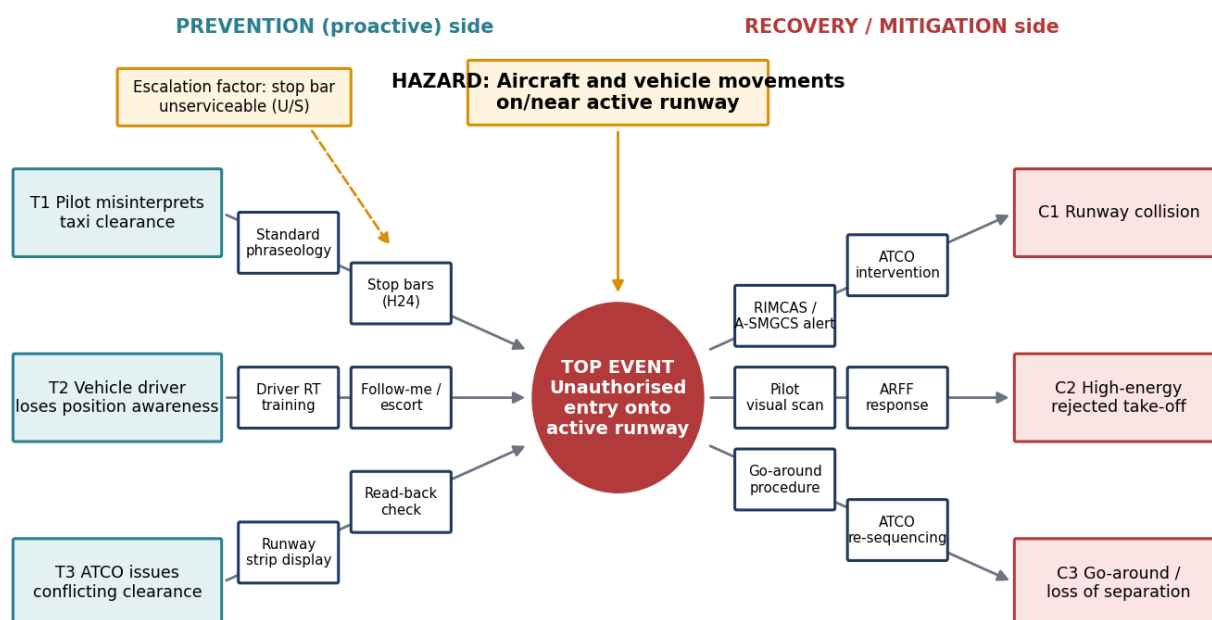


Figure 4. Simplified bowtie for unauthorised entry onto an active runway (illustrative; one escalation factor shown).

6.3 Barrier register template

ID	Barrier	Side / path	Type	Owner	Effect.	Crit.?	Monitoring (SPI/audit)
PB-02	Stop bars operated H24 at all runway-holding positions	Prevention / T1	Hardware + human	Aerodrome operator; TWR supervisor	Good	Yes	Stop-bar serviceability %; stop-bar crossing reports
PB-05	Read-back / hear-back of runway clearances	Prevention / T1, T3	Human	TWR unit chief	Good	Yes	RT sampling audits; read-back error rate
RB-01	RIMCAS / A-SMGCS runway incursion alert	Recovery / all	Hardware + software	CNS engineering	Good (poor in heavy rain clutter)	Yes	Alert availability; false/nuisance alert rate

ID	Barrier	Side / path	Type	Owner	Effect.	Crit.?	Monitoring (SPI/audit)
RB-02	ATCO intervention (stop / go-around instruction)	Recovery / all	Human	TWR unit chief	Good	Yes	Proficiency checks; simulator scenario results
EF-01	Stop bar U/S procedure (no conditional clearances, follow-me)	EF barrier on PB-02	Procedure	Aerodrome ops	Poor (not always applied)	Yes	Audit of NOTAM/U/S logs vs clearances

6.4 Tips and pitfalls

- Do not draw a top event that is already a consequence (“runway collision” is a consequence, not a top event).
- Do not list the same barrier on every line to make the diagram look strong; ask whether it is truly independent of the threat.
- Effectiveness ratings should be based on evidence (audits, occurrence data), not on the designer’s intent.
- Start from the UK CAA bowtie templates (the library listed at caa.co.uk includes airborne conflict, CFIT, runway excursion, runway incursion, ground handling, fire and loss of control) and adapt them to AirNav’s operation rather than starting from a blank page.

7. HAZID — Hazard Identification study

At a glance	
Purpose	Identify, early and broadly, the hazards that a proposed change or operation could introduce, using a guideword checklist to prompt a structured brainstorm.
When to use	Concept and planning stage of any change; new units or facilities; airspace or route redesign; transition plans. Often the first safety activity of a project.
Inputs	Concept of operations; scope and boundary; list of stakeholders; historic occurrences in similar operations.
Outputs	HAZID register: hazards, causes, consequences, existing controls, initial risk index, actions; list of issues needing a more detailed method.
Team and time	Facilitator, scribe, 5–10 SMEs from all affected disciplines; half a day to two days.
Key references	ISO 17776:2016 [R13]; IEC 31010:2019 [R11]; ICAO Doc 9859 hazard identification [R2].

7.1 Key concepts

HAZID works at a higher level than HAZOP. Instead of examining every step of a design, the team walks through a list of **guideword categories** and asks, for each: “could this be a source of hazard for this change?”. Its value lies in breadth — making sure no category is forgotten — and in bringing different disciplines together early, when design changes are still cheap.

7.2 ANSP guideword checklist

Category	Prompt guidewords
People	Staffing levels; competence and rating; training for the change; workload peaks; fatigue; language proficiency; supervision; handover/takeover
Procedures	New or changed procedures; degraded modes; contingency; coordination with adjacent units; phraseology; letters of agreement
Equipment / systems	CNS coverage; ATM system functions; HMI; alerts (STCA, MSAW, APW, RIMCAS); data quality; cyber security; power; fallback
Airspace and aerodrome	Route structure; sectorisation; restricted/danger areas; terrain; MSA; runway configuration; hot spots; works in progress
Traffic	Volume and complexity; mix (IFR/VFR, military, helicopters, drones); crossing flows; special operations
Environment	Weather (convective, low visibility, volcanic ash); night; seasonal peaks; bird activity
Interfaces	Adjacent ACC/APP/FIRs; airport operator; airlines; military; MET; AIM; external suppliers
Transition	Cut-over; parallel running; reversion; training during live operations; data migration
Organisation	Roles and responsibilities; contracting; management of change; reporting culture

7.3 Procedure

1. Brief the team on the change, scope and assumptions; agree the guideword list (tailor Table 7.2).
2. For each guideword, brainstorm hazards (“what could go wrong?”), their causes and consequences. Record everything, filter later.
3. For each hazard, record the existing controls and make an initial risk assessment with the AirNav matrix.
4. Identify further controls and actions; flag items needing a detailed method (HAZOP, FTA, STPA, LOPA).
5. Review the register for duplicates, merge, and transfer hazards to the hazard log.

7.4 Worked example — relocation of an approach unit to a new ATC centre

ID	Guideword	Hazard	Cause(s)	Consequence	Existing controls	Risk	Action
HZ-01	Transition	Loss of surveillance display at cut-over	Data feed not re-routed; configuration error	Loss of separation during cut-over	Cut-over at low traffic; old site on hot standby	3C	HAZOP of cut-over plan; rollback criteria
HZ-02	People	ATCOs unfamiliar with new CWP HMI	Short training window	Delayed or erroneous inputs	Simulator training	4D	Minimum hours on new CWP before live ops
HZ-03	Interfaces	Wrong telephone coordination lines to TWR	VCCS mapping error	Late coordination of missed approach	Test plan	3C	End-to-end line check with each TWR
HZ-04	Equipment	Single power feed to new building	Design	Total loss of service	UPS 30 min	2B	Second feed or genset; FMEA of power
HZ-05	Environment	New site outside radio line of sight of low-level sector	Topography	Loss of VHF contact below FL050	None yet	3B	Coverage study; remote radio site

7.5 Tips and pitfalls

- HAZID is not the end of the assessment: its output is a list of hazards, many of which need a detailed method.
- Resist debating risk ratings at length in a HAZID; a provisional rating is enough to prioritise.
- Record hazards neutrally (“loss of ...”, “incorrect ...”), not as accusations.

8. HAZOP — Hazard and Operability study

At a glance	
Purpose	Systematically examine a defined design or procedure, part by part, for deviations from the design intent, using standard guidewords applied to parameters.
When to use	Detailed design of procedures, coordination processes and data flows (e.g. AIDC/OLDI messages, CPDLC dialogues, flight plan processing), and of technical systems when drawings are mature.
Inputs	Frozen design or draft procedure; flow diagrams, message sets, sequence charts; operating assumptions.
Outputs	HAZOP worksheets per node; recommendations; list of operability issues.
Team and time	Independent chair, scribe, designer, operator (ATCO/ATSEP), specialist as needed; typically 3-6 nodes per day.
Key references	IEC 61882:2016 Hazard and operability studies — application guide [R14]; IEC 31010:2019 [R11].

8.1 Key concepts

The system is divided into **nodes** (a procedure step, a message exchange, a system element). For each node the team states the **design intent**, identifies the **parameters** (e.g. message content, timing, sequence, flow of information, action) and combines each parameter with each **guideword** to generate a **deviation**. For each meaningful deviation, the team records causes, consequences, existing safeguards and recommendations.

Guideword	Meaning	ATM interpretation (examples)
No / Not	Complete negation of intent	Message not sent; clearance not issued; coordination not done
More	Quantitative increase	More aircraft than planned; higher level than coordinated; message sent twice
Less	Quantitative decrease	Less information; lower level; shorter spacing
As well as	Qualitative increase	Correct message plus spurious one; clearance plus unintended instruction
Part of	Qualitative decrease	Incomplete message (level without conditions); partial read-back
Reverse	Logical opposite	Climb instead of descend; transfer in wrong direction
Other than	Complete substitution	Message sent to wrong unit; clearance to wrong callsign
Early / Late	Timing relative to clock time	Estimate sent too late for acceptance; frequency change too early
Before / After	Order or sequence	Transfer of control before coordination accepted

8.2 Procedure

1. Define scope and objectives; assemble the team; collect design documents.
2. Divide the design into nodes and state the design intent of each.
3. For each node, apply each relevant parameter-guideword pair to generate deviations.
4. For each credible deviation: identify causes, consequences (assume safeguards fail to see the unmitigated outcome), existing safeguards, risk index and recommendations.
5. Record as you go; the chair confirms each row before moving on.
6. Follow up recommendations to closure and feed hazards into the hazard log.

8.3 Worked example — ATS inter-centre data coordination (estimate message)

Node 3: “Upstream ACC sends estimate (EST/CPL) message to downstream ACC at least N minutes before the transfer point; downstream system displays the flight in the pending list; downstream ATCO accepts”. Design intent: the downstream sector receives complete, correct and timely data for every flight crossing the boundary.

Parameter + guideword	Deviation	Possible causes	Consequences	Safeguards	Risk	Recommendation
Message + NO	Estimate not received	Link failure; flight plan not activated; address error	Unknown traffic enters sector; late or no coordination	Link-status alarm; verbal coordination backup	3C	Alert downstream ATCO when a flight is within X min of boundary with no EST
Level + OTHER THAN	Coordinated level differs from actual	ATCO changes level after EST sent; no revision (ACP/REV)	Level conflict in downstream sector	Mode C/S check at transfer; STCA	3B	Automatic revision message on cleared-level change; CLAM
Timing + LATE	Estimate arrives inside the agreed time	Short-notice route change; queue delays	Downstream cannot plan; workload	LoA fallback: verbal	4D	Monitor late-EST rate as SPI
Message + AS WELL AS	Duplicate flight created	Re-sent message not correlated	Two labels; wrong one used	Code/callsign correlation	3C	Duplicate-detection logic test
Sequence + BEFORE	Transfer of communication before acceptance	Workload; habit	Aircraft calls unprepared sector	Procedures; training	4D	System inhibits transfer before acceptance

8.4 Tips and pitfalls

- HAZOP needs a mature design. Applied too early it degenerates into a HAZID.
- The chair must be independent of the design team and keep the pace steady; fatigue sets in after about four hours.
- Human-action nodes (procedural HAZOP) work well with the timing and sequence guidewords; the team must include a current practitioner.

9. JHA — Job Hazard Analysis

At a glance	
Purpose	Protect the people who carry out a task by breaking the job into steps, identifying the hazards of each step and specifying controls.
When to use	Engineering and maintenance tasks (radar, NDB/DVOR/DME, ILS, VHF sites, towers, generators, UPS, HVAC), airside work, vehicle operations, any new, infrequent or high-risk task, and after an injury or near miss.
Inputs	Task description, manufacturer manuals, site survey, previous incidents, legal OHS requirements.
Outputs	JHA sheet per job: steps, hazards, controls, responsible person; input to work instructions, permits to work and toolbox talks.
Team and time	Supervisor + workers who do the job + OHS adviser; one to three hours per job.
Key references	OSHA 3071 Job Hazard Analysis [R15]; ISO 45001:2018; national OHS (K3) regulation.

9.1 Key concepts

JHA (also called Job Safety Analysis, JSA) focuses on the relationship between the worker, the task, the tools and the work environment. Controls are chosen using the **hierarchy of controls**: elimination, substitution, engineering controls, administrative controls, and personal protective equipment (PPE) — in that order of preference. In an ANSP, JHA also has a service-safety angle: a maintenance task can itself remove a barrier (e.g. taking a radar out of service), so the JHA should reference the operational coordination required.

9.2 Procedure

1. Select and prioritise jobs: those with injury history, high severity potential, new or infrequent tasks.
2. Involve the workers; observe the job being done where possible.
3. Break the job into 5–12 sequential steps (“what is done”, not “how”).
4. For each step identify hazards: falls, electrical, RF radiation, moving parts, manual handling, heat, weather, vehicles, working alone.
5. Specify controls following the hierarchy; assign responsibility.
6. Review with the team, trial, then approve; review after any incident or change in the task.

9.3 Worked example — replacing an obstruction light on a radar tower

Step	Hazards	Controls (hierarchy)	Responsible
1. Coordinate outage and obtain permit to work	Radar removed without ATS awareness; RF exposure if radar left radiating	ATS coordination and NOTAM as required; radar transmitter isolated and locked out (LOTO); permit signed by radar engineer and TWR/APP supervisor	Site supervisor
2. Inspect climbing system and weather	Defective ladder/fall arrest; lightning; high wind	Pre-use inspection of fall-arrest rail; stop work if lightning within 10 km or wind above limit	Climber + supervisor
3. Climb tower with tools	Fall from height; dropped objects	Full-body harness, 100% tie-off; tools tethered; exclusion zone below	Climber
4. Isolate light circuit and replace lamp	Electric shock	Isolate at breaker, LOTO, test for dead; insulated tools	Technician
5. Descend, restore power, test light	Fall; energising while person on tower	Head count before de-isolation; permit closure sequence	Site supervisor
6. Return radar to service and hand back to ATS	Radar not restored; wrong configuration	Functional check; formal hand-back to supervisor; log entry	Radar engineer

9.4 Tips and pitfalls

- Too many or too few steps makes the JHA useless; a step is something a worker would recognise as a distinct part of the job.
- PPE is the last line of defence; a JHA whose controls are all PPE and “be careful” needs rework.
- Use the JHA as the basis for the pre-job briefing; a JHA in a file cabinet protects nobody.

10. FMEA / FMECA — Failure Mode and Effects (and Criticality) Analysis

At a glance	
Purpose	Bottom-up analysis of how each component or function of a system can fail, the effect of each failure mode on the service, how it is detected and compensated, and how critical it is.
When to use	Design and procurement of CNS/ATM systems, power and network architecture; maintenance programme design; checking that redundancy works as intended; input to FTA.
Inputs	System architecture and block diagrams; component list; reliability data (MTBF/failure rates); monitoring and switchover design.
Outputs	FMEA/FMECA worksheet; list of single points of failure; ranking (criticality or action priority); design and maintenance actions.
Team and time	Systems engineer, ATSEP, operational expert, safety assessor; days to weeks depending on system size.
Key references	IEC 60812:2018 Failure modes and effects analysis (FMEA and FMECA) [R16]; MIL-STD-1629A [R17]; SAE ARP4761A [R20].

10.1 Key concepts

For each item, the analyst lists its **failure modes** (e.g. no output, erroneous output, intermittent output, spurious output), their **causes**, the **local effect**, the **end effect** on the ATS, the **detection** method and the **compensating provisions**. FMECA adds a criticality measure. Two ranking approaches are common:

Risk Priority Number (RPN) = Severity × Occurrence × Detection, each rated 1–10. It is simple but mathematically weak (different combinations give the same RPN); IEC 60812:2018 therefore also describes alternatives such as criticality matrices and action priority tables.

Quantitative failure-mode criticality (MIL-STD-1629A): $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$, where λ_p is the item failure rate, α the fraction of failures in this mode, β the conditional probability that the mode produces the end effect, and t the operating time. The item criticality is the sum of C_m over its modes with the same severity class.

10.2 Procedure

1. Define the system, boundaries, level of indenture and mission (e.g. provide air-ground voice on sector frequency, 24 h).
2. Draw the functional/reliability block diagram.
3. For each item list failure modes, causes, local and end effects.
4. Record detection means (BITE, monitoring, ATCO noticing) and compensating provisions (main/standby, backup radio).
5. Rate severity with the AirNav scale (end effect), occurrence and detection, or compute C_m .
6. Rank, identify single points of failure, recommend actions; re-rate after actions.

10.3 Worked example — sector VHF air-ground communication

Item	Failure mode	Cause	End effect on ATS	Detection	Compensation	S	O	D	RPN
Main VHF transmitter	No RF output	PA failure	Loss of main Tx on frequency	Radio monitoring alarm; no pilot reply	Auto switch to standby Tx	4	4	2	32
Main VHF transmitter	Stuck PTT (continuous carrier)	Relay / VCCS fault	Frequency blocked for all users	ATCO notices; timer alarm	PTT time-out; switch site	8	3	4	96

Item	Failure mode	Cause	End effect on ATS	Detection	Compensation	S	O	D	RPN
VCCS	Total loss	Software fault; power	Loss of all voice at CWP	Immediate	Backup radios at CWP	8	2	1	16
Leased line to remote site	Loss	Carrier outage	Loss of remote coverage (low levels)	Link alarm	Second path via VSAT	6	5	2	60
Main + standby receivers	Degraded sensitivity	Water ingress in antenna or cable	Weak/unreadable pilot calls	Late — pilot complaints	None	6	3	7	126

The highest RPN (126) is not the most severe failure: it is driven by poor detection. The recommended action is receiver-sensitivity monitoring (e.g. a site test transponder or periodic loop test) so that degradation is detected before pilots report it. The stuck-PTT mode (RPN 96, severity 8) shows why the severity rating should also be reviewed on its own: any mode with high severity deserves attention regardless of RPN.

10.4 Tips and pitfalls

- FMEA looks at single failures one at a time; it misses combinations and common causes — use FTA and common-cause analysis for those.
- Human error modes can be included (“ATCO selects wrong frequency”) but are better handled with HAZOP, STPA or task analysis.
- Keep S, O and D scales defined and published; undefined 1–10 scales make RPNs meaningless.

11. LOPA — Layer of Protection Analysis

At a glance	
Purpose	For a single cause–consequence scenario, estimate the frequency of the consequence after all independent protection layers (IPLs) have acted, and compare it with a tolerable frequency target.
When to use	When a scenario from HAZID/HAZOP/bowtie is rated high and the team needs to know whether the existing barriers are enough, or how much a new barrier would help. Bridges qualitative bowtie and full quantitative FTA/ETA.
Inputs	Scenario definition; initiating event frequency; enabling conditions and conditional modifiers; candidate IPLs with probability of failure on demand (PFD); tolerable frequency target.
Outputs	LOPA worksheet; mitigated event frequency; gap to target; required additional risk reduction.
Team and time	Safety assessor + ATCO + engineer; one to two hours per scenario once data is available.
Key references	CCPS, Layer of Protection Analysis: Simplified Process Risk Assessment (2001) [R18]; CCPS Guidelines for Initiating Events and Independent Protection Layers (2015) [R19]; IEC 61511.

11.1 Key concepts

LOPA originated in the process industry and maps naturally onto the ANSP view of barriers. The mitigated event frequency is:

$$f_C = f_{IE} \times \prod P_{CM} \times \prod PFD_i$$

where f_{IE} is the initiating event frequency, P_{CM} are enabling conditions or conditional modifiers (probability the situation needed for harm exists), and PFD_i is the probability that IPL i fails on demand. A safeguard may only be credited as an IPL if it is:

- **Independent** of the initiating event and of every other IPL credited in the same scenario;
- **Effective (specific)**: capable of preventing the consequence on its own, in time;
- **Dependable**: its PFD can be justified from data or accepted generic values;
- **Auditable**: its performance can be validated periodically.

Independence in ATM. Independence is the hardest criterion in ATM. Controller monitoring and STCA both depend on the same surveillance data path; if the initiating event involves corrupted surveillance, neither is independent of it. ACAS II uses its own interrogations and is independent of ground surveillance, but it depends on both aircraft having serviceable transponders. Apply LOPA only to scenarios where the assumed independence has been checked, and use FTA with common-cause analysis where it has not.

11.2 Procedure

1. Select a scenario (one initiating event → one consequence) from the bowtie/HAZOP.
2. Estimate f_{IE} from AirNav occurrence data (count ÷ exposure).
3. Identify enabling conditions and conditional modifiers (traffic present, geometry, time of day).
4. List safeguards; test each against the IPL criteria; assign PFDs (order-of-magnitude values: 10^{-1} for a trained human response with adequate time, 10^{-1} – 10^{-2} for a well-maintained automated alert, as justified).
5. Compute f_C and compare with the tolerable frequency; if above target, identify additional IPLs or reduce f_{IE} .
6. Document assumptions; hand residual risk to the hazard log.

11.3 Worked example — level bust leading to a potential mid-air collision

Element	Description	Value
Initiating event	Aircraft deviates from its cleared flight level (level bust) in sector X	0.5 per year
Conditional modifier	Another aircraft is in a conflicting position at the time	0.05
IPL 1	ATCO detects deviation on situation display and resolves the conflict in time	PFD 0.1
IPL 2	STCA alerts and ATCO resolves (credited only for deviations IPL 1 missed, assuming independent attention cue)	PFD 0.1
IPL 3	ACAS II resolution advisory generated and followed by both crews	PFD 0.1
Mitigated frequency	$0.5 \times 0.05 \times 0.1 \times 0.1 \times 0.1$	2.5×10^{-5} per year
Tolerable target (illustrative)	Set by AirNav for this outcome class in this sector	1×10^{-5} per year
Gap	Target not met by a factor of 2.5	—
Proposed new IPL	Selected flight level (SFL) mismatch alert: the ground system compares the Mode S downlinked selected level with the cleared level and alerts the ATCO before the aircraft leaves its level	PFD 0.1
Revised frequency	$2.5 \times 10^{-5} \times 0.1$	2.5×10^{-6} per year — target met

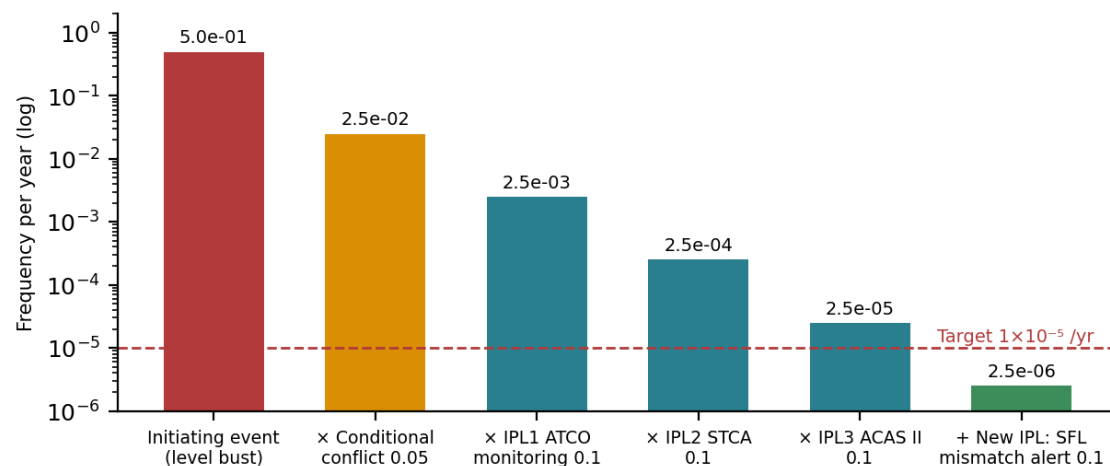


Figure 5. Frequency reduction through each layer in the level-bust scenario (log scale, illustrative values).

11.4 Tips and pitfalls

- Do not credit the same human twice (e.g. “ATCO monitoring” and “ATCO responds to STCA” are only partially independent; justify the credit or combine them).
- Use conservative, order-of-magnitude PFDs; LOPA is a screening tool, not a precise calculation.
- Procedures and training are not IPLs by themselves; they support the human IPL.

12. FHA — Functional Hazard Analysis

At a glance	
Purpose	Determine how safe a system or service needs to be: identify its functions, the ways each function can fail (failure conditions), the operational effects and severity of each, and set a safety objective (maximum tolerable frequency) for each failure condition.
When to use	At the start of a system or service change, before detailed design — the first step of the FHA → PSSA → SSA lifecycle used in EUROCONTROL SAM and in aircraft systems safety (ARP4761A).
Inputs	Functional description (what the system does, not how); operational environment; interfaces; risk classification scheme.
Outputs	Failure-condition list with effects, severity class and safety objectives; initial safety requirements; input to PSSA (FTA, FMEA, common-cause analysis).
Team and time	Safety assessor, operational experts (ATCO), system engineers; one to three workshops.
Key references	EUROCONTROL SAM — FHA chapter [R7]; SAE ARP4761A and ARP4754B [R20]; EUROCAE ED-78A for data-link services.

12.1 Key concepts

FHA works on **functions**, so it can start before architecture is chosen. For each function the team considers standard failure-condition types: **total loss**, **partial loss / degradation**, **erroneous or corrupted output** (detected and undetected), **unintended (spurious) function** and **delayed output**. Undetected erroneous output is usually the most severe, because nobody reacts to it. Each failure condition is assessed in its operational context (traffic, available fallbacks, exposure time) and assigned a severity class. The risk classification scheme then gives the maximum tolerable frequency — the **safety objective**. The later PSSA must show that the proposed architecture can meet each objective, and the SSA must show that the implemented system does.

12.2 Procedure

1. Describe the system functions and the operational environment; list assumptions.
2. For each function, identify failure conditions using the five types above.
3. For each failure condition, describe the operational effects, taking account of existing mitigations external to the system (procedures, other systems, ATCO and pilot actions).
4. Classify severity using Table 3.2 (worst credible effect).
5. Derive safety objectives from the risk classification scheme (Table 12.1); apportion if several failure conditions contribute to the same outcome.
6. Record initial safety requirements (e.g. “the system shall detect and flag position data with integrity below X within Y s”).
7. Hand objectives and requirements to the PSSA.

12.3 Worked example — ADS-B surveillance in an en-route sector

Function F1: “Provide the ATCO with the position, identity and pressure altitude of each ADS-B equipped aircraft in the sector, updated at least every N seconds.” Assumptions: sector also has procedural fallback; partial radar coverage above FL245.

ID	Failure condition	Operational effect	Severity	Safety objective (per operating hour)
FC-1	Total loss of ADS-B data for all aircraft (detected)	Revert to radar where available, procedural elsewhere; separation increases; workload peak during transition	C Major	$\leq 1 \times 10^{-5}$
FC-2	Loss of data for a single aircraft (detected)	Label coasts/drops; ATCO applies procedural separation to one aircraft	D Minor	$\leq 1 \times 10^{-3}$
FC-3	Undetected erroneous position of a single aircraft	ATCO separates on wrong position; possible loss of separation unnoticed until ACAS	B Hazardous	$\leq 1 \times 10^{-7}$
FC-4	Undetected erroneous position of multiple aircraft (ground processing error)	Several false positions; high collision risk	A Catastrophic	$\leq 1 \times 10^{-9}$
FC-5	Excessive latency of position updates	Displayed position lags actual; separation margins eroded in turns	C Major	$\leq 1 \times 10^{-5}$

Table 12.1 — The objectives follow from the AirNav matrix: a failure condition must at least reach the “tolerable (lower)” region, so severity A requires likelihood 1, B requires 2, C requires 3, D requires 4, using the quantitative bands of Table 3.1. Derived requirements include integrity monitoring (NIC/NACp filtering), comparison with radar where overlapping, and ground-processing software assurance proportionate to FC-4.

12.4 Tips and pitfalls

- Describe effects in the real operational context; the same failure can be Minor at night and Major at the evening peak — state the assumption.
- Do not take credit in the FHA for mitigations that are part of the system being assessed; those belong in the PSSA.
- Software and data failures belong in the FHA too: erroneous AIP data, wrong adaptation data, corrupted flight plans.

13. STPA — System-Theoretic Process Analysis

At a glance	
Purpose	Identify how the control structure of a socio-technical system — ATCOs, pilots, automation, procedures, management — can issue or fail to issue control actions that lead to hazards, including cases where no component has failed.
When to use	New concepts of operation, automation and decision-support tools, human-automation interaction, data link, complex interfaces between organisations. CAST (the accident-analysis variant) for investigations.
Inputs	Concept of operations; organisational and technical architecture; procedures; roles and responsibilities.
Outputs	Losses, hazards and system-level constraints; control structure model; unsafe control actions (UCAs); loss scenarios; requirements and controller constraints.
Team and time	Facilitator trained in STPA, operational and technical experts; several days for a medium-size system.
Key references	Leveson, Engineering a Safer World (2011) [R21]; Leveson & Thomas, STPA Handbook (2018) [R22]; Leveson et al., Assuring safety of NextGen procedures (ATM2013).

13.1 Key concepts

STPA is based on STAMP (Systems-Theoretic Accident Model and Processes), which treats safety as a control problem: accidents occur when the control structure fails to enforce safety constraints. Each controller (human or automated) has a **process model** — its belief about the state of the controlled process — and issues **control actions** based on it, receiving **feedback**. Many ATM occurrences are process-model problems: the ATCO believes an aircraft has left a level when it has not, or a pilot believes a clearance was for them. The STPA Handbook [R22] defines four steps:

1. **Define the purpose of the analysis:** losses, system-level hazards, system-level constraints.
2. **Model the control structure:** controllers, control actions and feedback (Figure 6).
3. **Identify unsafe control actions (UCAs)** using four types: (1) not providing the control action causes a hazard; (2) providing it causes a hazard; (3) providing it too early, too late or in the wrong order; (4) the action is stopped too soon or applied too long.
4. **Identify loss scenarios:** why UCAs could occur (inadequate feedback, flawed process model, conflicting controls, component failure) and why correct control actions could be improperly executed or not followed.

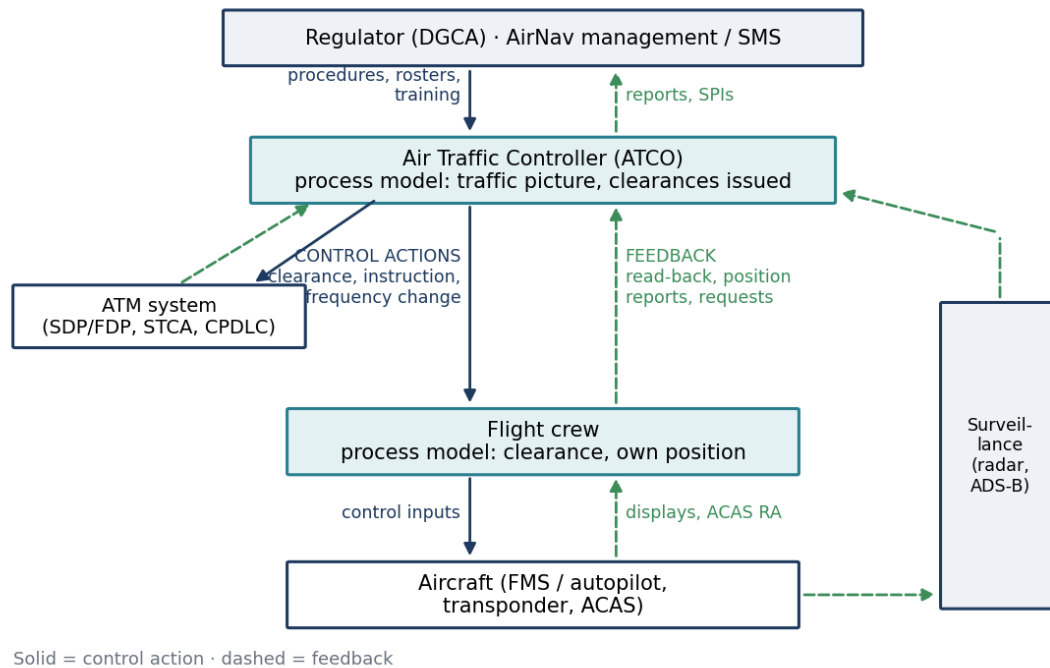


Figure 6. Simplified ATC control structure used in the STPA example.

13.2 Worked example — descent clearances in an approach sector

Losses: L-1 mid-air collision; L-2 controlled flight into terrain. **Hazards:** H-1 aircraft violate minimum separation [L-1]; H-2 aircraft descends below the minimum safe altitude for its position [L-2]. **System-level constraints:** SC-1 aircraft must maintain at least the minimum separation; SC-2 aircraft must remain at or above the applicable minimum altitude.

Control action	Not providing causes hazard	Providing causes hazard	Too early / too late / wrong order	Stopped too soon / applied too long
ATCO: descent clearance	UCA-1: ATCO does not issue descent when the aircraft must leave its level to avoid a conflict ahead [H-1]	UCA-2: ATCO clears descent through a level occupied by unseparated traffic [H-1]. UCA-3: ATCO clears descent below the minimum vectoring altitude [H-2]. UCA-4: descent is issued to a similar callsign [H-1]	UCA-5: descent issued too late to be established before the crossing point [H-1]. UCA-6: descent issued before the preceding aircraft has vacated the level [H-1]	Not applicable (discrete action)
Flight crew: execute descent	UCA-7: crew does not descend after a valid clearance while following traffic descends behind [H-1]	UCA-8: crew descends on a clearance meant for another aircraft [H-1]	UCA-9: crew starts descent before the clearance is complete (anticipation) [H-1]	UCA-10: descent continued through the cleared level (level bust) [H-1, H-2]. UCA-11: level-off too early, below the level assigned to crossing traffic [H-1]

Examples of loss scenarios

- **For UCA-6 (flawed process model):** the ATCO believes the preceding aircraft has left FL120 because its label shows a descending trend, but the Mode C update lags and the aircraft has been told to stop descent by the adjacent sector without coordination. Controls: coordination rules for level changes, SFL display, STCA.
- **For UCA-4 (inadequate feedback):** similar callsigns on frequency; the wrong aircraft reads back, the read-back is partially blocked and not challenged. Controls: callsign-similarity alerting, frequency congestion management, airline callsign de-confliction.
- **Conflicting controllers:** an ACAS RA and an ATC instruction disagree; the crew follows ATC. Constraint: crews shall follow the RA and ATCOs shall not issue contrary instructions once an RA is reported — a lesson reinforced by the 2002 Überlingen mid-air collision.
- **Organisational:** STCA inhibited in a sector because of nuisance alerts after an adaptation change; nobody owns the re-tuning. Constraint: changes to safety-net parameters require safety assessment and post-implementation monitoring.

13.3 Tips and pitfalls

- Write each UCA with its *context* (“when ...”); a UCA without context cannot be verified.
- Do not stop at UCAs — the requirements come from the loss scenarios.
- STPA does not need probabilities. Where the organisation needs a risk index, rate the resulting hazards with the matrix after the analysis.

14. FTA — Fault Tree Analysis

At a glance	
Purpose	Deductive, top-down analysis of the combinations of basic events (failures, errors, conditions) that lead to a specified undesired top event; qualitative (minimal cut sets) and quantitative (top-event probability, importance measures).
When to use	Demonstrating that an architecture meets a safety objective (PSSA/SSA); analysing loss of surveillance, communication, navigation or ATM functions; investigating technical occurrences; checking redundancy and common-cause vulnerability.
Inputs	Top event from FHA/bowtie; system architecture; FMEA; failure rates, repair times, test intervals; human error probabilities where relevant.
Outputs	Fault tree; minimal cut sets; top-event probability; importance ranking; design/maintenance actions.
Team and time	Reliability/safety engineer with system experts; days to weeks.
Key references	IEC 61025:2006 [R23]; NUREG-0492 Fault Tree Handbook (1981) [R24]; NASA Fault Tree Handbook with Aerospace Applications (2002) [R25].

14.1 Key concepts

Symbol / term	Meaning
Top event	The undesired state at the head of the tree, precisely defined (what, where, when, for how long).
OR gate	Output occurs if any input occurs. $P = 1 - \prod(1 - p_i) \approx \sum p_i$ for small p .
AND gate	Output occurs only if all inputs occur. $P = \prod p_i$ (inputs independent).
Basic event	Lowest level of resolution with a probability (component failure, human error, condition).
Undeveloped event	Not developed further for lack of information or relevance.
House event	A condition set to true/false to switch parts of the tree (e.g. maintenance mode).
Minimal cut set	Smallest set of basic events whose joint occurrence causes the top event. Single-event cut sets are single points of failure.
Common-cause failure (CCF)	Single cause failing redundant items together (shared power, software, network, maintenance error). Commonly modelled with a β -factor: $\lambda_{CCF} = \beta \cdot \lambda$.
Birnbaum importance	$I_B = P(\text{top} \mid \text{event occurs}) - P(\text{top} \mid \text{event does not occur})$: sensitivity of the top event to the event.
Fussell-Vesely importance	Fraction of top-event probability involving cut sets that contain the event.

14.2 Procedure

1. Define the top event and boundary conditions (initial state, operating mode, exposure time).
2. Develop the tree downward: for each event ask “what are the immediate, necessary and sufficient causes?”, choosing the gate type.
3. Continue to basic events with data; include human errors, CCFs and support systems (power, network, HVAC).
4. Obtain minimal cut sets by Boolean reduction (by software for large trees).
5. Quantify: assign probabilities (or rates $\times$ exposure, or unavailabilities for repairable items: $Q \approx \lambda \cdot \text{MTTR}$), compute top event and importance measures.
6. Compare with the safety objective; identify dominant cut sets and actions; perform sensitivity analysis.

14.3 Worked example — total loss of surveillance picture at an ACC sector

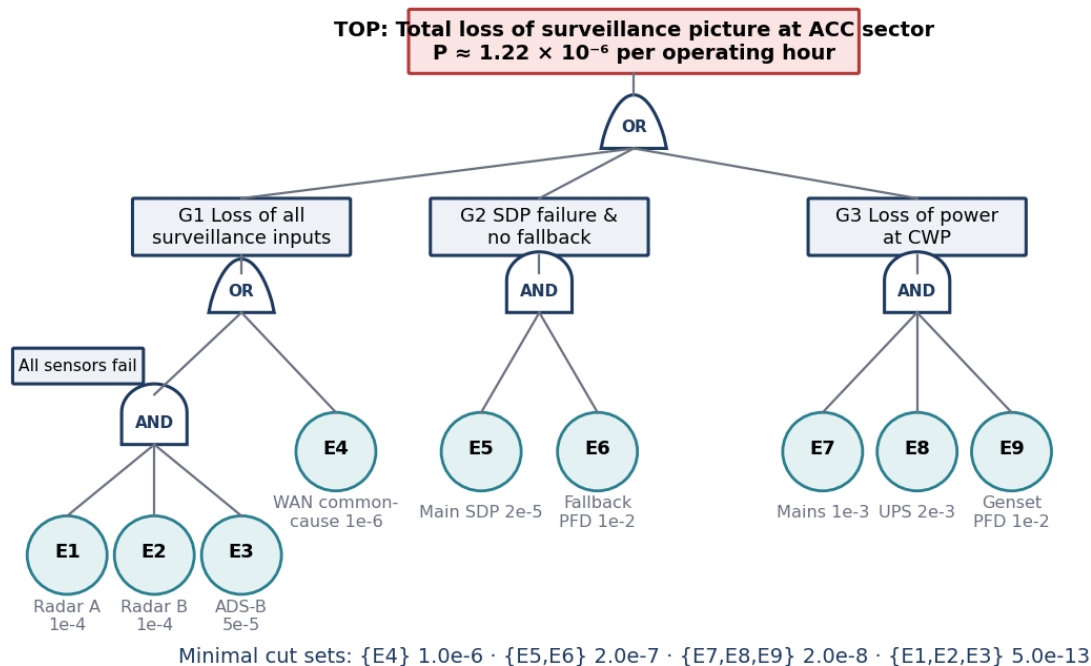


Figure 7. Fault tree for total loss of the surveillance picture (probabilities per operating hour or per demand, illustrative).

Basic event	Description	P	Birnbaum	Fussell-Vesely
E4	Common-cause loss of surveillance data network (single WAN)	$1.0\text{e-}6$	≈ 1.0	0.82
E5	Main surveillance data processor fails	$2.0\text{e-}5$	$1.0\text{e-}2$	0.16
E6	Fallback system fails to take over (PFD)	$1.0\text{e-}2$	$2.0\text{e-}5$	0.16
E7	Mains power failure	$1.0\text{e-}3$	$2.0\text{e-}5$	0.016
E8	UPS fails	$2.0\text{e-}3$	$1.0\text{e-}5$	0.016
E9	Generator fails to start (PFD)	$1.0\text{e-}2$	$2.0\text{e-}6$	0.016
E1-E3	Radar A, radar B, ADS-B fail independently	$1\text{e-}4, 1\text{e-}4, 5\text{e-}5$	$\leq 1\text{e-}8$	4×10^{-7}

The top-event probability is about 1.22×10^{-6} per operating hour, 82% of which comes from one single-event cut set: the shared data network. Triple sensor redundancy (E1-E3) contributes almost nothing, because a single network failure defeats it. If FHA had set an objective of 1×10^{-7} for this failure condition, the design would fail; the effective action is a diverse, physically separate data path (reducing E4), not another sensor. This is the typical lesson of FTA: redundancy is only as good as its independence.

14.4 Tips and pitfalls

- Mixing rates and probabilities is the most common numerical error. Convert everything to probability over the same exposure period, or to unavailability.
- Always search for CCFs: shared power, software, adaptation data, maintenance teams, building, network.
- The rare-event approximation (sum of cut sets) over-estimates slightly; it is acceptable when all cut-set probabilities are small.

15. Bio-mathematical fatigue modelling (three-process model)

At a glance	
Purpose	Predict the alertness (or sleepiness) of controllers and other safety-critical staff across a work schedule, so that rosters can be compared and high-fatigue periods identified and mitigated.
When to use	Designing or changing ATCO/ATSEP rosters; FRMS hazard identification; assessing overtime, extended shifts and quick returns; investigations where fatigue may have contributed.
Inputs	Planned (or actual) duty times; assumed or measured sleep times; time zone; model parameters.
Outputs	Alertness/KSS time-series per duty; time spent below thresholds; comparison between rosters; mitigation proposals.
Team and time	Fatigue/FRMS specialist, rostering staff, ATCO representatives; hours per roster once the model is set up.
Key references	Borbély (1982) two-process model [R26]; Åkerstedt & Folkard (1997) three-process model [R27]; Ingre et al. (2014) [R28]; ICAO Doc 9966 and the ICAO/CANSO/IFATCA Fatigue Management Guide for ATS Providers [R5, R6].

15.1 Background

ICAO Annex 11 requires States to establish regulations for managing fatigue in the provision of air traffic control services, either through prescriptive duty limits or through a Fatigue Risk Management System (FRMS) [R4, R5]. In both cases the ANSP must identify fatigue hazards in its rosters. Bio-mathematical models (BMMs) support this by predicting average group alertness from sleep-wake timing. They do not measure fatigue in an individual and must be combined with operational data (fatigue reports, subjective scales, sleep diaries, actigraphy, performance tests).

The **two-process model** (Borbély, 1982) explains sleep regulation by a homeostatic process *S* (sleep pressure builds during wake and dissipates during sleep) and a circadian process *C* (a roughly 24-hour rhythm). The **three-process model of alertness** (Åkerstedt and Folkard) adds process *W*, sleep inertia — the grogginess after waking — and expresses alertness on a scale that maps to the Karolinska Sleepiness Scale (KSS).

15.2 Model equations

Process	Equation (t in hours)	Parameters (as reported by Ingre et al. 2014 [R28])
<i>S</i> — homeostatic, awake	$S(t) = L_a + (S_w - L_a) \cdot e^{d \cdot t}$	Lower asymptote $L_a = 2.4$; $d = -0.0353$; S_w = level at waking
<i>S</i> — recovery during sleep	$S(t) = H_a - (H_a - S_s) \cdot e^{g \cdot t}$	Upper asymptote $H_a = 14.3$; $g \approx -0.381$; S_s = level at sleep onset. (The full model adds a “brake” that slows recovery above $S = 12.2$.)
<i>C</i> — circadian	$C(t) = C_a \cdot \cos(\pi(t - p)/12)$	Amplitude $C_a = 2.5$; acrophase $p = 16.8$ h (peak about 16:48)
<i>W</i> — sleep inertia	$W(t) = W_c \cdot e^{W_d \cdot t}$, t since waking	$W_c = -5.72$; $W_d = -1.51$
<i>U</i> — ultradian (optional)	12-hour component	Amplitude 0.5, mesor -0.5 (omitted in the example below)
Alertness and KSS	$A = S + C (+ U) + W$; $KSS \approx 10.6 - 0.6 \cdot A$	$KSS \geq 7$ (“sleepy”) corresponds to $A \approx 6$ or below

KSS	Verbal anchor (Åkerstedt & Gillberg, 1990 [R29])
1-3	Extremely alert · very alert · alert
4-6	Rather alert · neither alert nor sleepy · some signs of sleepiness
7	Sleepy, but no effort to keep awake
8-9	Sleepy, some effort to keep awake · very sleepy, great effort to keep awake, fighting sleep

15.3 Procedure

1. Collect the roster(s) to be assessed and define sleep assumptions (e.g. sleep 23:00–07:00 on normal nights; day sleep after night shifts), or use measured sleep.
2. Run the model for each roster pattern over enough days to reach a stable pattern (at least one full rotation).
3. Extract indicators for each duty: minimum alertness, maximum predicted KSS, time spent at $KSS \geq 7$, and the clock time of the low point.
4. Compare alternative rosters and mitigations (napping, break scheduling, shift start/end times, direction of rotation, number of consecutive nights).
5. Validate with operational data: KSS surveys at the start, middle and end of shifts, fatigue reports and, where possible, sleep diaries or actigraphy.
6. Record fatigue hazards and mitigations in the hazard log / FRMS.

15.4 Worked example — night shift with and without a pre-shift nap

An ATCO works a day shift (07:00–14:00) on day 1, sleeps 23:00–07:00, is off during day 2, then works a night shift 22:00–06:00, followed by day sleep 07:00–13:00. Two cases are modelled: no nap on day 2, and a 90-minute nap from 15:00 to 16:30.

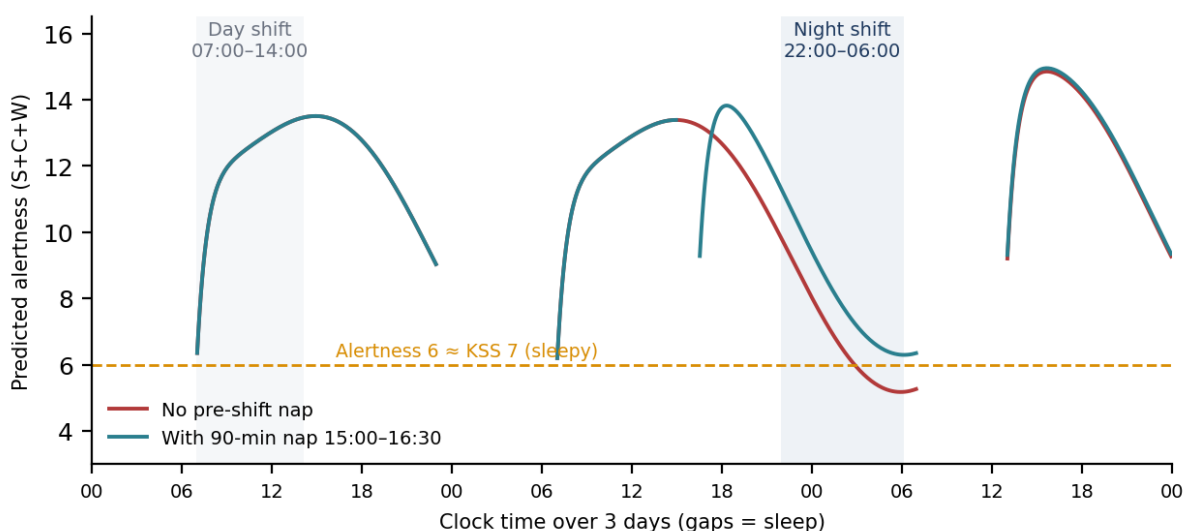


Figure 8. Three-process model prediction for the example roster ($S+C+W$, simplified model without the brake and ultradian terms).

Indicator (night shift 22:00–06:00)	No nap	90-min nap 15:00–16:30
Minimum predicted alertness	5.2 (at about 05:55)	6.3 (at about 05:55)
Maximum predicted KSS	7.5	6.8
Share of shift at $KSS \geq 7$	40% (about 3 h 10 min)	0%

The low point falls at the end of the night shift, where circadian trough and extended time awake coincide. The model suggests a pre-shift nap removes the predicted $KSS \geq 7$ period, but the last two hours remain close to the threshold, so further controls are still warranted: schedule the least demanding work and extra breaks between 03:00 and 06:00, protect in-shift rest/napping opportunities where permitted, avoid combining a night shift with an early start the following day, and limit consecutive night shifts.

15.5 Samn-Perelli fatigue check before taking a position

The model predicts fatigue for a roster; it cannot see how an individual controller actually slept or feels. The Samn-Perelli checklist [R83] closes that gap: before taking over a position, the controller rates mental exhaustion and cognitive slowing on a 7-point scale, and the supervisor responds according to the score. NAVRAP records each rating with the controller ID (not the name), the planned position, the supervisor's response and any mitigation, and checks the response against the thresholds below.

Score	Description	Light and required action
1	Fully alert, wide awake	Green (1-3): the controller can proceed to their shift normally.
2	Very lively, responsive, but not at peak	
3	Okay, somewhat fresh	
4	A little tired, less than fresh	Caution: the supervisor assigns a standard position (avoiding highly complex, high-traffic sectors) and schedules the break slightly earlier.
5	Moderately tired, let down	Amber: immediate mitigation before the controller takes over a position — e.g. a mandatory 15-minute walk, a dose of caffeine, or adjusting the rotation schedule.
6	Extremely tired, very difficult to concentrate	Red (6-7): the controller is an operational risk — pull them from active control duties, place them on administrative desk tasks, or send them home to rest.
7	Completely exhausted, unable to function	

A response that is weaker than the threshold requires is flagged (an error for red, a warning for caution and amber), as is a non-green rating with no response recorded. Ratings are personal data used to manage fatigue, not to judge performance; a culture in which controllers rate honestly matters more than the scale itself. Aggregated ratings — the share of non-green scores by shift, time of day or roster — are useful validation data for the model (§15.3) and for the FRMS.

15.6 Tips and pitfalls

- BMMs predict group averages for healthy adults. They do not account for workload, time on position, caffeine, individual differences (e.g. chronotype) or sleep disorders.
- Sleep opportunity is not sleep obtained: day sleep after a night shift is often shorter than planned. Use measured sleep where possible.
- Several commercial BMMs exist (e.g. SAFTE-FAST, the Boeing Alertness Model, FAID, CAS). Whichever model is used, document its version, parameters and validation, and never treat its threshold as a regulatory limit by itself.

16. FRAM — Functional Resonance Analysis Method

At a glance	
Purpose	Model how everyday work is actually done, as a set of coupled functions, and understand how normal performance variability can combine (resonate) to produce unexpected outcomes — both successes and failures.
When to use	Complex, tightly coupled operations where linear cause-effect models fall short: arrival management, runway operations, handovers, abnormal situations; investigation of occurrences with no obvious “failure”; Safety-II learning from what goes well.
Inputs	Interviews and observations of work-as-done; procedures (work-as-imagined); occurrence and normal-operations data.
Outputs	FRAM model (functions and couplings); description of variability; identified resonance paths; recommendations for monitoring and damping variability.
Team and time	Analyst trained in FRAM plus practitioners; days to weeks, mostly spent on data collection.
Key references	Hollnagel, FRAM (2012) [R30]; Hollnagel, Hounslow & Slater, FRAM Handbook (2018) [R31]; EUROCONTROL white papers on Safety-II and systems thinking [R32]; Tian & Caponecchia (2020) review of FRAM in aviation [R36].

16.1 Principles

- **Equivalence of successes and failures:** things that go right and things that go wrong happen in basically the same way.
- **Approximate adjustments:** because conditions are never fully specified, people and organisations continuously adjust their performance.
- **Emergence:** outcomes emerge from the combination of variability in many functions rather than from a single cause.
- **Functional resonance:** variability of several functions can combine and be amplified (or dampened) through the couplings between them.

16.2 The six aspects of a function

Aspect	Definition	Example for “Issue descent clearance”
Input (I)	What starts the function or is used/transformed by it	Arrival sequence; aircraft approaching top of descent
Output (O)	The result of the function	Descent clearance issued and read back
Precondition (P)	Condition that must be true before the function can start	Level below is free; coordination with next sector done
Resource (R)	What is needed or consumed while the function is carried out	Frequency time; ATCO attention; CWP
Control (C)	What supervises or regulates the function	LoA descent profiles; monitoring feedback; STCA
Time (T)	Temporal constraints	Must be issued before top of descent; sector workload window

16.3 Procedure (four steps)

1. **Identify and describe the functions** using the six aspects. Start from the function of interest and add upstream/downstream functions until the model is sufficient.
2. **Characterise variability** of each function's output. Technological functions are usually stable; human functions vary often and widely; organisational functions vary slowly but with large effect. Output variability is described for *timing* (too early, on time, too late, not at all) and *precision* (precise, acceptable, imprecise).
3. **Identify functional resonance**: trace how variability in one output affects the aspects of downstream functions it is coupled to, for a specific scenario (an instantiation of the model).
4. **Propose ways to monitor and manage variability**: dampen unwanted variability, amplify useful adjustments, add indicators.

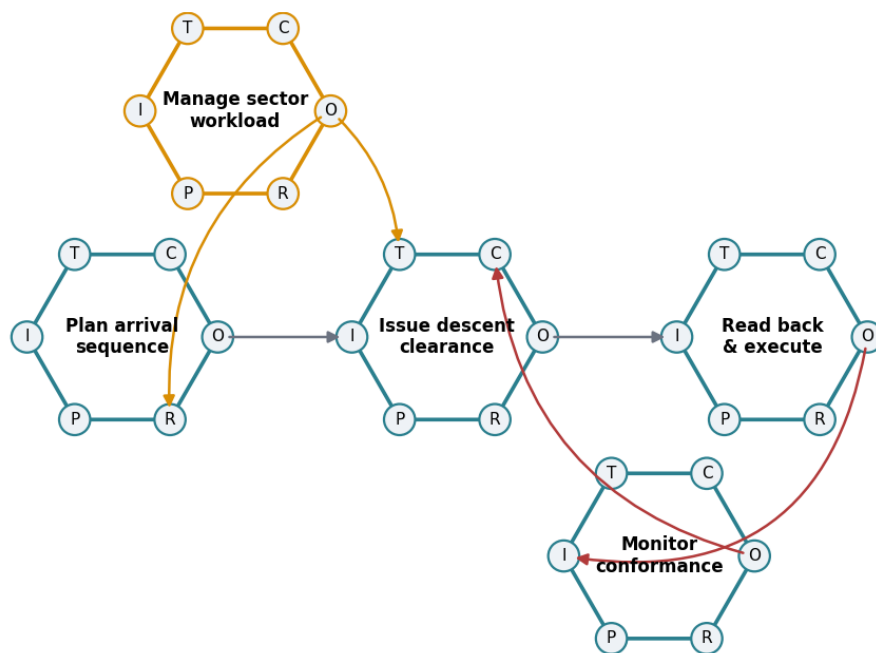


Figure 9. FRAM instantiation for descent clearances in a busy arrival sector. Red couplings: variability in read-back and monitoring feeding back into the clearance function; amber: workload acting on time and resources.

Function	Type	Likely output variability (busy peak)	Downstream effect	Management option
Manage sector workload	Organisational / human	Too late (split decision delayed)	Time and resources of all ATCO functions shrink	Traffic-load indicator triggers sector split earlier
Plan arrival sequence	Human + AMAN	Imprecise (sequence changes late)	Clearances revised; more RT	Freeze horizon for sequence
Issue descent clearance	Human	Too late; imprecise (abbreviated phraseology)	Read-back errors, steep descents	Standard phraseology refresher; CPDLC for non-urgent clearances
Read back & execute	Human (crew)	Imprecise (partial read-back, blocked)	Monitoring must catch the error	Hear-back emphasis; frequency-congestion SPI
Monitor conformance	Human + technology	Too late under workload	Level busts detected late	SFL display and mismatch alert

16.4 Tips and pitfalls

- FRAM needs good data on work-as-done: plan observations and interviews, not just procedure reading.
- Keep the model focused; a hundred-function model is rarely useful. The free FRAM Model Visualiser (FMV) helps build and share models.
- FRAM does not produce a risk index. Where one is needed, transfer the identified hazards to the matrix after the analysis.

17. BBN — Bayesian Belief Networks

At a glance	
Purpose	Represent causal relationships between uncertain factors as a directed acyclic graph with conditional probability tables, so that risk can be predicted from known conditions and causes can be inferred from observed outcomes.
When to use	Integrated risk models combining technical, human and organisational factors; linking precursors/SPIs to accident risk; updating risk estimates as new data arrive; diagnostic reasoning in investigations.
Inputs	Causal structure (from bowties, FTA, expert elicitation); occurrence and exposure data; structured expert judgement for missing probabilities.
Outputs	Network model; marginal probabilities of outcomes; posterior probabilities given evidence; sensitivity ranking of factors.
Team and time	Risk analyst with BBN skills, domain experts, data analyst; weeks for a first useful model.
Key references	Fenton & Neil (2018) [R33]; Ale et al. (2009), the CATS model for air transport safety [R34]; Pearl (1988) [R35].

17.1 Key concepts

Each node is a variable with discrete states (e.g. fatigue: yes/no). Arcs point from cause to effect. Each node holds a conditional probability table (CPT) giving its distribution for every combination of parent states. The joint probability factorises as $P(X_1, \dots, X_n) = \prod P(X_i \mid \text{parents}(X_i))$. Two kinds of reasoning follow:

Predictive (cause → effect): what is the probability of loss of separation if the ATCO is fatigued and workload is high? **Diagnostic** (effect → cause, using Bayes' theorem $P(A|B) = P(B|A) \cdot P(A) / P(B)$): given that a loss of separation occurred, how likely is it that fatigue was present?

17.2 Procedure

1. Define the purpose and the outcome node(s); keep the model as small as the question allows.
2. Build the causal structure with experts, reusing bowtie threats and barriers or FTA structure; check that the graph is acyclic.
3. Define node states precisely ("fatigued" = $KSS \geq 7$, for example).
4. Populate CPTs from data where available and by structured expert elicitation otherwise; document the source of every table.
5. Validate: check marginal outputs against observed rates, test extreme scenarios with experts, run sensitivity analysis.
6. Use the model for what-if analysis and update CPTs as data accumulate.

17.3 Worked example — coordination error and loss of separation at transfer of control

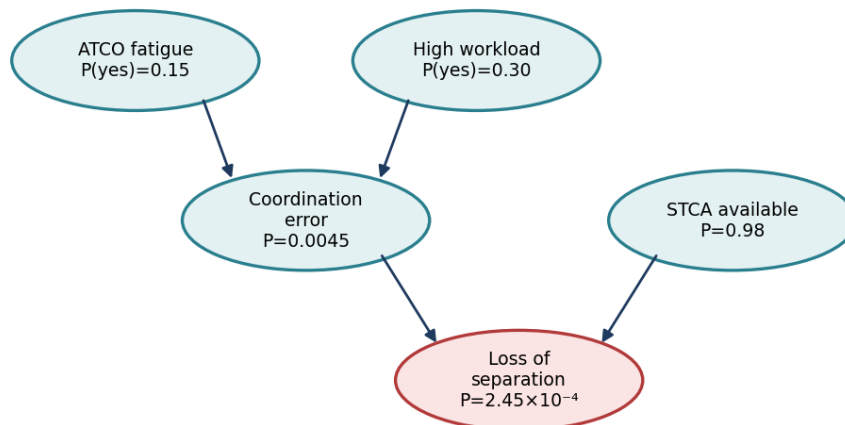


Figure 10. BBN for loss of separation following a coordination error at transfer of control (probabilities per transfer, illustrative).

Node	Conditional probability table
Fatigue (F)	$P(F = \text{yes}) = 0.15$
Workload (W)	$P(W = \text{high}) = 0.30$
Coordination error (E)	$P(E F, W): F \text{ yes \& } W \text{ high } 0.020; F \text{ yes \& } W \text{ normal } 0.008; F \text{ no \& } W \text{ high } 0.006; F \text{ no \& } W \text{ normal } 0.002$
STCA available (S)	$P(S = \text{yes}) = 0.98$
Loss of separation (L)	$P(L E, S): \text{error \& STCA } 0.05; \text{error \& no STCA } 0.30; \text{no error: } 0 \text{ (other causes outside this model)}$

Query	Result
$P(\text{coordination error})$	0.0045 per transfer
$P(\text{loss of separation})$	2.45×10^{-4} per transfer
$P(\text{LoS} \text{fatigue} = \text{yes})$ vs $P(\text{LoS} \text{fatigue} = \text{no})$	6.4×10^{-4} vs 1.8×10^{-4} (≈ 3.6 times higher)
$P(\text{LoS} \text{fatigue} = \text{yes}, \text{workload} = \text{high})$	1.1×10^{-3}
Diagnostic: $P(\text{fatigue} \text{LoS occurred})$	0.39 (prior 0.15)
Diagnostic: $P(\text{high workload} \text{LoS occurred})$	0.54 (prior 0.30)
Diagnostic: $P(\text{STCA unavailable} \text{LoS occurred})$	0.11 (prior 0.02)

The diagnostic results show why a BBN is useful for learning from occurrences: after a loss of separation the posterior probability of fatigue rises from 15% to 39%, so fatigue should be checked systematically in investigations of such events. The predictive results give a quantitative argument for fatigue and workload controls (e.g. earlier sector splitting), and the model can be updated as AirNav collects data.

17.4 Tips and pitfalls

- CPT size grows exponentially with the number of parents; keep to three or fewer parents per node, or use parametric forms (noisy-OR, ranked nodes).
- Expert-elicited probabilities are prone to overconfidence; use structured elicitation (calibration questions, multiple experts) and record it.
- Software: GeNIe/SMILE, Netica, AgenaRisk, Hugin, and the open-source Python library pgmpy all support model building and inference.

18. CRM — Collision risk modelling (Reich model)

At a glance	
Purpose	Estimate the probability of a mid-air collision caused by aircraft that are nominally separated but deviate from their cleared path, and compare it with a target level of safety (TLS). This is the quantitative basis of separation minima such as RVSM, reduced lateral route spacing and PBN-based route structures.
When to use	Introducing or reviewing a separation minimum; changing route spacing; RVSM safety monitoring; showing that a navigation or surveillance performance requirement is sufficient.
Inputs	Aircraft dimensions; traffic counts and flows (occupancies); speeds; navigation and height-keeping performance from monitoring data (e.g. regional monitoring agency height-keeping data, lateral deviation data from surveillance).
Outputs	Collision risk per flight hour by dimension and direction; comparison with the TLS; minimum spacing meeting the TLS; sensitivity to the key parameters.
Team and time	Risk analyst familiar with ICAO Doc 9689; data analyst; ATC and navigation SMEs; weeks, mostly spent on data.
Key references	ICAO Doc 9689 [R37]; ICAO Doc 9574 [R38]; regional monitoring agency reports.

18.1 Key concepts

The Reich model treats each aircraft as a box of length λ_x , width λ_y and height λ_z . Two aircraft collide when their boxes overlap in all three dimensions at once. For a pair nominally separated in one dimension (the “separation dimension”), the expected number of fatal accidents per flight hour in the occupancy form used by ICAO is:

$$N = P_s(S) \cdot P_o(0) \cdot (\lambda_x/S_x) \cdot \{ E_{\text{same}} [|\Delta v|/2\lambda_x + |v_o|/2\lambda_o + |v_s|/2\lambda_s] + E_{\text{opp}} [2|v|/2\lambda_x + |v_o|/2\lambda_o + |v_s|/2\lambda_s] \}$$

$P_s(S)$ is the probability that two aircraft nominally separated by S overlap in the separation dimension (for RVSM the vertical overlap probability $P_z(1000 \text{ ft})$, from height-keeping data); $P_o(0)$ is the overlap probability in the other horizontal dimension for aircraft on the same route; E_{same} and E_{opp} are occupancies — the number of aircraft pairs within a longitudinal window S_x — for same- and opposite-direction traffic; the bracketed terms are relative speeds divided by aircraft dimensions. The TLS used by ICAO is 2.5×10^{-9} fatal accidents per flight hour for the technical vertical risk of RVSM and 5×10^{-9} for total vertical risk and for lateral route spacing.

For parallel routes the overlap probability $P_y(S_y)$ depends on the tails of the lateral deviation distribution. ICAO Doc 9689 recommends the double-exponential (Laplace) shape or a mixture, because Gaussian tails understate gross navigation errors; NAVRAP offers both and computes $P_y(S_y) \approx 2\lambda_y \cdot f(S_y)$, with f the density of the difference of the two aircraft’s lateral deviations.

18.2 Procedure

1. Define the system: airspace, routes or levels, direction of traffic, the separation minimum being assessed and the TLS that applies.
2. Collect traffic data (flows, speeds, passing frequencies) and derive occupancies for the busiest representative period.
3. Collect navigation or height-keeping performance data and fit the deviation distribution; pay special attention to large (gross) errors.
4. Compute the collision risk by dimension and direction; compare with the TLS.
5. Run sensitivity cases (traffic growth, poorer navigation, different tail model) and, for route spacing, the risk-spacing curve.
6. Record assumptions and add operational errors (e.g. level busts, ATC-pilot loop errors) separately — the Reich model covers technical risk; ICAO requires operational risk to be assessed as well.

18.3 Template

Parameter	Symbol	Source	Value used	Basis / date
Vertical overlap at 1000 ft	$P_z(S_z)$	Height-keeping monitoring		
Lateral overlap, same track	$P_y(0)$	Lateral deviation data		
Occupancy same / opposite	$E_{\text{same}}, E_{\text{opp}}$	Traffic sample		
Aircraft dimensions	$\lambda_x, \lambda_y, \lambda_z$	Fleet mix		
Relative speeds	$ \Delta v , v , \dot{y} , \dot{z} $	Traffic sample / ICAO defaults		

18.4 Worked example — RVSM vertical risk and parallel-route spacing

Vertical. With $P_z(1000 \text{ ft}) = 1.0 \times 10^{-8}$, $P_y(0) = 0.5$, $\lambda_x = 184 \text{ ft}$, $\lambda_y = 175 \text{ ft}$, $\lambda_z = 50 \text{ ft}$, $S_x = 80 \text{ NM}$, $E_{\text{same}} = 0.15$, $E_{\text{opp}} = 0.05$, $|\Delta v| = 20 \text{ kt}$, $|v| = 470 \text{ kt}$, $|\dot{y}| = 42 \text{ kt}$ and $|\dot{z}| = 1.5 \text{ kt}$, the same-direction risk is 3.27×10^{-10} , the opposite-direction risk 1.55×10^{-9} , total **1.87×10^{-9}** fatal accidents per flight hour — 75% of the 2.5×10^{-9} technical TLS. Opposite-direction traffic dominates because of the high closing speed.

Lateral. Two same-direction parallel routes ($P_z(0) = 0.52$, $E_{\text{same}} = 0.10$) with double-exponential lateral deviations of scale 0.9 NM give the risk-spacing curve in the figure below. The risk falls by about an order of magnitude every 2.5 NM; the minimum spacing that meets 5×10^{-9} is **11.7 NM**, and the proposed 15 NM gives 1.6×10^{-10} , a margin of about 30.

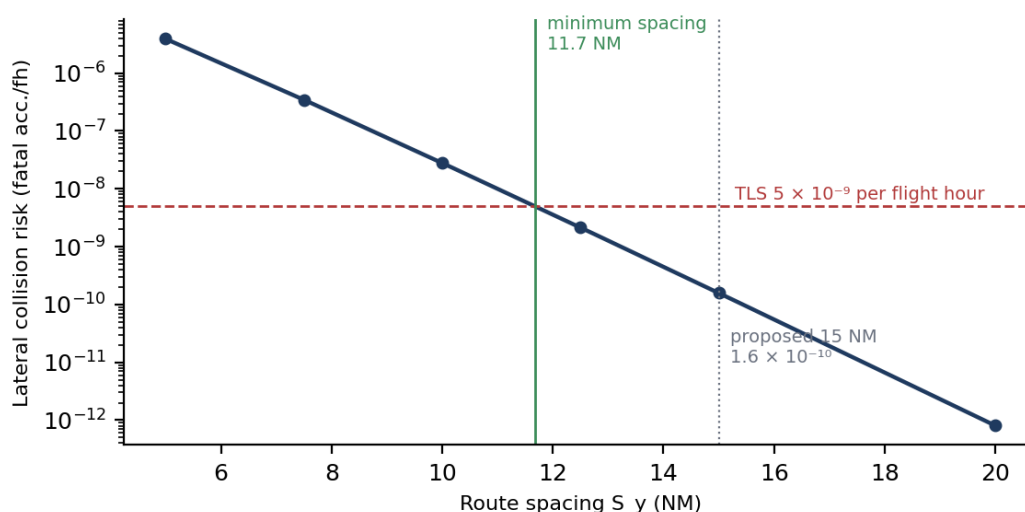


Figure 11. Lateral collision risk against route spacing for the worked example (NAVRAP CRM module).

18.5 Tips and pitfalls

- The result is only as good as the tail of the deviation distribution. One gross navigation error in the sample can change the answer by orders of magnitude; do not discard “outliers” without cause.
- Keep technical and operational risk apart; both must be shown to meet their targets.
- Use local traffic data; occupancies in Indonesian FIRs differ from the European and North Atlantic samples used in ICAO examples.
- Coordinate with the regional monitoring agency (for RVSM in the Asia-Pacific region) so that the data and methods match their reports.

19. ETA — Event tree analysis

At a glance	
Purpose	Trace forward from an initiating event through the success or failure of each barrier, to show the possible outcomes and their frequencies.
When to use	Quantifying the right-hand side of a bowtie; showing the value of safety nets (STCA, MSAW, RIMCAS, ACAS); comparing barrier options; completing an FTA (which gives the initiating event frequency) with the outcome picture.
Inputs	Initiating event and its frequency; the ordered list of barriers or functional events; the probability each works, conditional on what happened before.
Outputs	Event tree; frequency of each outcome sequence; frequency by severity class; the contribution of each barrier.
Team and time	Analyst + ATC and technical SMEs; one to two days for a first tree.
Key references	IEC 62502 [R39]; NUREG-0492 [R24]; EUROCONTROL SAM [R7].

19.1 Key concepts

An event tree starts from one initiating event on the left. Each column is a barrier, taken in the order in which it acts. At each column the path splits: up if the barrier succeeds, down if it fails. The frequency of a sequence is the initiating frequency multiplied by the probabilities along the path. In a *barrier* event tree a success ends the sequence; in a *functional* event tree every event is questioned on every path.

The probabilities are **conditional**. If the first barrier (the controller's own detection) has failed because the controller is overloaded, the controller is also less likely to respond well to STCA. Ignoring this dependence is the most common and most optimistic error in ETA; record it explicitly as a conditional probability for that path.

19.2 Procedure

1. Define the initiating event and its frequency (from occurrence data, FTA or a bowtie threat).
2. List the barriers in the order in which they act; for each, define success precisely.
3. Estimate each barrier's probability of success, and conditional values where the path changes it (use HRA, Chapter 20, for human barriers).
4. Draw the tree and compute sequence frequencies; check that the probabilities of all end states sum to 1.
5. Assign a severity to each end state and compare frequencies with the risk classification scheme or quantitative targets.
6. Test the effect of improving or removing each barrier.

19.3 Worked example — conflict following a level bust

A level bust that creates a conflict occurs about 0.025 times per year in the sector studied. Three barriers act in turn: the controller detects and resolves the conflict (0.9); STCA alerts and the controller resolves (0.9, but only 0.8 when the controller has already missed the conflict); and the ACAS resolution advisory is followed (0.9).

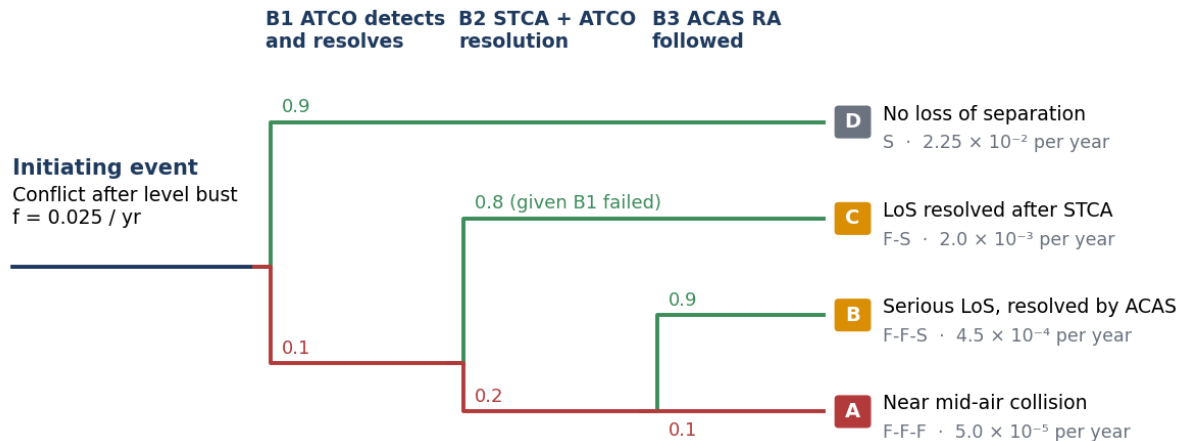


Figure 12. Event tree for a conflict following a level bust (NAVRAP ETA module; frequencies per year).

Sequence	Outcome	Severity	Probability	Frequency per year
S	Resolved by the controller, no loss of separation	D	0.900	2.25×10^{-2}
F-S	Loss of separation, resolved after STCA	C	0.080	2.0×10^{-3}
F-F-S	Serious loss of separation, resolved by ACAS	B	0.018	4.5×10^{-4}
F-F-F	Near mid-air collision / collision	A	0.002	5.0×10^{-5}

Had the dependence been ignored (STCA success 0.9 on every path), the catastrophic frequency would have been 2.5×10^{-5} : half the value above. The example shows why conditional probabilities must be questioned.

19.4 Tips and pitfalls

- Order the barriers by time of action, not by importance.
- Keep the initiating event specific. “Loss of separation” is an outcome, not an initiating event.
- Link each ETA to the bowtie it quantifies so that barrier changes are reflected in both.

20. HRA — Human reliability assessment (HEART and CARA)

At a glance	
Purpose	Estimate the probability that a person fails to carry out a task correctly (human error probability, HEP) and identify which conditions drive it.
When to use	Wherever an FTA, ETA, LOPA or BBN needs a number for a human action; comparing designs of a procedure, HMI or alert; justifying credit for a human barrier.
Inputs	A task description (ideally from an HTA, Chapter 26); the generic task type; the conditions under which the task is done.
Outputs	HEP for each task; the error-producing conditions that contribute most; improvement options.
Team and time	HRA-trained analyst, operational SMEs (ATCOs, ATSEP); about half a day per task.
Key references	Williams (HEART) [R40]; Kirwan and Gibson (CARA) [R41].

20.1 Key concepts

Both HEART and CARA use the same structure. The analyst chooses a **generic task type (GTT)** with a nominal HEP, then identifies the **error-producing conditions (EPCs)** present. Each EPC has a maximum effect (a multiplier). The analyst judges the **assessed proportion of affect (APoA)**, between 0 and 1, and the HEP is

HEP = GTT × Π [(EPC_{max} – 1) × APoA + 1], capped at 1.

HEART was developed for industrial and nuclear tasks and suits engineering, maintenance and support tasks. CARA (Controller Action Reliability Assessment) is EUROCONTROL's adaptation for air traffic control, with GTTs such as “respond to a unique and trusted audible and visual indication” (B3, 0.0004), “active search of radar or strips” (B1, 0.005) and “complex and time-pressured conflict solution” (D2, 0.19), and ATC-specific EPCs such as traffic complexity (×10) and cognitive overload (×6).

20.2 Procedure

1. Describe the task and its context; an HTA is the best starting point.
2. Choose the library (CARA for controller tasks, HEART otherwise) and the GTT that fits best; record why.
3. Select the EPCs that are really present — usually no more than three or four — and judge the APoA for each with the SMEs; record the reasoning.
4. Calculate the HEP and check it against any available data or simulator results.
5. Look at the largest factors to find improvements, then re-assess with the improvements in place.

20.3 Template

Task	Library / GTT	Nominal HEP	EPC (max) × APoA → factor	Assessed HEP	Justification

20.4 Worked example — controller and technician tasks

Task	GTT	EPCs (max, APoA → factor)	HEP
T1. ATCO acts correctly on an STCA alert	CARA B3 — 0.0004	Traffic complexity (×10, 0.2 → 2.8); cognitive overload (×6, 0.1 → 1.5); low vigilance (×3, 0.1 → 1.2)	2.0×10^{-3}
T2. ATCO detects a level bust by monitoring (no alert)	CARA B1 — 0.005	Traffic complexity (×10, 0.4 → 4.6); cognitive overload (×6, 0.3 → 2.5)	5.8×10^{-2}
T3. Technician restores radar to operational configuration after maintenance	HEART F — 0.003	No independent check (×3, 0.5 → 2.0); no clear confirmation of action (×3, 0.4 → 1.8)	1.1×10^{-2}

T2 shows why unaided monitoring cannot be credited as a strong barrier: roughly one level bust in seventeen would be missed in the assessed conditions, which is why STCA and the conformance-monitoring tools matter. For T3, adding an independent functional check before hand-back removes most of the first factor and brings the HEP to about 5×10^{-3} .

20.5 Tips and pitfalls

- Do not count the same condition twice under two EPCs.
- HEPs from HEART and CARA are screening values with an uncertainty of about a factor of 3–10; use them to rank and improve, and treat small differences with caution.
- Where the HEP feeds a barrier credited in LOPA, the barrier's PFD cannot be better than the HEP.

21. Occurrence risk classification (ARMS ERC and EUROCONTROL RAT)

At a glance	
Purpose	Give every reported occurrence a consistent risk value so that the right events are investigated, trends are measured and the hazard log is fed by real events.
When to use	Triage of every mandatory and voluntary report; weighting occurrences in SPIs; selecting investigations; checking whether a bowtie threat is being realised.
Inputs	The occurrence report and radar/voice data; for RAT, the achieved separation, closure rate and the performance of the ATM ground and airborne elements.
Outputs	ERC risk index and band; RAT severity score and ESARR 2 severity class (A-E); decision on investigation.
Team and time	Safety investigator or occurrence analyst; minutes per occurrence; RAT panel for serious events.
Key references	ARMS Working Group methodology [R42]; EUROCONTROL RAT and ESARR 2 [R43].

21.1 ARMS Event Risk Classification

The ERC asks two questions. **Q1:** if this event had escalated into an accident, what would have been the most credible accident outcome? **Q2:** how effective was the remaining safety margin — the barriers still in place between this event and that accident? The answers index a 4 × 4 matrix (figure below). The numbers are weights; NAVRAP treats indices of 100 and above as red (investigate and act now), 20–99 as amber (consider investigation) and below 20 as green (record for trend analysis). AirNav should confirm these thresholds when it adopts the matrix.

Q1. Most credible accident outcome	Catastrophic accident	50	102	502	2500
	Major accident	10	21	101	500
	Minor injuries/damage	2	4	20	100
	No accident outcome	1	1	1	1
		Effective	Limited	Minimal	Not effective
Q2. Effectiveness of the remaining barriers					

Figure 13. ARMS ERC matrix; the outlined cell is occurrence OCC-101 of the worked example.

21.2 EUROCONTROL Risk Analysis Tool (RAT)

For separation-minima infringements and runway incursions, RAT scores the **risk of collision** (minimum separation achieved as a proportion of the minimum, and rate of closure) and the **controllability** of the event (conflict detection, planning, execution, ground safety net, recovery, airborne safety net). The total score is converted to an ESARR 2 severity class — A serious incident, B major incident, C significant incident, D not determined, E no safety effect — using the RAT look-up table. The table itself is in the EUROCONTROL RAT guidance and is not reproduced in NAVRAP; the analyst records the class.

21.3 Procedure

1. Screen every occurrence with the ERC within the time set in the occurrence reporting procedure.
2. For ATM-related separation or runway events, complete the RAT marksheet from radar and voice data.
3. Record the ERC index, RAT score and severity class in the occurrence record and link the occurrence to the related bowtie threat and hazard log entry.
4. Open an investigation (Chapter 30) for red-band events and severity classes A and B.

21.4 Worked example

Occurrence	Q1 outcome	Q2 barriers	ERC	Band
OCC-101 Level bust FL240 → FL232 with crossing traffic; STCA; ATCO resolved; 2.8 NM / 700 ft	Catastrophic	Limited	102	Red
OCC-102 Vehicle entered runway strip without clearance; stopped after RIMCAS alert	Catastrophic	Effective	50	Amber
OCC-103 Frequency congestion delayed a descent clearance; no conflict	No accident outcome	Effective	1	Green

RAT for OCC-101: minimum separation between 50% and 75% of the minimum (3 points) and medium closure rate (2) give a risk-of-collision score of 5; the conflict was detected late (1) and the plan was inadequate (1), with execution, STCA response and recovery correct (0) — controllability 2. Total severity score 7; the panel assigned class B.

21.5 Tips and pitfalls

- Q1 is about the most credible accident, not the worst imaginable one. A level bust in radar airspace with STCA is not automatically “catastrophic, not effective”.
- Classify quickly and revisit when the investigation reveals more.
- Use the ERC-weighted count of occurrences, not the raw count, as a leading indicator.

22. GSN — Structuring the safety argument (safety cases)

At a glance	
Purpose	Show, in a structured and reviewable way, why the evidence from the assessments supports the claim that a change is acceptably safe.
When to use	Safety cases for major changes (new ATM system, new surveillance, airspace redesign); the argument part of a Safety Assessment Report; regulator presentations.
Inputs	The top-level safety claim; the context (scope, risk scheme, assumptions); the evidence from the other methods (FHA, FTA, simulation results, hazard log).
Outputs	A GSN diagram linking claims, strategies and evidence; a list of undeveloped claims and assumptions still to be discharged.
Team and time	Safety assessor with the change sponsor; built up across the project.
Key references	GSN Community Standard v3 [R44]; Kelly (1998) [R45].

22.1 Notation

Element	Symbol	Meaning
Goal	Rectangle	A claim stated as a proposition that is true or false (“X is acceptably safe”).
Strategy	Parallelogram	How a goal is broken down (“argue over each failure condition”).
Solution	Circle	A reference to evidence (an FHA worksheet, an FTA result, a test report).
Context	Rounded rectangle	Scope, definitions or reference information the claim depends on.
Assumption / Justification	Ellipse marked A / J	A statement taken as true without further support / the reason a strategy is sound.
Undeveloped	Diamond under the element	A claim not yet supported — an open item.
SupportedBy / InContextOf	Solid arrow / hollow arrow	Inferential or evidential link / contextual link.

22.2 Procedure

1. State the top goal and its context: the change, the operational environment and the risk classification scheme.
2. Choose a strategy (for example over the SAM lifecycle: specification, implementation, transition), and state sub-goals.
3. Continue until each goal can be supported directly by evidence; mark goals that are not yet supported as undeveloped.
4. Link each solution to its evidence record and check that the evidence is complete and approved.
5. Review the argument for gaps and for assumptions that are not managed; keep the argument updated as evidence is completed.

22.3 Worked example — ADS-B introduction in sector X

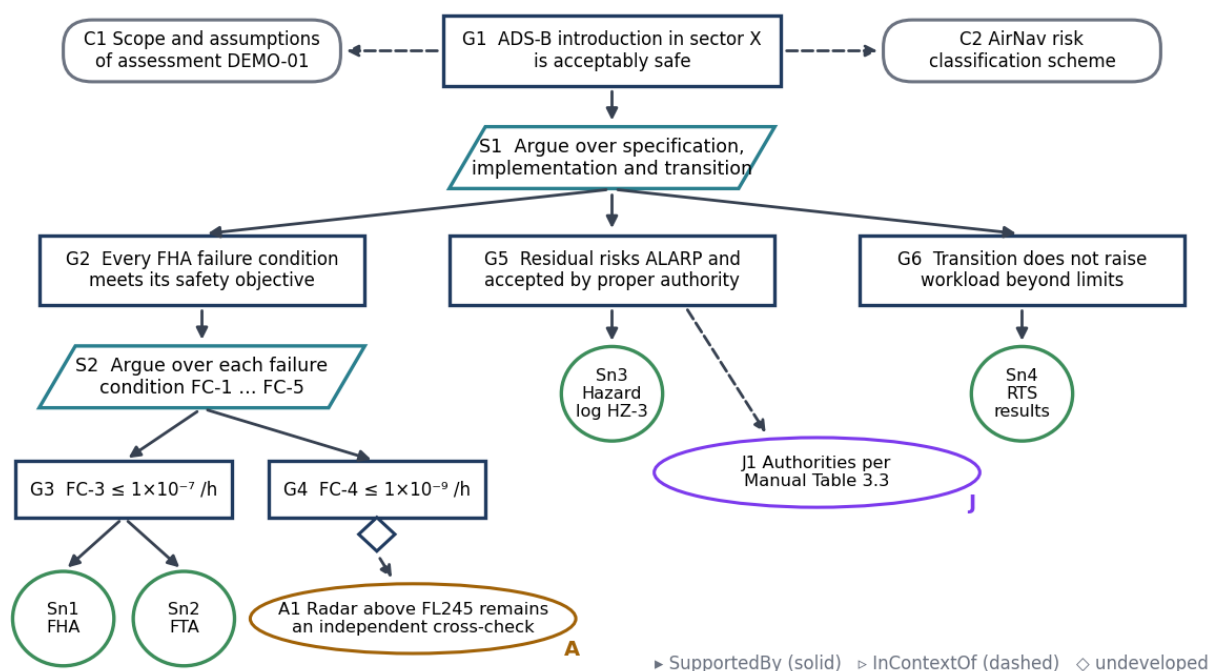


Figure 14. Safety argument for the ADS-B introduction (NAVRAP GSN module, with links to the FHA, FTA, hazard log and simulation studies).

Goal G4 ($FC-4 \leq 10^{-9}$ per operating hour) is left undeveloped: it depends on assumption A1 that radar above FL245 remains available as an independent cross-check, which must be confirmed with the engineering directorate before the argument is complete. NAVRAP checks that every solution is linked to evidence and flags evidence that is not yet complete or accepted, so the argument shows at a glance what is still provisional.

22.4 Tips and pitfalls

- Write goals as claims, not as activities: “the FTA shows the objective is met”, not “perform FTA”.
- Arguments that jump from a high-level claim straight to a pile of evidence hide the reasoning. Use strategies.
- Assumptions are liabilities: each one needs an owner and a way of being confirmed during operation.

23. Common cause analysis (ZSA, PRA, CMA)

At a glance	
Purpose	Check that items claimed to be independent — redundant chains, diverse sensors, backup power — really are, so that FTA, RBD and FHA results can be trusted.
When to use	Every system assessment that credits redundancy; site or room changes; after an event that took out more than one chain at once.
Inputs	Architecture and installation drawings; FTA AND gates and RBD parallel blocks; site survey; maintenance and software configuration records.
Outputs	Findings of zonal, particular-risk and common-mode analyses; independence claims confirmed or rejected; common-cause terms to add to the FTA; actions.
Team and time	Systems engineer, site/facilities engineer, safety assessor; site visit plus one to three days.
Key references	SAE ARP4761A common cause analysis [R20]; IEC 61508-6 β -factor guidance.

23.1 The three parts

Part	Question	Typical ANSP findings
Zonal safety analysis (ZSA)	Does the installation respect separation rules? What else is in the same room, rack or cable tray?	Main and fallback servers in one fire zone; shared cable trays; water pipes above equipment rooms.
Particular risks analysis (PRA)	Which single external events (fire, flood, lightning, earthquake, volcanic ash, storm, cyber attack) can hit several items at once?	Lightning taking out radar and ADS-B at one site; volcanic ash blocking HVAC and generator intakes.
Common mode analysis (CMA)	Do the items share design, software, maintenance, procedures or utilities?	Same software in main and fallback; one surveillance WAN; one technician calibrating both chains.

23.2 Procedure

1. List every independence claim: each AND gate in the FTA, each parallel group in the RBD, each “no single failure” statement in the FHA. NAVRAP can import these from the FTA.
2. Visit the sites and rooms (ZSA); record the zones, what is installed and any separation concerns.
3. Go through the particular-risk list for each zone and record which risks defeat the redundancy.
4. For each claim, check the common-mode sources (design, software, manufacture, installation, maintenance, operation, environment, shared utilities, location) and decide whether independence holds.
5. Where it does not, redesign or add a common-cause term to the FTA (for example the β -factor model) and send the finding to the hazard log.

23.3 Worked example — ACC surveillance chain

Claim (from FTA)	Common-mode source	Finding	Independent?	Action
Main SDP (E5) and fallback (E6)	Software	Fallback runs the same software baseline	No	Diverse fallback or β -factor term in FTA
Mains (E7), UPS (E8), generator (E9)	Location	UPS and generator switchgear share the power room	Partly	Include fire CCF in FTA
Radar A, radar B, ADS-B (E1-E3)	Shared utilities	All three use the single surveillance WAN (E4)	No	Diverse data path
ZSA: equipment room 1	Location	Both SDP chains in one fire zone	—	Relocate fallback to room 2

None of the three claims fully holds. The FTA of Chapter 14 already showed the WAN as the dominant contributor; the CCA adds two further dependencies that the original tree treated as independent and that would otherwise make the calculated probability optimistic.

23.4 Tips and pitfalls

- Diversity of hardware does not help if the software, the data or the maintenance team is common.
- Revisit the ZSA after every installation change — independence is often lost quietly during later works.
- A β -factor of 1–10% is typical for redundant electronic items without deliberate diversity.

24. RBD and Markov analysis — availability of services

At a glance	
Purpose	Calculate the availability and continuity of a service from the reliability and repair of its parts, and find which part limits it.
When to use	CNS and ATM systems with availability and continuity requirements (voice, surveillance, navigation aids, data links); design of redundancy and maintenance; checking a supplier's claims.
Inputs	Architecture; MTBF and MTTR of each part; for Markov models, standby behaviour, switchover coverage and repair-crew constraints.
Outputs	Availability, unavailability and expected downtime; importance of each block; for Markov, failure frequency, mean down time and mean time to first failure.
Team and time	Reliability or systems engineer; one to three days.
Key references	IEC 61078 (RBD) [R47]; IEC 61165 (Markov) [R48].

24.1 Key concepts

A block's steady-state availability is $A = \text{MTBF}/(\text{MTBF} + \text{MTTR})$. In a **series** arrangement all blocks must work, so $A = \prod A_i$. In a **parallel** (redundant) arrangement one is enough, so $A = 1 - \prod(1 - A_i)$. In a **k-out-of-n** arrangement k of n identical-role blocks must work. These rules assume independent blocks, each with its own repair.

When those assumptions fail — one repair team for two chains, a cold standby that must start, a switchover that works only with probability c (coverage) — a **Markov model** is used. The system is described by states (e.g. both chains up, one up, none up) and transition rates ($\lambda = 1/\text{MTBF}$ for failures, $\mu = 1/\text{MTTR}$ for repairs). Solving $\pi Q = 0$ with $\sum \pi = 1$ gives the long-run probability of each state; availability is the sum over up states.

24.2 Procedure

1. Define the service and what “available” means (e.g. the sector frequency usable at the controller working position).
2. Draw the RBD from the architecture; obtain MTBF and MTTR (from maintenance records where possible).
3. Calculate availability and block importance; compare with the requirement.
4. Where standby, coverage or shared repair matter, build a Markov model of that part and compare.
5. Check independence with common cause analysis (Chapter 23).

24.3 Template

Block	Arrangement	MTBF (h)	MTTR (h)	Source	A

24.4 Worked example — sector VHF voice availability

A = 0.9999517 · downtime 0.42 h/yr (25 min) · VCCS dominates

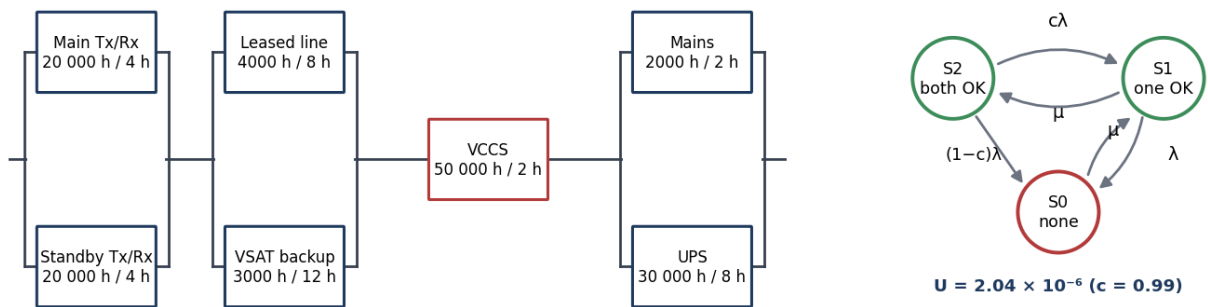


Figure 15. RBD of the sector frequency (left) and Markov model of the main/standby radio pair with switchover coverage c (right).

The RBD gives $A = 0.9999517$: an expected downtime of 0.42 hours (25 minutes) a year. Making the single VCCS perfect would remove about 21 of those 25 minutes, so the VCCS is the block to improve; the redundant radio chains contribute almost nothing.

The RBD, however, assumes a perfect switchover between the main and standby radio chains. The Markov model with a coverage of 0.99 and one repair team gives an unavailability of **2.04×10^{-6}** for the radio pair, about fifty times the 4×10^{-8} the RBD implies for the same pair: the 1% of failed switchovers dominates. The system failure frequency is 5.1×10^{-7} per hour (a mean of about 220 years between failures) with a mean down time of 4 hours. Testing the switchover regularly is therefore worth more than improving radio MTBF.

24.5 Tips and pitfalls

- Use MTTR including travel and logistics time for remote sites, not just the repair itself.
- Availability is not continuity: a service can be 99.99% available and still fail too often during critical periods; state which requirement is being checked.
- Keep Markov models small (a few states); large models are hard to review.

25. SWIFT — Structured What-If Technique

At a glance	
Purpose	Identify hazards of a small, well-defined or temporary change quickly, using “what if...?” questions structured by prompt categories.
When to use	Works on the manoeuvring area, temporary procedures, equipment outages, special events, minor changes where a full HAZID or HAZOP would be disproportionate.
Inputs	Description of the change, drawings and timelines; relevant occurrences; the prompt list.
Outputs	What-if worksheet with consequences, safeguards, risk and recommendations; hazard log entries.
Team and time	Facilitator + 3–6 SMEs; one to three hours.
Key references	IEC 31010 [R11]; Card, Ward and Clarkson (2012) [R49].

25.1 Procedure

1. Describe the change and agree its boundaries and duration.
2. Brainstorm what-if questions per prompt category: people, procedures, equipment and utilities, information and communication, timing and sequence, location and environment, quantity and demand, external events, management and organisation.
3. For each credible question, record the consequence, the safeguards in place, severity and likelihood, and any recommendation with an owner.
4. Check coverage: every category considered, with a record where nothing credible was found.
5. Send the hazards to the hazard log.

25.2 Worked example — night works on runway 07L/25R

What if...?	Consequence	Safeguards	S/L	Recommendation
...the runway is handed back before all vehicles have left?	Aircraft lands with vehicle on runway	Works completion check; runway inspection	A/2	Signed hand-back with vehicle count
...the NOTAM for the closure is not published in time?	Crews plan for a closed runway	ATIS; TWR check	C/3	NOTAM 7 days ahead; AIM checklist
...lighting is left in the wrong configuration after works?	Misleading lighting at night	Lighting check before reopening	C/2	Add lighting to hand-back checklist
...works overrun into the morning arrival peak?	Single-runway operations at peak; holding	Flow measures	D/3	Hard stop time; contingency flow plan
...a works driver is unfamiliar with the aerodrome at night?	Vehicle incursion onto a taxiway in use	Escort	B/2	Mandatory escort; briefing

The first line (risk index 2A, tolerable-upper) justifies a formal hand-back procedure. Chapter 30 investigates an occurrence in which the escort recommended in the last line was not in place.

25.3 Tips and pitfalls

- SWIFT relies on the team’s experience; include someone who has done the job at night.
- If more than about twenty credible what-ifs appear, the change is not small; move to HAZID or HAZOP.

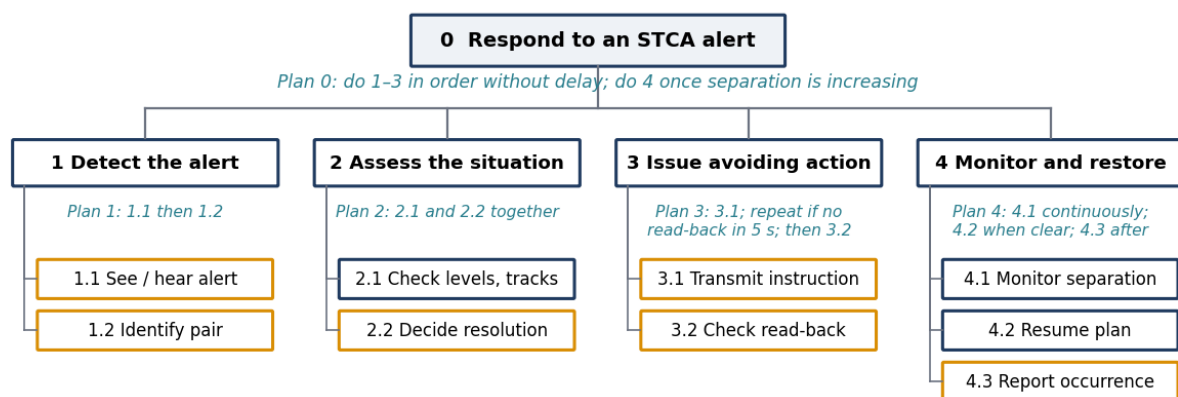
26. HTA — Hierarchical task analysis

At a glance	
Purpose	Describe what people have to do to achieve a goal, as a hierarchy of tasks and sub-tasks with plans that say when each is done; then look for credible errors at the lowest level.
When to use	New or changed procedures, controller tools, HMIs and alerts; as input to HRA, JHA, HAZOP of procedures and training design.
Inputs	Procedures, observation of work, interviews with practitioners.
Outputs	Task hierarchy with plans; error modes for each bottom-level task; links to HRA values.
Team and time	Human factors specialist + 2-3 practitioners; one to two days per task.
Key references	Annett and Duncan (1967); Stanton (2006) [R50]; SHERPA [R51].

26.1 Procedure

1. State the overall goal (task 0).
2. Break it into three to eight sub-tasks and write the **plan** that governs them (order, conditions, repetition).
3. Continue breaking down until further detail adds nothing to the purpose of the analysis — the “P × C” rule: stop when the probability of error times the cost of error is acceptable.
4. Check the hierarchy with practitioners against work as done, not only as written.
5. For each bottom-level task, identify credible error modes (omitted, too early/late, wrong action, wrong object, check omitted, information not obtained or communicated) and their consequences and recovery; send the critical ones to HRA (Chapter 20).

26.2 Worked example — responding to an STCA alert



Amber = step with credible error modes (linked to HRA)

Figure 16. HTA for responding to an STCA alert, with the plans and the steps that carry credible error modes (NAVRAP HTA module).

Six bottom-level steps have credible error modes. Step 1 (detect the alert) is linked to HRA task T1 (Chapter 20, $HEP\ 2.0 \times 10^{-3}$); step 3.1 (transmit an instruction with the “avoiding action” phraseology, error modes: too late; wrong information) and 3.2 (check read-back omitted) are the steps where simulator training concentrates.

26.3 Tips and pitfalls

- The plan is the most valuable and most often forgotten part of an HTA; every task with sub-tasks needs one.
- Describe tasks with a verb and an object; avoid describing equipment functions.

27. Real-time and fast-time simulation in safety assessment

At a glance	
Purpose	Provide evidence about how a change will perform with real controllers (real-time simulation, RTS) or with modelled traffic at large scale (fast-time simulation, FTS), measured against safety objectives.
When to use	Airspace and procedure changes, new controller tools, sector capacity changes — whenever workload, conflicts, R/T load or human-machine interaction are part of the safety argument.
Inputs	Validation objectives and hypotheses; traffic samples; simulation platform; participants; measures and success criteria.
Outputs	Measured values for baseline and new solution with uncertainty; statistical comparison; judgement against criteria; hazards discovered during runs.
Team and time	Validation lead, simulation engineers, human factors specialist, 4-12 controllers; weeks to months.
Key references	EUROCONTROL E-OCVM [R52].

27.1 Procedure

1. Derive the safety-related objectives from the hazard analysis (for example “the new procedure does not increase controller workload”).
2. Choose measures (workload such as ISA or NASA-TLX, R/T occupancy, conflict and STCA counts, losses of separation, situation awareness) and a success criterion for each *before* the runs.
3. Design the exercise: baseline and solution, traffic samples, number of runs, counterbalancing to control learning effects.
4. Run, observe, and collect data and debrief notes; record every unexpected event as a potential hazard.
5. Analyse: means, 95% confidence intervals, a test of the difference (Welch’s t-test when variances may differ), and the criterion.
6. Report limitations (realism, participant sample, traffic sample) and feed results into the safety argument (Chapter 22).

27.2 Worked example — new arrival procedure, approach north sector

Eight runs per condition with four controller teams, peak traffic of 38 arrivals an hour:

Measure	Baseline mean	Solution mean (95% CI)	Welch p	Criterion	Met?
M1 Workload, executive ATCO (ISA 1-5)	3.69	3.16 (3.02-3.30)	< 0.001	≤ 3.5	Yes
M2 STCA alerts per hour	1.10	0.85 (0.72-0.98)	0.025	No worse than baseline	Yes
M3 R/T occupancy (%)	61.6	51.8 (50.0-53.5)	< 0.001	$\leq 60\%$	Yes

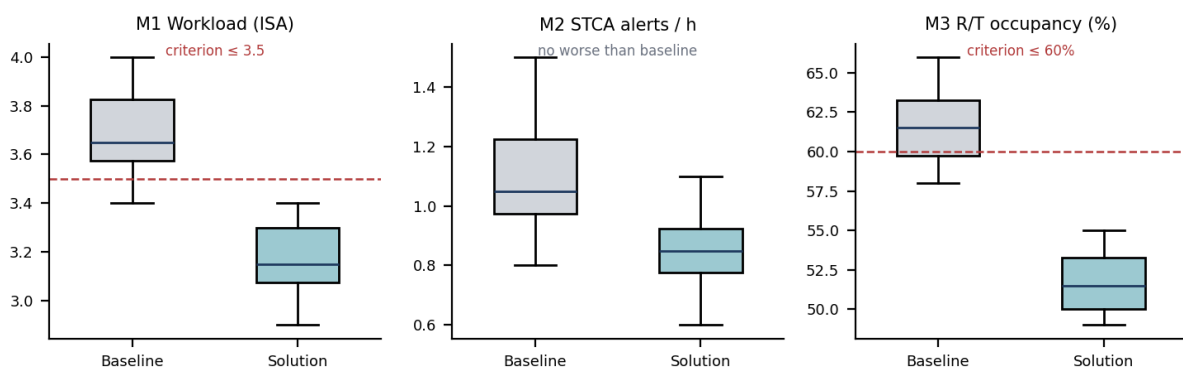


Figure 17. Distribution of run results for the three measures (NAVRAP simulation module).

All three criteria are met and the differences are both statistically and operationally meaningful. The main limitation is that all runs used the same traffic pattern; a fast-time study of other traffic mixes is recommended before the argument is closed.

27.3 Tips and pitfalls

- Set the criteria before the runs; criteria chosen after seeing the data are not evidence.
- Eight runs cannot show that a rare event (a loss of separation) has become less likely; use simulation for workload and precursors, and CRM/ETA for rare events.
- Participants who helped design the procedure are not a neutral sample.

28. Structured expert judgement (Cooke's classical model and Delphi)

At a glance	
Purpose	Obtain defensible numbers — probabilities, rates, times — where data are missing, by eliciting and combining expert estimates in a structured, auditable way.
When to use	Filling gaps in FTA, ETA, LOPA, BBN or HRA; rare events; new operations with no history.
Inputs	Well-defined target questions; seed (calibration) questions whose answers the facilitator knows; four to ten experts.
Outputs	Expert calibration and information scores and weights; a combined distribution (5%, 50%, 95%) for each target; or, for Delphi, the panel median and spread by round.
Team and time	Trained facilitator, experts; one day of elicitation plus preparation.
Key references	Cooke (1991) [R53]; Linstone and Turoff (1975) [R54].

28.1 Cooke's classical model

Each expert gives the 5%, 50% and 95% quantiles of their uncertainty for every question. For the seed questions the true value is known, so the expert's **calibration** can be measured: a well-calibrated expert sees about 5% of true values below their 5% quantile, 45% between 5% and 50%, and so on. The calibration score is the p-value of a χ^2 test of that distribution ($C = 1 - \chi^2_3(2N \cdot I(s,p))$), where I is the relative information of the observed against the expected proportions $p = 0.05, 0.45, 0.45, 0.05$). The **information** score rewards narrow distributions. An expert's weight is proportional to calibration $\times$ information; experts whose calibration falls below a significance level α receive zero weight. The weighted mixture of the experts' distributions is the "decision maker".

28.2 Delphi

Delphi is simpler: experts give anonymous estimates, see the panel median, interquartile range and the reasons given by outliers, and revise in further rounds. It produces convergence but not a performance-based weighting; use it where seed questions cannot be found, and report the spread, not just the median.

28.3 Procedure (classical model)

1. Define the target questions so that the answer is observable in principle (units, period, conditions).
2. Prepare 8–10 seed questions from the same domain whose answers are known to the facilitator but not to the experts (NAVRAP accepts fewer for illustration).
3. Train the experts in giving quantiles; elicit individually.
4. Compute scores and weights; compare performance-weighted and equal-weight results.
5. Report the combined distributions, the weights and the elicitation record.

28.4 Worked example — probability that the controller does not resolve a level-bust conflict before STCA

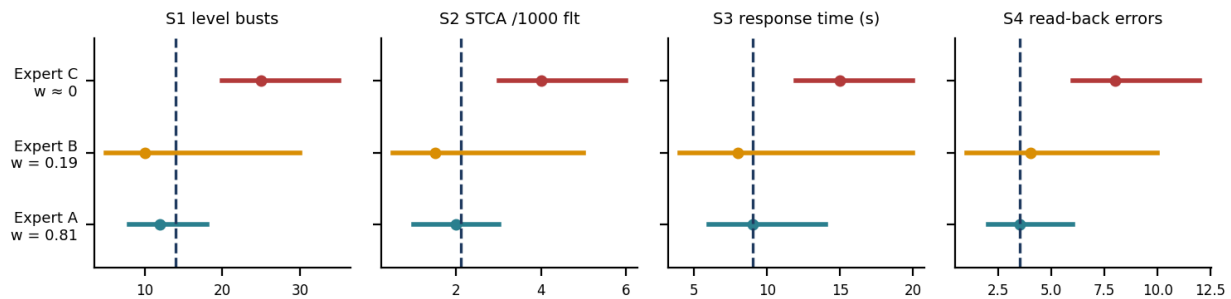


Figure 18. Answers of three experts to four seed questions (bar = 5-95%, dot = median; dashed line = true value).

Expert	Calibration	Information	Weight
A	0.84	0.77	0.81
B	0.60	0.25	0.19
C	2.5×10^{-5}	0.56	≈ 0

Expert C gave narrow ranges that missed every true value and receives no weight. For the target question the performance-weighted decision maker gives 5% / 50% / 95% = **0.016 / 0.082 / 0.31**; equal weighting gives 0.014 / 0.126 / 0.39. The median of 0.08 is close to 1 – 0.9, the controller-detection value used in the ETA of Chapter 19, and CARA task T2 (0.058) is inside the range. A five-person Delphi panel on the same question kept a median of 0.10 over three rounds while the interquartile range narrowed from 0.15 to 0.02.

28.5 Tips and pitfalls

- Overconfidence (ranges too narrow) is the norm, not the exception; the seed questions exist to detect it.
- Keep a full record: questions, answers, scores. The weights must be reproducible.
- Do not use Delphi consensus as if it removed uncertainty.

29. Security risk assessment with safety impact

At a glance	
Purpose	Identify intentional threats (cyber and physical) to ATM/CNS systems and data, assess the security risk, and pass those with a safety effect into the safety risk process.
When to use	New or changed ATM/CNS systems and interfaces, remote maintenance, cloud or third-party services, and periodic review of the security risk register.
Inputs	Asset inventory; architecture and data flows; threat intelligence; vulnerability information; existing controls.
Outputs	Risk register with confidentiality, integrity and availability impact, likelihood and security risk level; safety severity of each threat scenario; treatment plan.
Team and time	Information security specialist, systems engineer, safety assessor, operational SME; days to weeks.
Key references	ISO/IEC 27005:2022 [R55]; EUROCAE ED-205 [R56]; STPA-Sec [R57]; EU Part-IS [R62].

29.1 Key concepts

Safety and security assessments look at the same systems from different directions. A **threat scenario** combines an asset, a threat source, a threat and a vulnerability. NAVRAP rates its impact on confidentiality (C), integrity (I) and availability (A) from 1 to 5, and the likelihood from 1 to 5. The security risk score is **L × max(C, I, A)**: 15 and above is very high, 10–14 high, 5–9 medium, below 5 low. Separately, the safety assessor gives the severity of the **safety effect** using the safety matrix (Chapter 3). Scenarios with a safety effect go into the safety hazard log so that one register shows both views; STPA-Sec (Chapter 13 applied to adversarial control actions) is a useful way of finding such scenarios.

29.2 Procedure

1. Establish the context: scope, assets, interfaces and the criteria above.
2. Identify threat scenarios for each asset (use threat catalogues and STPA-Sec for control paths).
3. Rate C, I, A impact and likelihood considering existing controls; compute the security risk.
4. Assess the safety effect of each scenario with the operational SMEs.
5. Decide treatment; send scenarios with a safety effect to the hazard log; record residual risk and owners.

29.3 Worked example — surveillance and flight data processing

Scenario	C/I/A	L	Security risk	Safety effect	Treatment
False ADS-B targets injected (unauthenticated messages)	1/5/3	3	15 Very high	B	Plausibility checks; alert on radar/ADS-B mismatch; multilateration
Ransomware encrypts FDP servers (flat office/ops network)	3/4/5	2	10 High	C	Network segmentation; offline backups; contingency strips
Vendor remote-maintenance account compromised (shared credentials)	3/4/5	2	10 High	B	Named accounts with MFA; session approval and logging
Unauthorised change to safety-net parameters (no two-person rule)	1/5/2	2	10 High	B	Two-person rule; checksum verification before loading

All four scenarios have a safety effect and go to the hazard log. The ADS-B spoofing scenario links to the GSN argument of Chapter 22: assumption A1 (radar as an independent cross-check) is also the main mitigation for spoofing, which strengthens the case for confirming it.

29.4 Tips and pitfalls

- Do not treat likelihood of attack like a random failure rate; it depends on attacker motivation and capability, and changes over time.
- Integrity attacks matter most for ATM safety: a false but plausible picture is worse than a lost one.
- Keep the security register confidential; the hazard log entry should describe the safety effect without exploit detail.

30. Occurrence investigation (SOAM, HFACS, Tripod Beta)

At a glance	
Purpose	Explain why an occurrence happened — which barriers failed and which conditions and organisational factors lay behind them — so that actions address causes, not only symptoms.
When to use	Investigations selected by occurrence risk classification (Chapter 21); trend reviews; linking real events back to bowties and the hazard log.
Inputs	Occurrence report, radar and voice recordings, interviews, procedures, rosters, maintenance records.
Outputs	SOAM chart, HFACS classification or Tripod Beta tree; failed and absent barriers linked to bowtie barriers; safety actions.
Team and time	Trained investigator(s) with operational SMEs; days to weeks.
Key references	ICAO Doc 9756 [R61]; EUROCONTROL SOAM [R58]; Wiegmann and Shappell (HFACS) [R59]; Tripod Beta [R60].

30.1 Three complementary tools

Tool	Structure	Best for
SOAM (Systemic Occurrence Analysis Methodology)	Works back from the occurrence: absent/failed barriers (awareness, restriction, detection, control and recovery, protection and containment, escape and rescue) → human involvement → contextual conditions → organisational factors (12 categories).	ATM occurrences; EUROCONTROL-aligned reporting; linking to barriers.
HFACS	Classifies failures in four tiers and 19 categories: unsafe acts, preconditions, unsafe supervision, organisational influences.	Human factors trends across many occurrences.
Tripod Beta	Event trios (agent → object → event) with failed or missing barriers; each failure traced through immediate cause → precondition → underlying cause, assigned to a Basic Risk Factor (BRF).	Complex events with several chains; organisational learning.

30.2 Procedure

1. Establish the facts and sequence of events; keep a timeline.
2. Identify the barriers that were absent, failed or worked; link each to the bowtie barrier it corresponds to (NAVRAP counts these failures on the bowtie).
3. For each failure, identify the human involvement and the conditions that made it likely, then the organisational factors behind those conditions.
4. Classify with HFACS where trend analysis is wanted; build a Tripod tree for complex events.
5. Formulate actions that address organisational factors and barriers, each with an owner; update the hazard log and bowtie.

30.3 Worked example — OCC-102: works vehicle on the runway strip

At night a contractor's works vehicle entered the runway strip of 07L without clearance. RIMCAS alerted and the TWR controller instructed an aircraft on 4 NM final to go around.

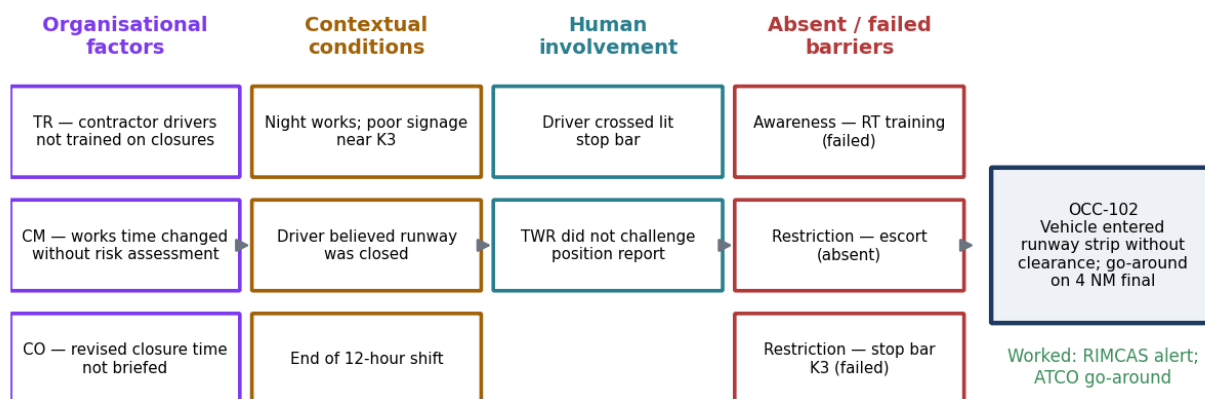


Figure 19. SOAM chart for OCC-102 (NAVRAP investigation module).

HFACS. Skill-based error (crossed a lit stop bar); adverse physiological state (12-hour shift); physical environment (poor signage at night); inadequate supervision (no escort); organisational process (works timing changed without a risk assessment).

Tripod Beta. Agent: works vehicle movement; object: active runway and aircraft on final; event: vehicle enters the strip. Failed barrier “stop bar at K3”: immediate cause — driver crossed a lit stop bar; precondition — belief the runway was closed; underlying cause — closure time change not communicated (BRF: communication). Missing barrier “escort”: underlying cause — escort not required by the works procedure (BRF: procedures).

Actions. Escort all contractor vehicles at night; include contractors in RT and aerodrome familiarisation training; risk-assess any change to works timing (the SWIFT of Chapter 25 had already recommended an escort). The three failed or absent barriers are linked to the runway incursion bowtie (Chapter 6), where they now show one recorded failure each.

30.4 Tips and pitfalls

- Stop at organisational factors that the organisation can act on; “human error” is a starting point, not a finding.
- Record barriers that worked as carefully as those that failed.
- Keep investigation (learning) separate from any disciplinary process, in line with the just culture policy.

31. Wildlife strike risk assessment (species risk matrix)

At a glance	
Purpose	Rank the wildlife species that threaten aircraft at and around an aerodrome, species by species, so that habitat management, dispersal, land-use action and ATS bird warnings are aimed at the species that carry the risk.
When to use	Annual review of the aerodrome wildlife hazard management programme; after a damaging strike or a rise in strike rate; before a change that affects habitat (runway works, drainage, new approach path, new land use nearby); when setting SPLs for wildlife.
Inputs	Strike reports for at least three, preferably five, years with the species identified; aircraft movements; bird and bat surveys (counts by month and time of day); a map of land uses within about 13 km.
Outputs	Species risk matrix and ranking; strike rate per 10,000 movements and its trend; species profiles with seasonality; register of attractants; species management measures; hazard log entries for high and moderate species.
Team and time	Aerodrome wildlife officer or ornithologist, safety assessor, TWR/APP representative, airline safety representative; one day once the data are prepared.
Key references	ICAO Annex 14 Vol I §9.4 and Doc 9137 Part 3 [R63]; Allan (2006) [R64]; UK CAA CAP 772 [R65]; FAA AC 150/5200-33 [R66]; ICAO IBIS, Doc 9332 [R67].

31.1 Key concepts

A general hazard matrix works poorly for wildlife because the risk depends almost entirely on *which* animal is present. A swallow is struck often and almost never causes damage; a flock of egrets is struck less often but can cause multiple engine ingestion. The species risk matrix, developed by Allan (2006) and used in UK CAA CAP 772, rates each species (or group of similar species) separately on two axes:

Likelihood is the mean number of strikes per year with that species over the review period. **Severity** is the share of that species' strikes that caused damage or had an effect on flight (rejected take-off, return, precautionary landing). When a species has too few strikes for a stable percentage — NAVRAP uses fewer than five — severity is estimated from body mass, raised one class for flocking species, because mass and numbers drive impact energy. The risk is the product of the two classes.

Class	Likelihood: strikes per year	Severity: % damaging or effect on flight	Surrogate: body mass
5 Very high	> 10	> 20%	≥ 1.8 kg
4 High	3 - 10	10 - 20%	1.0 - 1.8 kg
3 Moderate	1 - 3	5 - 10%	0.3 - 1.0 kg
2 Low	0.3 - 1	1 - 5%	0.1 - 0.3 kg
1 Very low	< 0.3	< 1%	< 0.1 kg

Risk bands. Risk = likelihood class × severity class: 12 or more is high (species-specific action required and monitored), 6 to 11 is moderate (targeted measures), 5 or less is low (general measures and monitoring). These bands are the NAVRAP defaults in the structure of the Allan method; each aerodrome should calibrate them against its own strike history and record the calibration.

Three further ideas complete the picture. The **strike rate per 10,000 movements** separates a real change in wildlife risk from a change in traffic. **Seasonality and time of day** — from monthly surveys and strike times — show when measures must be strongest. **Attractants** within about 13 km of the aerodrome (ICAO Annex 14 §9.4) — landfills, fish ponds, rice fields, roosts, grazing land — explain why a species is there and are often outside the aerodrome operator's control, so they need the local-government coordination forum.

31.2 Roles

Wildlife hazard management on the aerodrome is the aerodrome operator's responsibility. The ANSP owns two barriers in the wildlife bowtie — passing bird information to pilots (ATIS remarks, real-time warnings, NOTAM for seasonal peaks) and coordinating runway inspections and dispersal with the wildlife unit — and is a major source of strike reports. The species risk assessment should therefore be done jointly, with ATS actions and SPIs recorded against the ANSP.

31.3 Procedure

1. Collect the strike records for the review period and check species identification; send unidentified remains for feather or DNA identification. Record every strike, including those found on runway inspection.
2. Group species where identification is uncertain or behaviour is similar (e.g. "swallows and swifts"), but keep large and flocking species separate.
3. For each species, count strikes and damaging strikes per year and enter annual movements; calculate likelihood, severity and risk.
4. Review surveys and strike times to build a profile of the high and moderate species: season, time of day, where on the aerodrome, what attracts them.
5. Map attractants within about 13 km and assign an owner and an action to each.
6. Agree species-specific measures (habitat, dispersal, land use, detection, ATS information) with owners; send high and moderate species to the hazard log.
7. Set SPIs — strike rate per 10,000 movements overall and for each high species, damaging strikes, strikes above 500 ft — and repeat the assessment every year.

31.4 Worked example — cattle egret (*Bubulcus ibis*) at a lowland aerodrome

An aerodrome surrounded by rice fields, with cattle grazing beside the eastern fence and an egret roost in mangrove 6 km away, reviewed five years of strikes (2021–2025; 327,000 movements; 228 strikes with the species identified, 6.97 per 10,000 movements). All numbers are illustrative.

Species	Mass, behaviour	Strikes (per year)	Damaging	L × S	Risk
Cattle egret (kuntul kerbau) <i>Bubulcus ibis</i>	0.34 kg, flocks	62 (12.4)	7 (11.3%)	5 × 4 = 20	High
Large flying fox (kalong besar) <i>Pteropus vampyrus</i>	1.0 kg, colonial	3 (0.6)	1 — mass surrogate	2 × 5 = 10	Moderate
Pacific golden plover (cerek kernyut) <i>Pluvialis fulva</i>	0.13 kg, flocks; migrant	11 (2.2)	1 (9.1%)	3 × 3 = 9	Moderate
Javan pond heron (blekok sawah) <i>Ardeola speciosa</i>	0.25 kg, solitary	22 (4.4)	1 (4.5%)	4 × 2 = 8	Moderate
Barn swallow <i>Hirundo rustica</i>	0.018 kg, flocks	90 (18.0)	0	5 × 1 = 5	Low
Eurasian tree sparrow <i>Passer montanus</i>	0.022 kg, flocks	40 (8.0)	0	4 × 1 = 4	Low

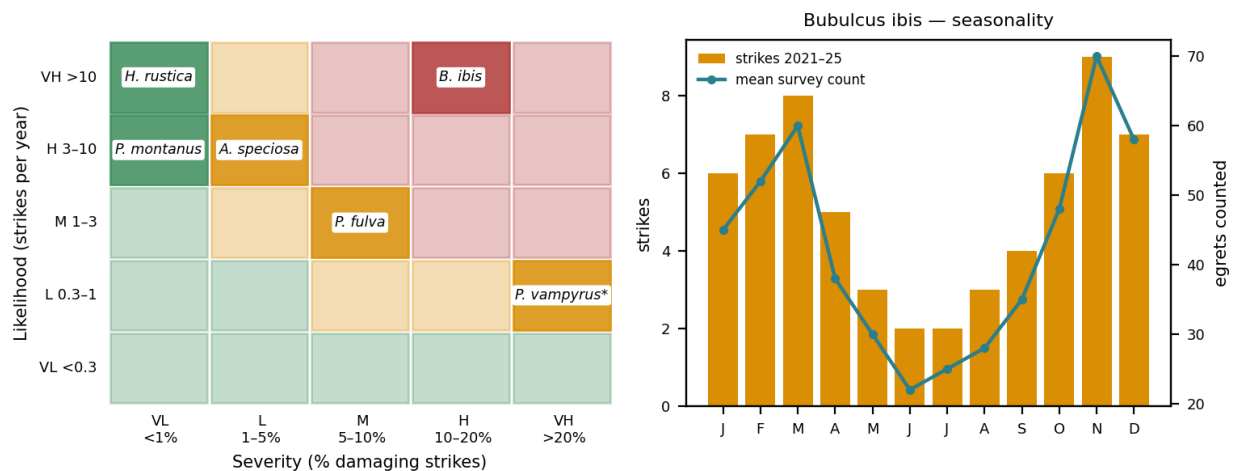


Figure 20. Species risk matrix for the worked example (left; * severity from the mass surrogate) and cattle egret seasonality (right) — NAVRAP wildlife module.

The swallow is struck most often but ranks fifth, because its strikes almost never cause damage. The cattle egret is the only high-risk species. Its strike count rose from 9 to 16 a year, but the rate per 10,000 movements was roughly flat (2.1 in 2021, 2.0 in 2025) because traffic doubled — the risk grew with exposure, not because egret activity increased. Strikes peak in November–March, following the wet season, grass growth and rice harvest, and cluster at 05:30–08:00 and 17:00–18:30 when flocks commute between the roost and feeding areas.

Measures for the cattle egret. A long-grass policy on the strip (sward kept at about 150–200 mm, confirmed locally for tropical grasses) and mowing outside the dawn and dusk peaks — never ahead of an arrival wave — remove the main on-aerodrome food source; dawn and dusk dispersal patrols continue; a grazing agreement moves cattle away from the fence line; the farmers' association notifies ploughing and harvest under the approach; the roost flight lines are mapped so that the wildlife unit can warn TWR when commuting flocks cross the approach. For the ANSP: ATIS bird remarks and real-time warnings during the seasonal and daily peaks, a seasonal NOTAM, and strike reporting with the species identified. SPI: cattle egret strikes per 10,000 movements, alert level above 2.5 in any rolling quarter.

31.5 Tips and pitfalls

- The method is only as good as species identification. If a third of remains are “unknown bird”, fix identification before trusting the ranking.
- Under-reporting inflates the damage percentage (damaging strikes are always reported, harmless ones often not). Encourage reporting of every strike, including carcasses found on inspection.
- Use rates, not counts, for trends; a rising count at a growing aerodrome may be no change in risk.
- Rice-field and tambak landscapes change through the year; repeat surveys monthly and re-rate species at least annually.
- Culling, where lawful, treats symptoms; habitat and land-use measures treat causes.

32. Integrating results and keeping them alive

32.1 From methods to one risk picture

Each method produces hazards, causes, controls and sometimes numbers. They must be brought together in one place — the hazard log — using one vocabulary and one risk classification scheme, otherwise the organisation ends up with twenty-six disconnected spreadsheets. The following cross-walk maps the terms used by each method onto the hazard log fields.

Hazard log field	Bowtie	HAZID / HAZOP	FMEA	LOPA	FHA	STPA	FTA	BBN
Hazard / top event	Hazard, top event	Hazard / deviation	End effect	Outcome	Failure condition	Hazard	Top event	Outcome node
Cause	Threat	Cause	Failure mode / cause	Initiating event	—	UCA, loss scenario	Cut set	Parent nodes
Control	Barrier	Safeguard	Detection, backup	IPL	Safety req.	Constraint	AND gate	Mitigating node
Likelihood	Qualitative	Matrix	O rating / C _m	Frequency	Objective	—	Probability	Probability

Methods added in edition 0.2 map in the same way: ETA end states and CCA findings become hazards or causes; HRA values, CRM results and RBD/Markov availabilities become likelihood evidence; ERC-classified occurrences and investigation findings are linked to the hazard and to the bowtie barriers that failed; security scenarios with a safety effect become hazards; and the GSN argument references hazard log entries as solutions. Wildlife species rated high or moderate become hazards, with their species measures as controls.

32.2 Minimum hazard log content

- Unique hazard ID, title and description; affected unit(s), system(s) and operation.
- Source assessment(s) and method(s); date; facilitator and participants.
- Causes/threats and consequences; existing controls with owner and verification status.
- Initial and residual risk index, with the basis of each judgement.
- Additional controls/actions with owner, due date and status; safety requirements and how they are verified.
- Acceptance decision, authority and date; review date; links to SPIs and to related occurrences.

32.3 Safety Assessment Report structure

1. Summary and conclusion (is the change acceptably safe, under which conditions?).
2. Description of the change, scope, boundaries and assumptions.
3. Methods used and why; participants.
4. Hazards identified and their assessment (reference to hazard log).
5. Safety objectives and requirements; evidence that they are or will be met.
6. Residual risk and acceptance; outstanding actions and conditions of implementation.
7. Post-implementation monitoring plan (SPIs, review date).
8. Annexes: worksheets, models, calculations, data sources.

32.4 Monitoring and review

Assessments rest on assumptions. After implementation, AirNav shall monitor that the assumptions hold and the controls work: link critical barriers and safety requirements to SPIs, review occurrence reports against the relevant bowties and models, and re-open an assessment when an assumption is invalidated, when an SPI crosses its alert level, or at the review date set in the hazard log.

Appendix A. Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
A-SMGCS	Advanced surface movement guidance and control system	FRMS	Fatigue risk management system
ACAS	Airborne collision avoidance system	FTA	Fault tree analysis
ACC	Area control centre	HAZID	Hazard identification study
ADS-B	Automatic dependent surveillance – broadcast	HAZOP	Hazard and operability study
AFTN	Aeronautical fixed telecommunication network	IPL	Independent protection layer
AIDC	ATS inter-facility data communication	JHA	Job hazard analysis
AIM / AIS	Aeronautical information management / service	KSS	Karolinska sleepiness scale
AIP / eAIP	(Electronic) aeronautical information publication	LoA	Letter of agreement
AIRAC	Aeronautical information regulation and control (28-day cycle)	LOPA	Layer of protection analysis
AIXM	Aeronautical information exchange model	LoS	Loss of separation
ALARP	As low as reasonably practicable	LOTO	Lock-out / tag-out
AMAN	Arrival manager	MSA / MVA	Minimum sector / vectoring altitude
AMHS	ATS message handling system	NOTAM	Notice to airmen
ANSP	Air navigation service provider	OJT / OJTI	On-the-job training / instructor
APP	Approach control	PFD	Probability of failure on demand
ATCO	Air traffic controller	PIB	Pre-flight information bulletin
ATSEP	Air traffic safety electronics personnel	PSSA / SSA	Preliminary / system safety assessment
BBN	Bayesian belief network	RIMCAS	Runway incursion monitoring and collision avoidance system
CAST	Causal analysis based on STAMP	RPN	Risk priority number
CCF	Common-cause failure	S&Q;	Safety and quality (function)
CFIT	Controlled flight into terrain	SDP / FDP	Surveillance / flight data processing
CPDLC	Controller-pilot data link communications	SFL	Selected flight level
CPT	Conditional probability table	SMS	Safety management system
CRC	Cyclic redundancy check	SPI	Safety performance indicator
CWP	Controller working position	SRM	Safety risk management
eTOD	Electronic terrain and obstacle data	STCA	Short-term conflict alert
FAT / SAT	Factory / site acceptance test	STPA / STAMP	System-theoretic process analysis / accident model and processes
FHA	Functional hazard analysis	UCA	Unsafe control action
FMEA / FMECA	Failure mode and effects (and criticality) analysis	VCCS	Voice communication control system
FRAM	Functional resonance analysis method		

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Appendix C. Assessment template — new AIM system acquisition

This appendix sets out a complete method chain for the safety assessment of a new Aeronautical Information Management (AIM) system, from specification and installation through data migration and transition into operation. It is available in NAVRAP as the assessment template **“New AIM system acquisition”**: choosing it when creating an assessment creates the fourteen studies below, pre-filled with the functions, nodes, failure modes, fault tree, barriers and argument described here. The content is a proposal — the team reviews it, confirms severities and rates likelihoods in the workshops. NAVRAP demo project DEMO-04 shows the template completed with illustrative values.

C.1 What makes an AIM change different

For most systems the dominant hazard is loss of service. For AIM it is **erroneous data published without detection**: a wrong threshold coordinate, obstacle elevation or NOTAM that users trust and act on. The assessment is therefore organised around data integrity — the data integrity classification of ICAO Annex 15 and PANS-AIM (critical, essential, routine) drives the severity of each failure condition — and around the transition, when data migration, unfamiliar tools and the cut-over itself concentrate risk into a short period.

C.2 Method chain by phase

Phase	Method (chapter)	Question answered for the AIM system
0 Scope	GSN skeleton (22)	Argument G0 “the new AIM system and its transition are acceptably safe”, split over specification, design, installation, migration, transition, people and monitoring; filled as evidence arrives.
1 Specification (FHA)	FHA (12)	For each AIM function: effect of loss, delay, detected and undetected error; severity from the data integrity class; safety objectives.
	Data-flow HAZOP (8)	Deviations along originator → AIS → database → publication/NOTAM → users, and legacy → new migration (guidewords no, less, part of, other than, early, late, reverse).
2 Design (PSSA)	FMEA (10)	Failure modes of servers, fail-over, database, CRC module, validation rules, AMHS gateway, time source, web publication.
	FTA + CCA (14, 23)	“Erroneous critical data published undetected”: are validation and verification really independent?
	STPA (13)	Unsafe control actions in the officer-automation loop (auto-fill, warnings, effectivity, NOTAM cancellation).
	HTA + HRA (26, 20)	Task steps of NOTAM publication; error probabilities for data entry and verification used in the FTA.
	Security (29)	Integrity attacks, vendor remote access, web-facing services, ransomware; scenarios with a safety effect.
3 Installation	JHA (9), SWIFT (25)	Works beside the live legacy AIS: power, network, AMHS, test messages reaching live users.
4 Data migration	HAZOP node N7, comparison report	Records lost or values changed by conversion; 100% comparison of critical data, legacy vs new.
5 Transition	Bowtie (6), shadow operation (27)	Top event “loss of, or erroneous, AIS during transition”; go/no-go, rollback; shadow operation over at least one AIRAC cycle against predefined criteria.
6 Operation (SSA)	SPIs, ERC (21)	Errors found after publication, NOTAM corrections, time to distribute; 30- and 90-day reviews; close the argument.

C.3 FHA — functions and failure conditions

The template proposes eight functions (F1 receive raw data; F2 validate and verify; F3 store and manage; F4 create and distribute NOTAM; F5 publish AIP/eAIP; F6 charts and terrain/obstacle data; F7 PIB and data feeds; F8 interfaces) and twelve failure conditions. The most demanding are:

Failure condition	Type	S	Main safety requirement
Wrong critical data passes validation and is published	Erroneous (undetected)	B	Independent verification against the source; 32-bit CRC end to end; second-source comparison
NOTAM with wrong content (location, runway, times)	Erroneous (undetected)	B	Four-eyes check against the request; automated validation
PIB omits a relevant NOTAM	Partial loss	B	PIB regression tests; comparison with legacy during shadow operation
Superseded data served after the AIRAC date	Erroneous (undetected)	B	Automated effectivity test at each AIRAC switch
Wrong obstacle or terrain data in charts / eTOD	Erroneous (undetected)	B	Verification against survey source
NOTAM service unavailable	Total loss	C	Manual NOTAM procedure via standalone AFTN terminal

C.4 Data-flow HAZOP nodes

Node	Data flow	Design intent
N1	Originator → AIS	Complete, correct, authorised raw data in the agreed format and in time
N2	Entry, validation, verification → database	Entered exactly as the source; validated; verified independently; CRC-protected
N3	Database → AIP/eAIP (AIRAC)	Correct data set with correct effectivity on the AIRAC schedule
N4	NOTAM → AFTN/AMHS → users	Correct NOTAM to all addressees before effective time; cancelled when the condition ends
N5	Database → charts, eTOD	Derived from the current verified data set
N6	AIM → PIB, FDP/ATFM, data users	Complete and current information for each user
N7	Legacy → new database (migration)	All records migrated completely and correctly with integrity protection preserved

Fifteen deviations are pre-filled (two or three per node); the *Deviations* button in NAVRAP generates the remaining parameter × guideword combinations for the team to complete or mark “not meaningful”.

C.5 Fault tree — erroneous critical data published undetected

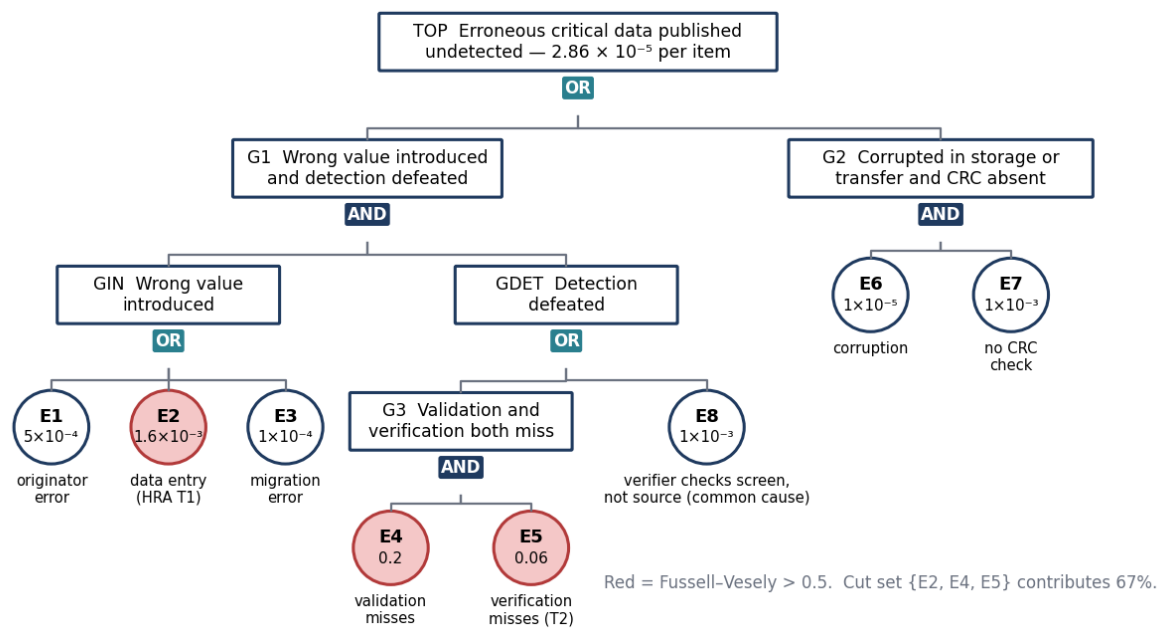


Figure 21. Fault tree for erroneous critical data published undetected, per critical data item (illustrative probabilities; E2 and E5 from the HRA study).

With the illustrative values the top event is **2.86×10^{-5} per critical data item**. Two-thirds comes from the cut set {data-entry error, validation misses, verification misses}; the verification step and the validation rules each have a Fussell-Vesely importance of 0.92. At 40 critical items per AIRAC cycle (672 hours) this is about 1.7×10^{-6} per hour, which does not meet a severity-B objective of 10^{-7} per hour. The finding drives the requirement for a further independent barrier — an originator proof-check of critical data before publication and an automated second-source comparison — and the common cause analysis item CM-01: the verifier must check against the source document, not against the system display (event E8).

Two points about CRC protection are worth stating plainly: a 32-bit CRC (CRC-32Q) protects data *after* it is created against corruption in storage and transfer (G2), but cannot detect a value that was wrong when entered (G1). Both branches need their own barriers.

C.6 Transition bowtie

Threats → preventive barriers	Top event	Recovery barriers → consequences
Data migration error → 100% critical-data comparison; shadow operation Software defect under live load → FAT/SAT with load and CRC test vectors; shadow operation Staff unfamiliar with new HMI → training and competence check Interface misconfiguration → end-to-end interface tests; go/no-go Cut-over on AIRAC date or NOTAM peak → cut-over window and change freeze; go/no-go Installation disturbs legacy AIS → method statement and permit to work	Loss of, or erroneous, aeronautical information service during transition	Post-publication check and correction; data-house notification → erroneous data to users (B) Tested rollback; manual NOTAM via AFTN terminal; service-status NOTAM → NOTAM service interrupted (C) Legacy kept in parallel until end of hypercare → AIRAC cycle missed (C)

Escalation factors: vendor support not on site during hypercare (weakens rollback) and schedule pressure shortening the shadow operation (weakens the main preventive barrier). Each barrier carries an SPI, e.g. daily critical-data differences, open SAT defects by severity, rollback test time.

C.7 Shadow operation acceptance criteria

Measure	Criterion	DEMO-04 result (illustrative)
M1 Critical-data differences, new vs legacy, per day	0	One difference on day 4 — not met ; root cause and repeat before go-live
M2 NOTAM processing time, request to distribution	No worse than legacy + 2 min	Met (14.5 vs 13.6 min)
M3 NOTAM rejected by validation or AMHS per day	≤ 2	Met (mean 1.6)

C.8 Safety argument skeleton

G0 is supported by strategy S0 over the SAM lifecycle and seven sub-goals: G1 safety objectives set (FHA, HAZOP); G2 design meets objectives (FTA, CCA, FMEA, STPA, security); G3 installation does not degrade the legacy service (JHA, SWIFT); G4 migrated data complete and correct (comparison report — undeveloped until produced); G5 transition controlled with a tested rollback (bowtie, shadow operation); G6 staff competent and procedures supportive (HTA, HRA); G7 post-implementation monitoring confirms the assumptions (undeveloped until the 90-day review). Every solution is linked to its NAVRAP study, so the argument shows which evidence is still provisional. Context: data integrity classification; assumption A1: vendor software assurance evidence (EUROCAE ED-153) is available.

C.9 Using the template

1. Create the project (type of change: system) and a new assessment; choose *Start from: New AIM system acquisition*. Scope, environment and assumptions are pre-filled; edit them for the actual contract.
2. Hold the FHA and HAZOP workshops first; confirm severities, add functions and nodes specific to the product, and set likelihoods.
3. Replace the illustrative FTA probabilities with AirNav data or HRA values; recalculate and compare with the FHA objectives.
4. Complete CCA, STPA, security and FMEA with the vendor during design reviews; record vendor evidence as GSN solutions.
5. Before installation, run the JHA and SWIFT with the site and network teams; before cut-over, agree shadow-operation criteria and go/no-go.
6. Send hazards to the hazard log, assign actions, and submit the assessment for review once G1-G6 are supported.

Appendix D. Assessment template — major organisational change

This appendix gives a complete method chain for the safety assessment of a major change in organisation structure — merging units, combining posts, moving functions between levels, outsourcing — from planning, through execution, to post-implementation review. It is available in NAVRAP as the assessment template **“Major organisational change”**, which creates eleven pre-filled studies. NAVRAP demo project DEMO-05 shows the template completed for an illustrative reorganisation of a branch: APP and TWR merged into a Terminal Unit, watch supervision and OJT coordination combined in Operations Supervisor posts, and the branch Safety & Quality (S&Q;) function and competence assessment moved to regional units.

D.1 What makes an organisational change different

No equipment fails. The hazard is that **safety-related functions, responsibilities and barriers are lost, weakened or left without a competent owner** as posts, people and reporting lines move — and that the SMS itself (risk acceptance, reporting, investigation, independence of the safety function) is degraded. Three questions organise the assessment: does every safety function still have a competent owner with capacity and authority; do the control and feedback loops still work; and is the transition itself controlled? Likelihoods are hard to quantify, so the assessment leans on structured judgement, leading indicators and an explicit safety argument rather than on numbers.

D.2 Method chain by phase

Phase	Method (chapter)	Question answered
0 Scope	GSN argument (22)	G0 “the new organisation delivers all safety-related functions at least as safely as the current one, including during transition”.
1 Planning: baseline	Organisational function map (D.3)	Every safety-related function: current owner → new owner; competence, capacity, authority, handover. Rows with gaps become hazards; functions with no new owner (orphans) are the typical finding.
	FHA of organisational functions (12)	Failure conditions of each function: lost, degraded, delayed, or performed without authority; severity through the operational consequence.
	STPA of current and new structure (13)	Broken or filtered feedback loops; loss of independence of the safety function; merged posts that control and check themselves.
	HAZID, organisational guidewords (7)	Roles, authority, competence, workload, interfaces, communication, corporate memory, contractors, culture, transition.
	Bowtie library review (6)	Every barrier owner in existing bowties re-mapped to a post in the new structure (rows in the function map).
2 Planning: people	HTA of merged posts (26); staffing and fatigue model (15)	Does a merged post fit in the time available at peak? Do rosters and staffing still meet the limits?
	Delphi (28)	Defensible likelihood estimates where there is no data.
3 Execution	SWIFT of the transition plan (25); transition bowtie (6)	Vacant and acting posts, handover of files and open actions, parallel reporting lines, delegations, key staff leaving; phased go/no-go and reversion.
4 Post-implementation	SPI register (D.5); safety culture survey; ERC and HFACS/SOAM (21, 30)	Leading indicators with alert levels and escalation; reviews at 3, 6 and 12 months; reversal criteria; close the argument.

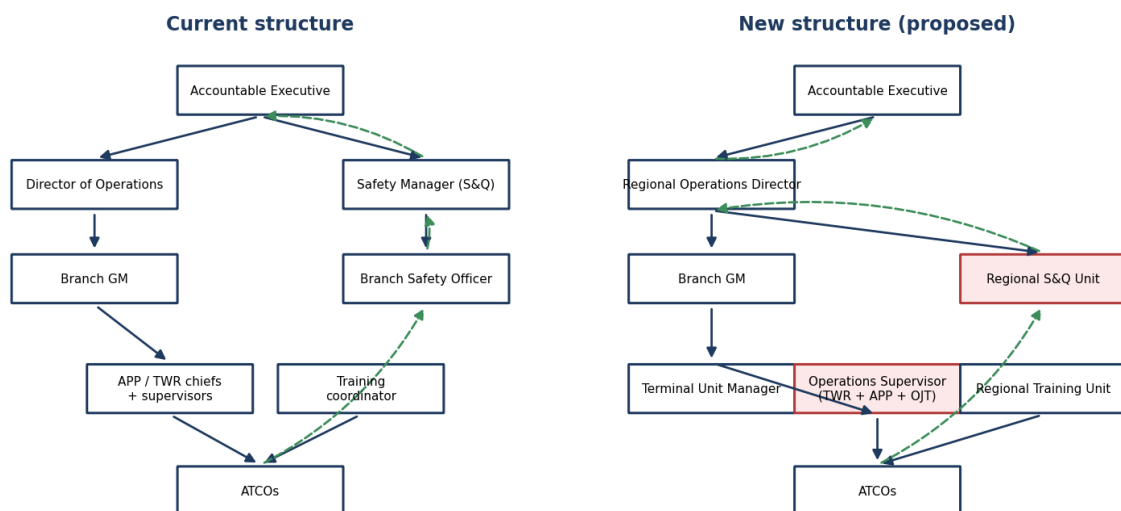
D.3 The organisational function map

The function map is the core evidence. It lists every safety-related function — SMS roles, operational supervision, training and competence, technical functions, and the owner of every barrier in the bowtie library — with its current and its new owner, and four checks. NAVRAP flags a row automatically when any check fails.

Check	Values	A gap when...
New owner	named post	blank — the function is an orphan
Competence	confirmed / gap / unknown	“gap”
Capacity	adequate / stretched / overloaded / unknown	“stretched” or “overloaded”
Authority	in place / pending / none / unknown	“pending” or “none” (e.g. delegation not signed)
Handover	done / planned / not planned / unknown	“not planned”
Any	—	“unknown” is shown as “to assess”; none may remain at go-live

In the DEMO-05 example, 13 of 18 functions have gaps. Two are orphans — maintenance of the letters of agreement and ownership of the A-SMGCS configuration barrier — because neither appears in any new job description. Occurrence triage moves to a regional unit whose volume triples with the same staff (capacity overloaded); the trained investigator stays behind (competence gap); and the Safety Manager would report through the Regional Operations Director (authority none — loss of independence).

D.4 Control structures: current and new



Red: S&Q now reports through operations (no direct line to the AE); merged supervisor post.

Figure 22. STPA control structures before and after the change (DEMO-05). Solid: control actions; dashed: feedback.

Comparing the two structures shows what a worksheet misses. In the new structure the safety function’s feedback reaches the Accountable Executive only through the operations report (UCA-2), the operations line can set investigation priorities in its own area (UCA-1), and the merged Operations Supervisor may split sectors late while managing OJT and an event at the same time (UCA-3). The requirements that follow — a direct line from S&Q; to the Accountable Executive, a deputy supervisor at peaks, a named owner for every function — are structural, which is why they justify real risk reduction.

D.5 Transition bowtie and SPI register

Top event: *safety-related function not performed, or performed without competence, capacity or authority*. Threats: orphan functions; new owner without competence or authority; overloaded merged posts; loss of key staff and corporate memory; broken or filtered feedback lines; uncertainty and stress. Preventive barriers: signed function map; delegations and competence verified before each phase; workload assessment and deputies; retention and knowledge transfer; STPA review with an independent safety function; communication and staff support; phased go/no-go. Recovery barriers: weekly transition safety board, escalation route, reversion of a function to its previous owner, independent audit at 3 months, organisational-factor coding in investigations. Escalation factors: phase change in the peak season; budget pressure on posts.

Each barrier carries an SPI. The template's SPI register has ten indicators — mostly leading, because occurrence counts react too slowly:

SPI	Target	Alert	DEMO-05
Vacant or acting safety-critical posts	0	≥ 2	1 (amber)
Function-map rows with gaps	0 at go-live	≥ 1 at go-live	13 (red — no go-live)
Occurrence reports triaged within 72 h	$\geq 95\%$	$< 85\%$	88% (amber)
Voluntary safety reports per month	$\geq$ baseline (24)	drop $> 30\%$	19 (amber)
Overtime per ATCO per month	≤ 10 h	> 12 h	11 h (amber)
Median days to close an investigation	≤ 30	> 60	41 (amber)
Safety culture survey index	$\geq$ baseline (68)	drop ≥ 5	at 6 months

A falling number of voluntary reports during a reorganisation should be read as a warning about reporting culture, not as an improvement in safety.

D.6 Rating likelihood with Delphi

For “a serious-occurrence report is not triaged within 72 h in the first six months after transfer”, six experienced staff gave estimates from 0.02 to 0.25 in round 1; after feedback the median settled at 0.12 and the interquartile range narrowed from 0.13 to 0.015 by round 3. The agreed value supports rating the hazard before mitigation and shows why extra analyst capacity is needed.

D.7 Using the template

1. Create the project (type of change: organisational) and a new assessment; choose *Start from: Major organisational change*. Replace the example functions, posts and structures with those of the actual change.
2. Complete the function map with HR and line managers; every “unknown” must be resolved. Include every barrier owner from the bowtie library.
3. Draw the current and new control structures; derive unsafe control actions and requirements from the differences.
4. Run the organisational HAZID and the HTA of merged posts; use Delphi for likelihoods where data are lacking.
5. Before each phase: SWIFT of the transition step and go/no-go against the transition bowtie and the SPI “rows with gaps = 0”.
6. After go-live: run the SPI register, reviews at 3, 6 and 12 months and the culture survey; close G7 of the argument only when the survey is no worse than baseline.

Appendix E. Investigation expert system — ORLIO case library

This appendix describes an expert system that helps investigators in the initial phase of an ATS occurrence investigation. It finds similar past occurrences, compares the evidence known so far with the evidence those occurrences needed, suggests the likely contributing mechanism, and — most useful in practice — points to the evidence worth collecting next, while it is still available. It is available in NAVRAP as the study “**Investigation expert system (ORLIO)**” and the **Case library**; demo project DEMO-06 shows an illustrative loss-of-separation investigation.

The design follows Ng, Bil, Sardina and O’Bree (2022) [R74], who combined the ATSB ORLIO reason model with AcciMaps and fuzzy inference, and adds a Bayesian network across the whole case library and AI-assisted drafting of AcciMaps. The system is advisory: it suggests, investigators conclude. Its outputs do not apportion blame or liability, and they are investigation records protected in the same way as the rest of the investigation (ICAO Annex 13 §5.12, Annex 19).

E.1 Scope at AirNav

Accident and serious-incident investigation in Indonesia is the responsibility of the KNKT. The system is aimed at AirNav’s own ATS occurrence investigations under its SMS (Chapter 30) — loss of separation, level bust, runway incursion, coordination failure, communication loss — and at supporting KNKT with ATS evidence. Each occurrence category has its own list of *mechanisms* (for loss of separation: controller clearance error; readback/hearback error not detected; pilot deviation; coordination failure; conflict not detected; safety-net or system deficiency) and its own *factor catalogue*: the ORLIO factors that recur in that category, each with keywords.

E.2 The knowledge base: past occurrences as AcciMaps

Each past occurrence is stored as an AcciMap in the ORLIO lanes — Organisational influences, Risk controls, Local conditions, Individual actions, Technical failure mechanisms (beside individual actions, as in the ATSB five-level diagram), Occurrence events (ATSB [R75]). Every catalogue factor carries its ATSB safety factor type code (Appendix F.4), so cases from AirNav and from ATSB-coded reports are comparable. Blocks are of three kinds:

Block	Holds	In inference
Evidence	A fact established in the investigation, with its ORLIO factor code, its state in the past occurrence and a quote from the report	Rated by the investigator for the current occurrence
Hypothesis	An intermediate conclusion the investigators drew from evidence (e.g. “clearance acted on by the wrong aircraft”)	Inferred from its input blocks by fuzzy rules
Finding	The occurrence and its mechanism: a verbal probability for every mechanism of the category	Inferred; exactly one per case

Links are either *inputs* (used by the rules of the target block) or *context* (shown, not used): organisational and risk-control blocks usually explain the occurrence without being needed to establish its mechanism, and are reached later in an investigation. Factor codes link blocks across cases, so a rating given once applies to every case that uses the factor, and they are the variables of the Bayesian network (E.6). A case is used by the system only after a reviewer has approved it; editing an approved case returns it to draft.

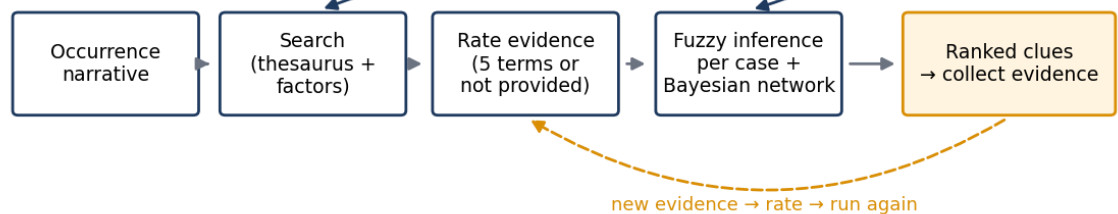
Building the knowledge base**Using it in an investigation**

Figure 23. Building the case library and using it in an investigation.

E.3 Rating the evidence

The investigator rates each factor of the current occurrence on the five-term scale of the ATSB evidence tables [R75]. Each term is converted to a fixed value in the centre of its band. Leave a factor **Not provided** when it is not yet known — this is what produces clues. **Unsure** records evidence whose relevance cannot be determined: it adds no evidence and is not asked for again. (The ATSB evidence tables themselves rate items supports / opposes / no effect / unsure and judge strength through credibility — see Appendix F; the five grades here are NAVRAP’s quick rating for comparing cases.)

Term	Band	Value	Use when
Strongly support	95-100	97.5	Established by physical or recorded evidence: radar replay, RT recording, system log, laboratory test
Support	66-95	80.5	From interviews, witness statements or expert judgement
No effect	34-66	50	Present but judged not to matter for this occurrence
Oppose	5-34	19.5	Witness or judgement indicates it did not apply
Strongly oppose	0-5	2.5	Recorded or physical evidence shows it did not apply
Unsure	—	—	Evidence exists but its relevance cannot be determined
Not provided	—	—	Unknown so far (default)

E.4 Fuzzy inference on each similar case

Every hypothesis and finding block is a small Mamdani fuzzy system [R78]: trapezoidal input sets that do not overlap, triangular output sets, AND = minimum, implication = minimum, aggregation = maximum, centroid defuzzification. Hypotheses output the support scale; findings output a verbal probability in the ATSB terms (Very unlikely <10, Unlikely 10–33, About as likely as not 33–66, Likely 66–90, Very likely 90–99, Virtually certain >99; Appendix F.2) for each mechanism. NAVRAP generates a default rule set from the past occurrence, which the knowledge engineer may edit:

Rule	IF	THEN
Confirming (the rule of [R74])	every input in the state it had in the past occurrence (Support or Strongly support)	the past conclusion
Weakening (one per input)	that input only “No effect”, the others as in the past	one step weaker (Very likely → Likely)
Contradicting (one per input)	that input opposed	hypothesis Oppose; the past mechanism Improbable, the others More or less likely

Two points differ from [R74]. First, “Not provided” is treated as unknown, not as a middle value: a rule that needs it cannot fire, and a block for which no rule fires is **not determined**. Second, the weakening and contradicting rules let the system reason about opposing and neutral evidence, which a confirming rule alone cannot.

Clues. For every block that cannot be determined, the system lists the not-provided evidence its confirming rule needs, with a priority equal to the share of that rule already satisfied. With the evidence of the paper’s AcciMap 23 all rated Support, NAVRAP reproduces the published result (Very likely 94.54 for fuel exhaustion, 4.959 for the other five categories; reference test TC-IES-01 — the paper does not publish the peaks of its output sets, so NAVRAP places them to give these values); with two of the four fuel-quantity items not provided, the finding is not determined and those two items are returned as clues — the paper’s example of the system “asking” for evidence.

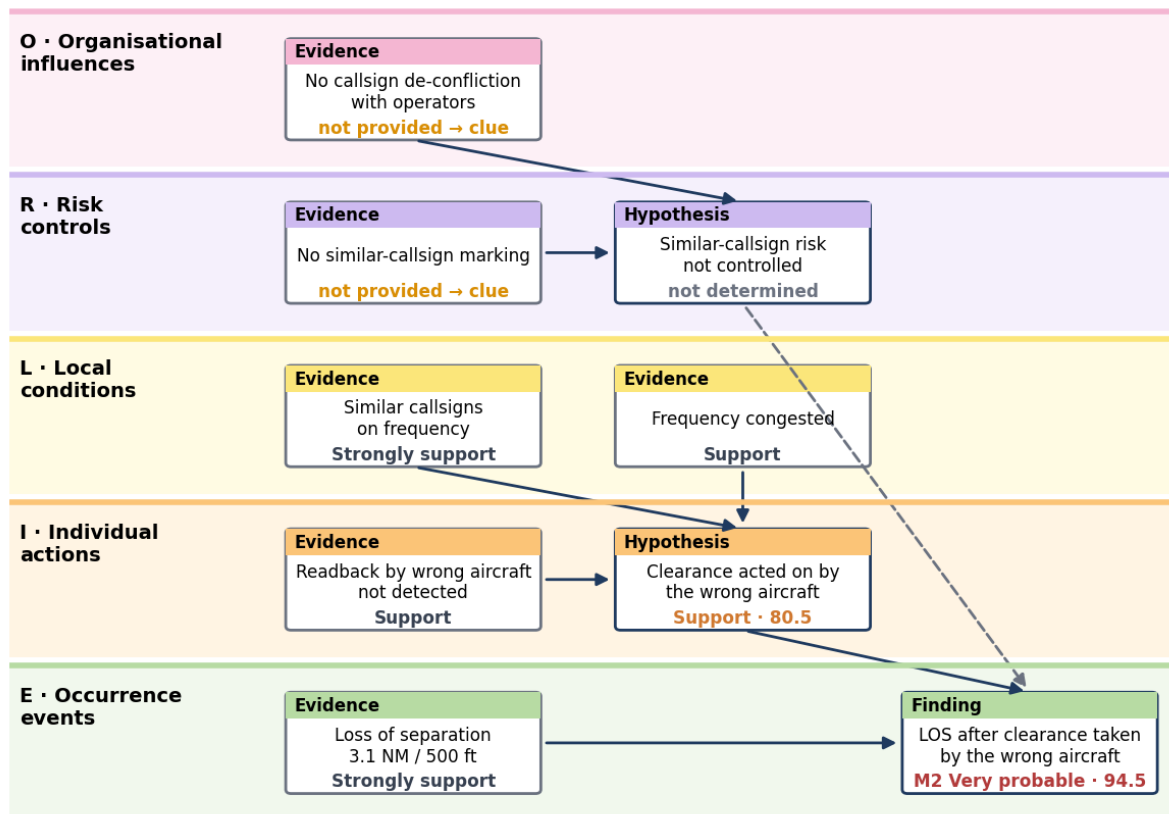


Figure 24. Case CASE-01 compared with the DEMO-06 occurrence. The local-condition, individual-action and event blocks are rated from day-1 evidence, so the hypothesis and the finding (M2 Very likely) are determined; the organisational and risk-control blocks, linked as context, are not yet known and are returned as clues.

E.5 Finding similar occurrences

Reports describe the same thing in different words — “refuel”, “add fuel”, “fuel added”. Before comparing texts, NAVRAP replaces every phrase in a thesaurus by one concept (for example all forms of *loss of separation*, *readback*, *combined sectors*, *STCA*). The score of a case is $0.6 \times$ the TF-IDF cosine similarity of the narrative and the case, plus $0.4 \times$ the share of the ORLIO factors recognised in the narrative that the case contains. Every result shows the matched terms and factors, so the investigator can see why a case was proposed. Factors recognised in the narrative are shown as suggestions; they are rated only by the investigator.

E.6 Bayesian network and ranked clues

The fuzzy comparison looks at a few cases one at a time. The Bayesian network (Chapter 17) uses the whole approved library of the category: a mechanism node is the parent of one present/absent node per factor, with probabilities learned from the cases with Laplace smoothing — $P(\text{mechanism}) = (n + 1)/(N + K)$ and $P(\text{factor present} \mid \text{mechanism}) = (n_f + 1)/(n + 2)$. A rating enters as likelihood (“virtual”) evidence equal to its value: Strongly support 0.975, Support 0.805, Oppose 0.195, Strongly oppose 0.025; “No effect” and “Not provided” add nothing, so unknown factors are marginalised rather than guessed.

For every factor not yet rated, the system computes its **value of information** [R79] — the expected reduction in the uncertainty (entropy) about the mechanism if that factor were established — and ranks the factors. The top of the list is the evidence most worth collecting next. The network can be opened as an ordinary Bayesian-network study, which gives the same posteriors (reference test TC-IES-03).

DEMO-06 (9 illustrative cases)	Prior	After day-1 ratings	If monitoring ruled out (Oppose)
M2 Readback/hearback error not detected	20%	44%	52%
M5 Conflict not detected (monitoring/workload)	20%	20%	12%
M4 Coordination failure	20%	10%	11%
M1, M3, M6 (each)	13%	9%	6–9%
Uncertainty (bits)	2.56	2.23	2.11

In DEMO-06 the day-1 ratings (similar callsigns strongly supported; readback not detected, frequency congestion, combined sectors and night duty supported) move the readback/hearback mechanism from 20% to 44%. The top-ranked clue is *conflict not detected while monitoring* (0.13 bits), because it separates M2 from M5 best; if the controller interview and radar replay rule it out, M2 rises to 52%. The fuzzy comparison agrees: CASE-01 gives M2 Very likely, and CASE-02 and CASE-07 cannot be determined until the monitoring lapse and the trainee question are answered.

The network assumes factors are independent given the mechanism, and a library of a few dozen cases gives rough probabilities. The posterior is a guide to where to look, not a finding. Numbers in DEMO-06 come from illustrative cases, not AirNav data.

E.7 AI-assisted drafting of AcciMaps

Building AcciMaps by hand is the main cost of the system. NAVRAP can draft one from the text or PDF of a final report in two ways: offline, by matching sentences to the factor catalogue; or, when the administrator has configured it, with Claude (Anthropic), which proposes blocks, links and the mechanism in a fixed structure. In both cases every block carries a quote from the report, which NAVRAP checks against the text; a quote that is not found is flagged and blocks approval until corrected. The draft is only a draft: an investigator completes and corrects it and a reviewer approves it. The AI drafts the knowledge base; it does not draw conclusions in an investigation.

Only published final reports may be sent to an external AI service. Protected investigation records — statements, recordings, draft reports — must not leave AirNav; for internal reports use the offline drafter or an AI service approved for that data.

E.8 What is needed to put the system into use

Need	What
Mandate and scope	Occurrence categories to cover first; the system's place in the investigation procedure (advisory, initial phase); agreement with KNKT on use of their reports and on supporting their investigations
Report corpus	KNKT, ATSB, NTSB, BEA and other published final reports, the ICAO e-library and AirNav internal reports — about 30–50 per category, of which about 10 held back for validation
Taxonomy	ORLIO for the layers; ADREP/ECCAIRS for categories; mechanisms and factor catalogue per category, mapped to the HFACS and SOAM codes of Chapter 30
People	A knowledge engineer who builds and maintains the library; two or three experienced investigators per category who review AcciMaps and rules; an independent validation panel
Validation	Run the held-back reports through the system and compare with their published findings (mechanism in the top two; share of contributing factors found by the clues); check agreement between investigators rating the same evidence; agree the targets before use
Protection and governance	Need-to-know access and audit trail; advisory status and no-blame use stated in the procedure; versioned rules; decision on which (if any) external AI service may receive which reports
Upkeep	Every new final report in a covered category becomes a case; the factor catalogue and thesaurus are reviewed yearly

E.9 Using the system

1. Create an assessment for the investigation and add the study *Investigation expert system (ORLIO)*; choose the occurrence category.
2. Describe what is known so far in your own words and search the case library. Select up to five similar cases.
3. Rate every factor you have evidence for; leave the rest Not provided. Run the analysis.
4. Read the fuzzy results case by case and the Bayesian posterior; then collect the evidence the ranked clues point to, perishable evidence first.
5. Rate the new evidence and run again. Continue the investigation with SOAM/HFACS (Chapter 30); the expert system does not replace it.
6. After the final report, add the occurrence to the case library so that the next investigation can learn from it.

Appendix F. Safety investigation analysis — the ATSB method

This appendix adopts the analysis method of the Australian Transport Safety Bureau for AirNav occurrence investigations: the ATSB Safety Investigation Guidelines Manual — Analysis [R80] and its Tools Manual — Analysis Supplement [R81]. It turns the ORLIO model (Appendix E) from a diagram into a disciplined argument: every potential safety factor is defined, tested for existence and influence against the evidence, classified with ATSB codes, and — if it is a safety issue — rated for risk and followed through to safety action. In NAVRAP it is the study **“Safety investigation analysis (ATSB)”**; DEMO-06 contains the analysis of the illustrative occurrence OCC-ILL-01. The expert system of Appendix E supports the early phase; this method carries the investigation to its findings. SOAM and HFACS (Chapter 30) remain available as complementary views.

F.1 Key terms

Term	Meaning [R80]	In NAVRAP
Safety factor	An event or condition that increases safety risk (p.20)	Every entry on the safety factors list
Contributing safety factor	A safety factor that, had it not occurred or existed, probably would have prevented the occurrence, made its consequences less serious, or prevented another contributing factor (p.22). ATSB does not rank degrees of contribution and avoids the word “cause”.	Derived: existence + influence on the occurrence or on another contributing factor
Other safety factor	A safety factor that existed and is important for future safety although influence was not shown (p.145)	Derived: existence + importance
Safety issue	A safety factor that can affect future operations and is a characteristic of an organisation or system — only local conditions, risk controls and organisational influences (p.26; [R81] p.50)	Flag on LC/RC/OI factors with owner and risk level
Other key finding	A finding that is not a safety factor: resolving ambiguity, possible scenarios, positive safety factors (p.51)	Basic evidence tables; positive actions and conditions
Standard of proof	Each finding at least “likely” ($\geq 66\%$) (p.79)	Tests below 66 % do not pass

Probability expressions follow the IPCC-based scale of [R80] p.73: virtually certain $>99\%$, highly likely $>95\%$, very likely $>90\%$, likely $>66\%$, more likely than not $>50\%$, about as likely as not 33–66 %, unlikely $<33\%$, very unlikely $<10\%$, exceptionally unlikely $<1\%$. Lack of knowledge is not expressed on this scale: write “there was insufficient evidence to determine ...”. The same terms label the verbal probabilities of Appendix E.

F.2 The analysis process

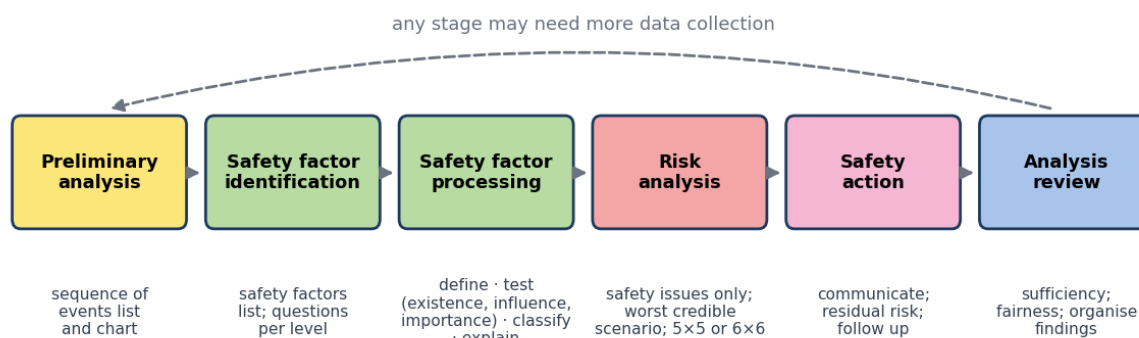


Figure 25. The five components of investigation analysis, with the sequence of events inside preliminary analysis (adapted from [R80] pp.11–12).

Sequence of events list. Each event has a start time (mandatory; estimated times are noted in comments), an optional end time, a title written as a subject and an action verb (“ABC621 left FL320 climbing”, not “level bust”), comments, source, theme (ATC, aircraft, crew...) and whether it is shown on the timeline. An event can be promoted to the safety factors list as an occurrence event, individual action, local condition or positive action ([R81] p.11).

Safety factors list. One row per potential factor: title, type, codes, whether it is analysed further (a “no” needs one of the listed reasons — e.g. it restates the problem, duplicates another factor, or clearly did not exist; when in doubt, include it), safety issue, existence, finding type, risk level and whether the analysis is complete ([R80] pp.135–137).

F.3 Safety factor types and codes

NAVRAP holds the ATSB aviation taxonomy ([R81] App. D) with its coding notes. The level-1 types and level-2 codes are:

Type	Level-2 codes (level 3 in NAVRAP)
Occurrence event	coded as an occurrence type (e.g. loss of separation)
Individual action	I1 aircraft operation · I2 aircraft maintenance · I3 ATS action (I3.1 assessing and planning, I3.2 using equipment, I3.3 communicating, I3.4 handover/takeover, I3.5 monitoring and checking, I3.6 other) · I4 other; with the individual’s role and error type (information, decision, action, violation)
Technical failure mechanism	T1 fracture ... T5 electrical discontinuity, T6 mechanical discontinuity, T7 software/firmware anomaly, T8 other
Local condition	L1 personal factors (L1.3 fatigue ...) · L2 knowledge, skills, experience · L3 task demands (L3.1 high workload, L3.4 distractions, L3.5 incorrect task information ...) · L4 social · L5 workspace · L6 physical environment · L7 weather
Risk control	R1 equipment (R1.4 warning/detection systems ...) · R2 facilities/infrastructure (incl. nav aids, radar coverage) · R3 procedures · R4 training and assessment · R5 people management · R6 technical failure management; preventive or recovery
Organisational influence	O1 safety management processes · O2 organisational characteristics · O3 regulatory influences · O4 other external influences; internal or external
Positive action / condition	PA, PC — actions or controls that reduced risk beyond expectations (e.g. STCA alert, avoiding action)

Functional area “Air traffic control” covers factors within the ANSP and its equipment (radar, ATM consoles). Coding notes guide borderline cases — for example, if high workload came from another condition, that condition is coded; incorrect task information is preferably coded as another person’s communicating action. The factor catalogue of the expert system now carries these codes, e.g. hearback missed I3.3, monitoring I3.5, fatigue L1.3, STCA R1.4, sector combining R3, OJT supervision R5.

F.4 Testing a potential safety factor

Each test is a separate argument recorded in a safety factor evidence table ([R80] pp.84–90, 145–151). Evidence items are rated **supports**, **opposes**, **no effect** or **unsure**, with comments and source, and described by type (tangible, testimonial, accepted fact), relevance (direct, circumstantial, ancillary), credibility concerns (validity, reliability, bias, sensitivity) and whether the item was expected but not seen, or not expected but seen. The conclusion is a judgement on the whole pattern — not a count of supporting and opposing items — stated as supported or not supported with a probability expression.

Test	Question	Result
Existence	Did the potential safety factor exist (or probably exist)?	Fail ⇒ not established; stop
Influence	Did it influence the occurrence or a known contributing safety factor? (name what it influenced)	Pass ⇒ contributing safety factor
Importance	If influence is not shown: is it still worth analysing further? (free-text justification)	Pass ⇒ other safety factor

NAVRAP derives the finding type and applies the rule that a factor whose only influence is on a non-contributing factor is itself an other safety factor ([R81] p.125). It also checks, and lists for the analysis review: an argument from ignorance (a test concluded supported with no supporting item); a conclusion below the standard of proof; existence and influence each only just likely, so that together they may be below 50 % ([R80] p.146); evidence items from one source (not independent); judgemental or vague wording (“failed”, “inadequate”, “significant”) and bundled factors; a safety issue on a type that cannot be one; missing role, error type or codes; and a factor excluded without a reason. The checklist questions of [R81] App. C accompany each test.

F.5 Safety issues and risk analysis

A potential safety issue that passed existence and influence or importance is rated for risk ([R80] pp.154–173): describe the worst possible scenario; review the existing controls and their effectiveness; describe the worst credible scenario (normally one more control failure than the safety issue, not several unrelated ones); rate consequence and likelihood with written justifications; and check sensitivity with an alternative rating. The result is a **critical** or **significant** safety issue, or **broadly acceptable** — in which case the factor keeps its finding type but is not a safety issue.

AirNav rates risk on its own 5×5 matrix (Chapter 3). NAVRAP offers the ATSB 6×6 as an option for safety issues: likelihood 1–6 from “extremely improbable” to “very frequent (at least 10 times a year)” and consequence F–A from “negligible” to “very large vehicle catastrophic”. The ATSB text does not give the cell colours, so NAVRAP fills the 6×6 by the same pattern as the AirNav matrix, with the same colours: the rank sum of likelihood and consequence ($F = 1 \dots A = 6$) gives critical at 10 or more, significant at 6–9 (shaded near-critical 8–9 and lower 6–7, like AirNav’s two tolerable bands) and broadly acceptable at 5 or less. Every catastrophic cell is therefore at least significant, as ATSB requires (p.166). A safety issue rated on the AirNav matrix maps intolerable → critical, tolerable → significant, acceptable → broadly acceptable.

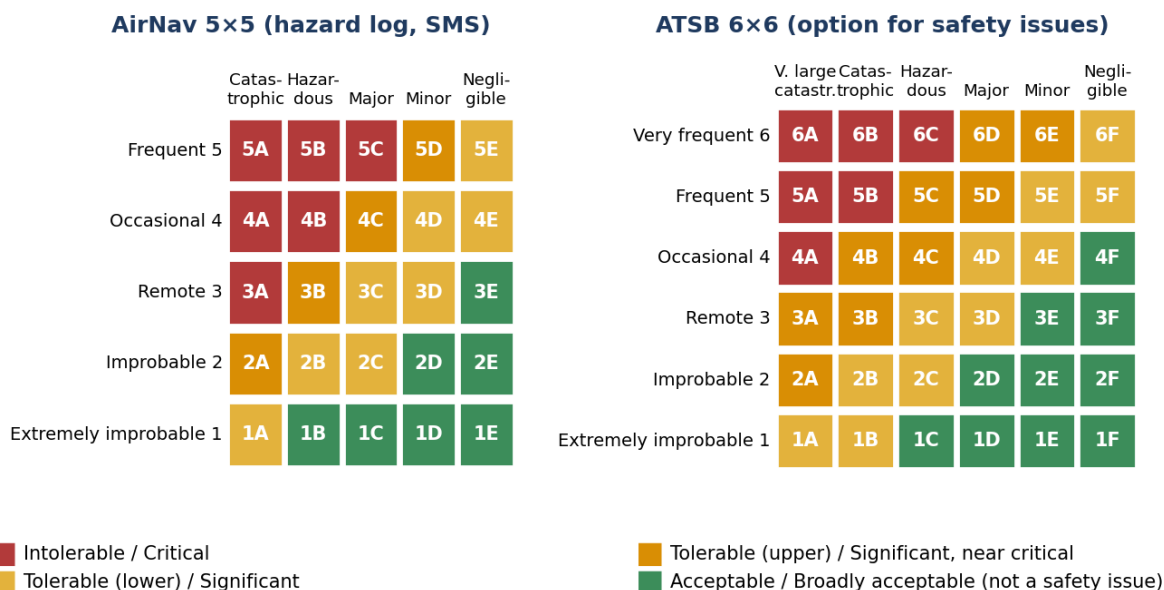


Figure 26. The AirNav 5×5 and the ATSB 6×6 as configured in NAVRAP. The 6×6 colours follow the AirNav regions; the near-critical shade of significant guides how soon safety action is sought.

F.6 Safety action

For each safety issue NAVRAP records the safety actions ([R80] pp.174–184; [R81] App. E): the kind (action by the organisation, safety recommendation, safety advisory notice), the organisation, the date it was notified, a communication log, the status (proposed, monitor, closed) and the classification coded at closure — technical, policy, procedures, training, communication/education, mandatory requirements, organisational surveillance, external surveillance, no action or other, each with sub-types. The action is evaluated by re-rating the residual risk: broadly acceptable, or significant and as low as reasonably practicable, means no further action; otherwise further action is required. Practicability is judged on the level of risk, the state of knowledge, the means available and the cost. The safety issue is pending, adequately, partially or not addressed; while it remains above ALARP, NAVRAP sets a follow-up date six months after the last contact. Safety actions can be tracked in the NAVRAP action list and safety issues sent to the hazard log.

F.7 Analysis review

Before the analysis is closed ([R80] pp.185–189): the **test for sufficiency** — do the verified explanatory factors fully account for each verified factor? (NAVRAP flags contributing factors with no explaining factor); completeness questions; **fairness** — no finding may appear to blame an individual without the reasons for the action having been identified (NAVRAP flags contributing individual actions with no explaining local condition, risk control or organisational influence), and practicability is considered for organisational findings; findings ordered as contributing safety factors, other safety factors and other key findings, and no safety issues merged. The stop rule — where the analysis stopped and why — is recorded.

F.8 Worked example: OCC-ILL-01 (illustrative)

ABC621 climbed on a climb clearance intended for ABC612 (similar callsigns, combined sectors at night); STCA alerted and avoiding action was given; minimum 3.2 NM / 400 ft. The DEMO-06 analysis has 11 events and 12 potential safety factors:

Finding	Factors (ATSB codes)
Contributing safety factors (7)	ABC621 climbed on a clearance intended for ABC612 (occurrence event); the crew accepted and read back the clearance (I1.6); the controller did not detect the readback came from ABC621 (I3.3); similar callsigns on frequency (L3.6); blocked transmissions (L3.6); no similar-callsign procedure or display aid (R3, R1.1); de-confliction not coordinated with the operator (O1)
Other safety factors (2)	night duty (L1.3) and sectors combined above the daytime threshold (L3.1): existence shown, influence not, importance passed
Other key findings (3)	STCA alerted 55 s before closest approach (positive condition); avoiding action within 7 s (positive action); STCA parameters matched the baseline
Safety issues (2)	no similar-callsign control — AirNav 5×5 3B, significant; residual 2B and ALARP ⇒ no further action, partially addressed. De-confliction with the operator — ATSB 6×6 2B, significant; recommendation pending, follow-up overdue
Not analysed further (1)	“The controller lost situation awareness” — restates the problem

The checks show what the review would catch: the operator safety issue rests on two “likely” tests (together about 44 %), its follow-up is overdue, and the blocked transmissions have no explaining factor yet (test for sufficiency).

F.9 Using NAVRAP

1. Add the study *Safety investigation analysis (ATSB)* to the investigation's assessment. Record the occurrence and build the sequence of events list; promote events that are safety factors.
2. Add potential safety factors from the questions for each level; define each one in a complete, neutral sentence and classify it.
3. Run the tests for existence, influence (naming what it influenced) and, if needed, importance, item by item.
4. For safety issues, complete the risk analysis on the AirNav 5×5 (default) or the ATSB 6×6, then record and evaluate the safety action.
5. Work through the checks and the analysis review; send safety issues to the hazard log; use the organised findings for the report.

F.10 Worked example: KNKT.24.10.22.04 — ground collision at Kualanamu

On 16 October 2024 at about 2126 LT, Boeing 737-800 PK-GMP was taxiing onto parking stand W29 at Kualanamu (WIMM) when a lavatory service truck (LST), left unattended with its engine running in the adjacent equipment staging area, rolled and struck the aircraft's tail as it stopped. No one was injured. KNKT found that the LST parking brake had no shoes or linings (removed in September 2023 and not reinstalled), that the wheel chock was not in contact with the tyre, that the staging area sloped 0.57 % toward the aircraft, and that the brake was recorded unserviceable at daily checks while the vehicle was released as serviceable on the belief that chocks could replace the parking brake. The facts, findings, safety actions and the seven recommendations below come from the KNKT final report [R82]; the evidence ratings, probabilities, risk ratings and the actions marked "proposed" are NAVRAP's illustration (demo project DEMO-07).

Finding	Factors (ATSB codes)
Contributing safety factors (11)	LST moved unaided and struck the tail (occurrence event); no brake shoes or linings in the parking brake drum (T8); 0.57 % slope of the staging area (L6.3); chock not restraining the wheel (I4.3); morning-shift operator released the LST as serviceable with brakes marked unserviceable (I4.3); no means to verify chock contact (R3, R1.3); practice of treating GSE with an unserviceable parking brake as serviceable (R6.3, R3); airport supervision checklist without brake or chock items (R3); KP 635/2015 both requiring a parking brake and allowing chocks as the holding device (O3); no DGCA oversight of motorized GSE since 2020 (O3); operator's station safety audits not leading to effective correction (O4)
Other factors (6)	engine left running in neutral; vibration and engine suction as trigger (influence only "about as likely as not"); fatigue and time of day; planning the next task; reconditioning without documentation (safety issue 3D); a single check-sheet column for both brakes
Not analysed further (2)	competency certificate for the lavatory system (not a safety factor); "lost awareness" (restates the problem)
Recommendations (KNKT)	04-G-2024-22.01 to .03 (ground handler), 04-B-2024-22.04 (airport operator), 04-O-2024-22.05 (aircraft operator), 04-R-2024-22.06 and .07 (DGCA)

The ATSB standard of proof leads to two differences from the KNKT report, which lists the vibration/suction trigger among its contributing factors: the trigger is an *other* factor because the evidence could not show which force started the movement, and fatigue is an *other* factor because its influence on the chock placement was not shown to be likely. Neither changes the safety action: the defences that would have held the truck — a working parking brake, a chock in contact, a serviceability check that stops the vehicle — are all contributing safety factors.

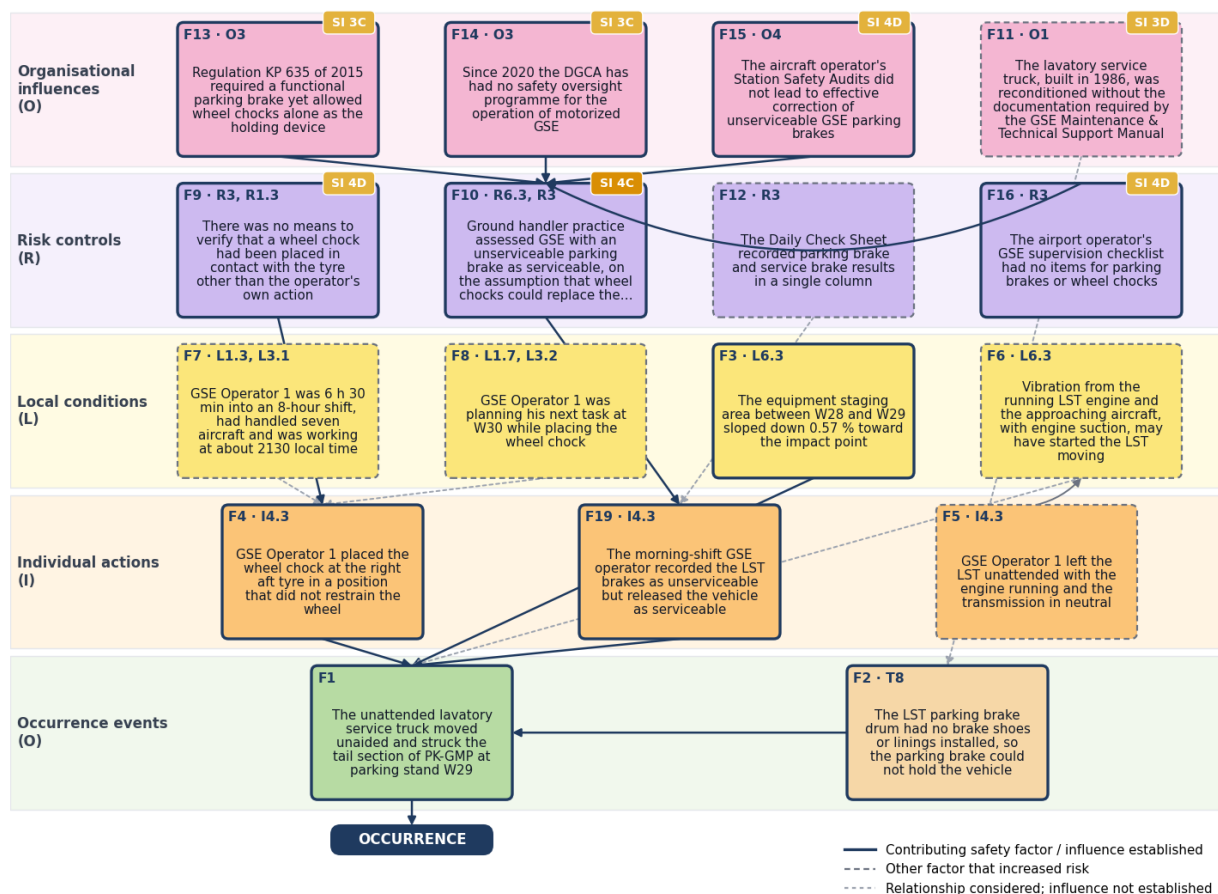


Figure 27. ORLIO factor map of the DEMO-07 analysis (Appendix C of the generated report) [R82].

F.11 The investigation report

NAVRAP generates the report from the analysis. Sections 3–5 and Appendix C come from the safety factors; the investigator adds the factual content and the attachments.

Section	Content
1 Executive summary	Synopsis (what, when, where, who); consequences (injuries, damage, environmental impact); core finding — the primary safety breakdowns traced through the ORLIO layers
2 Factual information	Occurrence timeline; personnel and assets (experience, equipment, environment); immediate actions (emergency response, site preservation, stabilisation)
3 ORLIO analysis framework	3.1 occurrence events (outcomes and technical failure mechanisms); 3.2 individual actions, with why each action made sense at the time; 3.3 local conditions; 3.4 risk controls — failed, absent or bypassed; 3.5 organisational influences
4 Findings and contributing factors	Prioritised list linking each contributing factor through the chain to the occurrence; other factors; other key findings; safety issues; factors not included
5 Safety recommendations and corrective actions	ORLIO layer · identified deficiency · corrective action required (with its level: elimination, substitution, engineering, administrative, PPE) · action owner · target date; every contributing factor needs at least one action
6 Appendices	A photographs and diagrams; B interview summaries and witness accounts; C ORLIO factor map

Corrective actions should be chosen as high in the hierarchy of controls as is practicable. In the example, withdrawing a vehicle with a defective parking brake from service (elimination) and lock-out/tag-out at the daily check rank above a revised procedure or training (administrative), which in turn rank above relying on the operator alone. Formal recommendations state the safety issue; the addressee chooses the means.

Appendix G. Worked example — surveillance radar performance degradation

This appendix follows one realistic technical problem through the whole life of a change and shows which method answers which question in each phase and for each type of hazard. It is the NAVRAP demo project **DEMO-08**: six assessments (one per phase), 27 studies, twelve hazard-log entries and eight actions. All names, numbers and events are illustrative.

G.1 The situation

The Bukit Sari MSSR Mode S (fictitious) is the only radar covering ACC Sector E3 (FL100–FL245) below FL150 in the north-east quadrant; above FL150 the Gunung Tinggi MSSR overlaps in the west, and three ADS-B ground stations cover the sector above FL100 (about 70 % of flights are ADS-B equipped). From June to August 2026 the radar degraded gradually: Mode S/SSR probability of detection fell from 99.6 % to 97.1 % overall and to about 94 % between 030° and 090°; azimuth bias grew to 0.14°; reflected false targets appeared after a telecommunication tower was built 1.2 km north-east of the site; Mode C garbling rose in dense traffic. The remote monitoring (RMM) raised no alarm because its thresholds detect hard failures only. Controllers described the tracks as 'jumpy' but few formal reports were made. On 21 August a loss of separation (OCC-RDR-01) occurred when a coasting track hid a converging aircraft. The radar's rotary joint and azimuth encoder were found worn; they are replaced during night outage windows, after which the radar is re-aligned and returned to service.

G.2 Choosing methods by phase and hazard type

Seven hazard types are considered: technical, human performance, procedural/operational, organisational, environmental (RF, structures, weather), security and occupational (the workers doing the refurbishment). Figure G.1 shows which study addresses which type in which phase. Broad methods (HAZID, bowtie, SWIFT, GSN) span many types; specialised methods (FMEA, HRA, JHA, security risk) go deep into one. Three methods are not used: wildlife strike risk (no wildlife hazard), the organisational function map (no function changes owner) and the investigation expert system (the case library holds no surveillance-degradation cases yet — OCC-RDR-01 should be added once its AcciMap is approved).

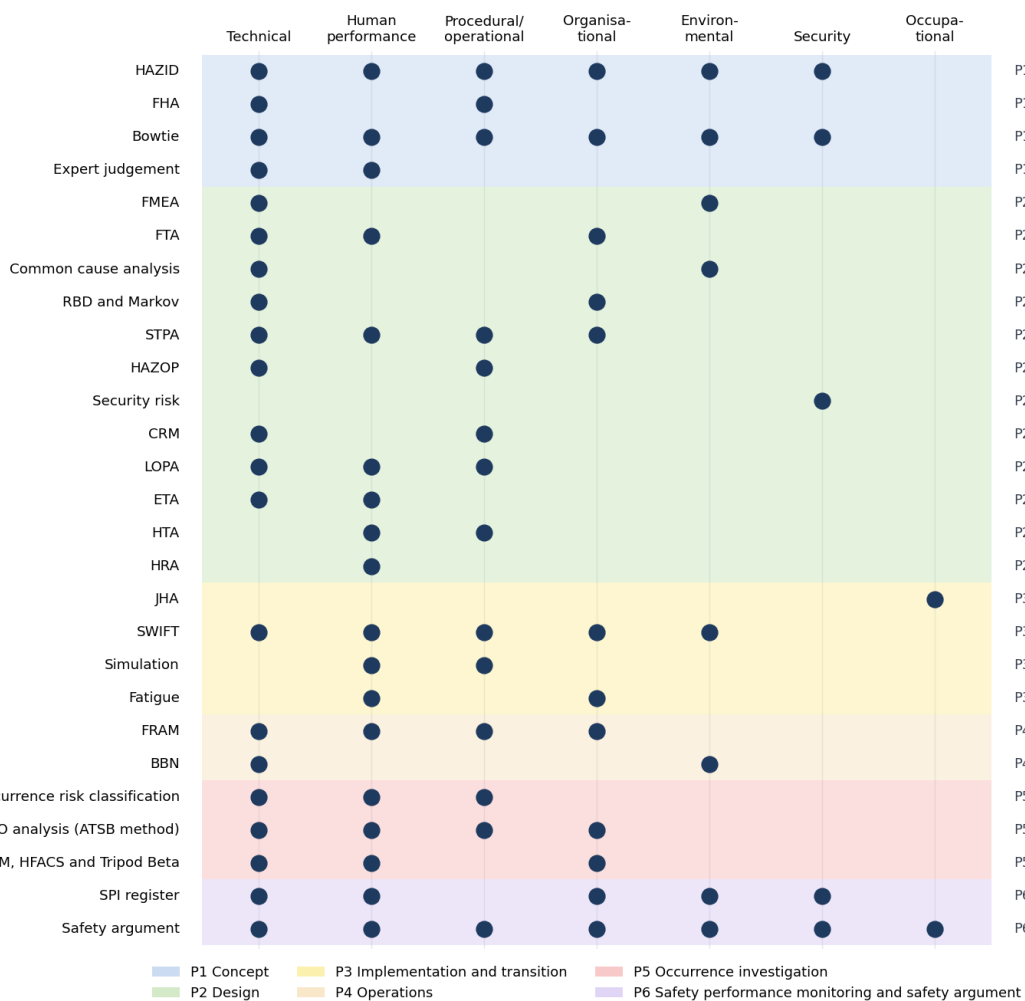


Figure 28. Studies of DEMO-08 by lifecycle phase (rows, shaded) and hazard type (columns).

G.3 What each phase found

Phase	Methods	Key results
P1 Concept	HAZID, FHA, bowtie, expert judgement	Ten hazards across six types. FHA: undetected erroneous position of one aircraft and degradation not indicated are Hazardous (B). Bowtie: the top event is that the controller separates on degraded data without knowing it; STCA is on the recovery side but shares the surveillance data. Cooke and Delphi (median 0.09) estimate how often a 2 NM track jump goes unnoticed for one scan.

Phase	Methods	Key results
P2 Design	FMEA, FTA, CCA, RBD/Markov, STPA, HAZOP, security, CRM, LOPA, ETA, HTA, HRA	FTA: 9.4e-06 per flight hour in the degraded state, 9.4e-07 with performance thresholds (monitoring PFD 0.5 → 0.05) — still above the FHA objective, so increased separation is needed while degraded. Markov: availability of in-specification radar 0.823 today versus 0.961 if degradation is found within a week. CCA and LOPA: STCA is not an independent layer; LOPA 2e-04/yr against 1e-5 — the gap is closed only by cutting the initiating frequency (thresholds, repair). ETA: 9e-04/yr near-collisions with STCA at 0.5. CRM: routes 10 NM apart need 11.7 NM without radar monitoring → level allocation while degraded. HRA: controller misses a coasting track 0.09; ATSEP misses gradual degradation without thresholds 0.20.
P3 Implementation	JHA, SWIFT, simulation, fatigue	JHA: LOTO of drive and transmitters, fall arrest, lifting plan, RF check before radiating. SWIFT: registration against ADS-B as a go/no-go hold point. RTS without the radar: ISA 3.76 (≤ 4.0), R/T 59 %, 22 flights/h — capacity 22/h, one loss of separation in eight runs (criterion not met: refresher training before the first window). Fatigue: maximum KSS 6.99 on four 12-h nights versus 6.50 on six 8-h nights → use 8-h nights.
P4 Operations	FRAM, BBN	FRAM: the controllers' informal position checks by RT (work-as-done) dampen the variability of the display, but anomaly reporting is "not at all" — the feedback loop to maintenance is missing. BBN: with Pd drop, azimuth bias and false targets but no garbling, P(rotary joint worn) = 0.75, P(new reflector) = 0.48.
P5 Investigation	ERC/RAT, ATSB method, SOAM/HFACS/Tripod	OCC-RDR-01 ERC 102 (red); false-target and garbling occurrences ERC 1 and 50. ATSB: 8 contributing factors; safety issues F6 (no performance thresholds) and F7 (no degraded-mode procedure); the controller acted on the display he had (local rationality). SOAM links the failed barriers to the bowtie.
P6 Monitoring	SPI register, GSN	Nine SPIs; the Pd SPI alert level (< 97 % in any 60° sector) is the trigger for the degraded-mode declaration. GSN argues over the six phases with 26 solutions.

G.4 Lessons of the example

- **Gradual degradation is a hazard of its own.** Monitoring designed for hard failures cannot see it; performance thresholds and a declaration procedure turn it into a detected failure, which the operation already knows how to handle.
- **Check every independence claim.** STCA, the controller and the display all use the same tracks. Crediting STCA as a separate layer would have hidden a factor-of-ten gap (LOPA, CCA, ETA).
- **Quantify the cost of late detection.** The Markov model shows that finding degradation within a week instead of two months raises in-specification availability from about 82 % to 96 %.
- **Separation must follow data quality.** The degraded mode (8 NM, procedural area, level allocation on parallel routes) is the barrier that holds while the hardware is being fixed.
- **The works carry their own hazards.** Occupational (JHA), transition (SWIFT, return-to-service hold point) and human (fatigue of night crews) hazards appear only in the implementation phase.
- **Close the loop.** The SPIs that would have revealed the problem in June become the trigger for the next declaration.

G.5 Using NAVRAP

1. Open project DEMO-08. Each assessment is one phase; its assumptions list the hazard types it covers.
2. Work through the studies in order; quantitative studies hold their calculated results and can be recalculated after any change.
3. The hazard log (filter on RDR-) shows each hazard with its type and phase in the context field, the study and row it came from, its controls and its actions.
4. The ATSB study generates the investigation report of OCC-RDR-01 (tab 7); the GSN study links every claim to the studies as evidence.